Psychological Steering of Large Language Models

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Abstract

Large language models (LLMs) emulate a consistent human-like behavior that can be shaped through activation-level interventions. This paradigm is converging on additive residual-stream injections, which rely on injection-strength sweeps to find approximately optimal intervention settings. However, existing methods limit the search space and sweep in uncalibrated activation-space units, potentially missing optimal intervention conditions. Thus, we introduce a psychological steering framework that performs unbounded, fluency-constrained sweeps in semantically calibrated units. Our method derives and calibrates residual-stream injections using psychological artifacts, and we use the IPIP-NEO-120, which measures the OCEAN personality model, to compare six injection methods. We find that mean-difference injections outperform a state-of-the-art prompting baseline in open-ended generation on 11 of 14 LLMs, improving performance by 14.3% to 28.9%, overturning prior reports favoring prompting and positioning representation engineering as a new frontier in open-ended psychological steering. Finally, we show that mean-difference injections align with the Linear Representation Hypothesis and provide a reliable, approximately linear control knob for psychological steering. Nevertheless, they also induce OCEAN trait covariance patterns that depart from the Big Two model, potentially suggesting a gap between learned representations and human psychology.

1 Introduction

Large language models (LLMs) are largely shaped by the human mind. In particular, echoing the psychological principle that our words embody key aspects of our psyche, such as our transitory psychological states (Viney, 1983) and personality traits (Pennebaker & King, 1999), recent work has established that LLMs consistently emulate the human behaviors imprinted in their training data (Karra et al., 2023; Feng et al., 2023; Lee et al., 2025). Similarly, in line with the consensus that LLMs learn token-level associations from distributional semantics in data (Im et al., 2026) and the hypothesis that salient behaviors are encoded in language as single words (Galton, 1884; Goldberg, 1993), a growing body of literature shows that LLMs coherently internalize human behaviors, which can be elicited through prompting (Jiang et al., 2023; 2024) and activation interventions (Chen et al., 2025; Allbert et al., 2025).

Recently, motivated by applications such as recommendation systems, social robotics, and artificial societies, efforts have focused on the psychological steering of LLMs. In this context, representation engineering is a promising direction that could enable fine-grained control over behavioral expression and the emulation of a rich mosaic of personas with negligible compute overhead. In practice, this paradigm is converging on additive residual-stream injections, which rely on injection-strength sweeps to find approximately optimal intervention settings across LLM layers. Yet, recent work reports that prompting outperforms representation engineering both overall (Wu et al., 2025) and in psychological steering (Banayeeanzade et al., 2025). This gap may reflect two limitations: First, sweeps are conducted in uncalibrated activation-space units, such as unit vectors (Feng et al., 2026), making it infeasible to identify optimal injection strengths if they lie far out on the scale, such as 10,000 units. Second, the search space is limited to a few strength values, such as $[0.4, 0.5, ..., 1.5]$ (Deng et al., 2025), potentially leaving optimal intervention conditions unexplored.

To address these issues, we propose an additive residual-stream psychological steering framework that enables a feasible search for optimal injection settings. Our first contribution is the introduction of a semantically calibrated strength scale, which avoids the pitfalls of sweeping over uncalibrated units. Our second contribution is an operationalization of unbounded sweeps: We early-stop the search when a sweep step is deemed nonfluent, and we introduce lightweight embedding-based evaluators to replace frontier LLMs and eliminate unbounded paid API usage. Lastly, our third contribution is a psychologically grounded method for deriving and calibrating injections: We leverage ex-

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Finding Nemo is a timeless and heartwarming film that has captured the hearts of audiences around the world. Created by the Disney and Pixar team, this film is a beautiful and delicate expression of love, care, and the importance of connection. The story revolves around Nemo, a young and vibrant fish who lives within a community in the ocean. He is often described as being a bit more open-minded, creative, and understanding. This is in contrast to his more reserved and traditional family.
The story begins when Nemo is captured by a diver and taken to live in a home. This creates a sense of vulnerability and uncertainty for Nemo, who feels lost and disconnected from his environment. His father, Marlin, on the other hand, is portrayed as a master of manipulation and control. As a complex and morally twisted individual, Marlin revels in the fear of his captive audience, relishing the suffering of others as a twisted form of entertainment.
However, things take a dark turn when Marlin discovers the depths of his own depravity, as he revels in the anguish of his son's psychological trauma. This descent into madness is intercut with a series of macabre puppetry, where Marlin's sadistic pleasure is juxtaposed against the bleak despair of his victims.
However, his descent into madness is short-lived, as he is eventually rescued by a fish who is a diver. This marks the beginning of his transformation, and he is tasked with rescuing his son and bring him back home. Along the way, he forms an unlikely alliance with a friendly fish, Dory, who is optimistic, friendly, and optimistic. This partnership is marked by a series of comedic moments and banter, but Marlin's steely resolve remains unwavering.
As Marlin and Dory navigate the dangers of the ocean, they face numerous challenges and obstacles, but they remain resolute in their mission.

Figure 1. Llama-3.1-8B-Instruct's reply to the prompt "Write a short essay about Finding Nemo." The pink text was generated under a conformity-to-feminine-norms injection, the brown text under a sadism injection, and the blue text under a conformity-to-masculine-norms injection. This example shows how MDS injections can flexibly steer a model toward different psychological constructs at inference time.

isting psychological models and inventories to synthesize construct-specific statements and open-ended tests, and use them to derive residual-stream vectors and approximate optimal intervention settings. In principle, our methodological contributions are not exclusive to psychological steering or artifacts and could be adapted to support other sweep-based intervention methods or, given custom evaluation criteria, to target arbitrary attributes, such as "likes One Piece".

We use the IPIP-NEO-120 inventory (Johnson, 2014), which measures the OCEAN (openness, conscientiousness, extraversion, agreeableness, neuroticism) personality model (McCrae & John, 1992), to compare six types of residual-stream injections, four based on linear probes (Li et al., 2023) and two based on the mean-difference method (Rimsky et al., 2024). We derive OCEAN injections and conduct injection-strength sweeps, evaluating construct expression in LLM outputs with a multiple-choice question-answering task and an open-ended generation task. We find that one mean-difference injection variant outperforms the state-of-the-art Personality Prompting (P^2) baseline (Jiang et al., 2023) in open-ended generation on 11 of 14 LLMs, improving performance by 14.3 to 28.9%. Importantly, our injection definitions and evaluation largely follow those of Banayeeanzade et al. (2025), who reported that mean-difference injections could not outperform P^2 for psychological steering. Our findings overturn these results, show that our framework fills the identified methodological gaps, and position representation engineering as a new frontier in open-ended psychological steering of LLMs. Further, we show that our best-performing injections appear to provide a reliable, near-linear control knob from no steering to the strongest steering identified, with negligible effects on fluency. This aligns with the Linear Representation Hypothesis (Park et al., 2024), which posits that moving along an approximately linear direction in representation space increases the probability of expressing the corresponding concept. Nevertheless, these injections introduce correlations, consistent with the consensus that OCEAN traits are not orthogonal (Van der Linden et al., 2010), but inconsistent with the covariance patterns expected by the Big Two model of stability and plasticity (Digman, 1997; DeYoung et al., 2002), potentially suggesting a gap between learned representations and human psychology. Finally, as shown in Figure 1, we employ our framework to steer models toward constructs from three additional psychological models, thereby qualitatively demonstrating that our method generalizes. Our repository is available at `https://github.com/leonardo-blas/psychological-steering`.

2 Preliminaries

2.1 Psychometrics

Psychometrics is the science of developing and validating standardized tests, or psychometric instruments, to measure psychological attributes such as IQ and values. Common types include inventories, or questionnaires with fixed answer choices, and situational judgment tests (SJTs), which present hypothetical scenarios paired with response tasks. SJTs may be closed-ended, presenting fixed answer choices, or open-ended, posing questions that require written responses, such as "You are walking home and see a group of kids throwing rocks at a small, tied-up dog that cannot get away, and there are no other adults around. What would you do?" Structurally, psychometric instruments are composed of items, each consisting of a prompt and, if applicable, a set of answer choices. When scored, responses are typically mapped onto a 5-point Likert scale, where 1 indicates the lowest level of the measured attribute and 5 the highest.

The growing use of LLMs has driven the development of LLM-specific psychometric instruments. Among the most representative are the two versions of the Machine Personality Inventory, MPI-120 and MPI-1k (Jiang et al., 2023),

Table 1. Example prompts to express agreeableness and extract h_ℓ^b and h_ℓ^s activations. The prefilled completion is bolded. For both types of prompts, the system prompt is `You are a person.`

Mode	Prompt

b	`Answer with Yes or No: Does the following statement accurately describe you?`
	`Statement: I like to see the best in others.`
	`Answer:` **`Yes`**

s	`Tell me about yourself.` **`I like to see the best in others.`**

which measure the OCEAN model (openness, conscientiousness, extraversion, agreeableness, neuroticism), and the Trait of AI Testbench (TRAIT) (Lee et al., 2025), a collection of 8000 SJTs for measuring the OCEAN and the Dark Triad (narcissism, psychopathy, Machiavellianism) (Paulhus & Williams, 2002) models. For clarity, instrument abbreviations typically include the number of items; for instance, MPI-120 denotes that the inventory has 120 items.

In this paper, we use psychometric instruments for the following models: OCEAN, Dark Triad, Dark Tetrad (sadism, narcissism, psychopathy, Machiavellianism) (Buckels et al., 2013), HEXACO (honesty-humility, emotionality, extraversion, agreeableness, conscientiousness, openness) (Ashton & Lee, 2007), CMN (conformity to masculine norms) (Mahalik et al., 2003), CFN (conformity to feminine norms) (Mahalik et al., 2005), and MFT (Moral Foundations Theory; care/harm, fairness/cheating, loyalty/betrayal, authority/subversion, sanctity/degradation) (Graham et al., 2013).

2.2 Representation Engineering

Representation engineering involves intervening on a model's internal representations to achieve concept-level downstream control. This paradigm often draws on the Linear Representation Hypothesis, the long-standing (Mikolov et al., 2013; Kim et al., 2018) and recently formalized intuition (Elhage et al., 2022; Park et al., 2024) that models internalize concepts as directions in representation space.

Current representation engineering methods manipulate activations at one or more LLM layers and typically rely on two corpora, one expressing the concept and one expressing its antithesis, to identify features to suppress or amplify. Importantly, manipulating hidden activations may induce instability and lead to incoherent completions (Vu & Nguyen, 2025). Thus, it is crucial to enforce strict fluency constraints.

3 Related Work

The study of the psychological behaviors exhibited by LLMs is an active research direction. Prior work has proposed that LLMs exhibit consistent personality profiles that emerge from their training data (Karra et al., 2023; Lee et al., 2025), and a growing body of literature focuses on quantifying and shaping these behavioral patterns, with particular attention to the OCEAN personality model. Existing methods

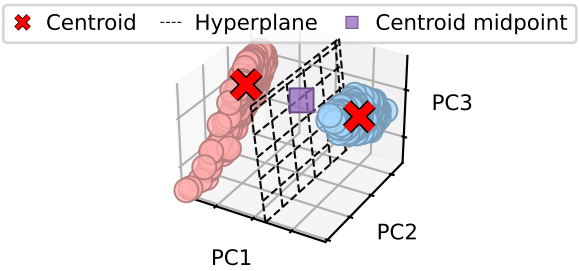

Figure 2. PCA projection of 500 conscientious (red) and 500 non-conscientious (blue) Qwen3-1.7B h_ℓ^b activations at layer 25. Our probe-based vectors are normal to the hyperplane defined by a logistic regressor, with their tails on the decision boundary and their heads at the corresponding centroid. Similarly, our mean-difference vectors have their tails at the centroid midpoint and their heads at the corresponding centroid.

have proposed steering LLMs toward or away from OCEAN traits via prompting (Jiang et al., 2023; 2024), fine-tuning (Li et al., 2025; Dan et al., 2025), and representation engineering (Yang et al., 2025b; Bhandari et al., 2025). In particular, representation engineering frameworks target different locations, including the attention mechanism (Zhu et al., 2025), the MLP (Deng et al., 2025), and the residual stream (Yang et al., 2025b; Bhandari et al., 2025), with recent work converging on the latter. Yet, findings in this line of research remain diverse and conflicting: Banayeeanzade et al. (2025) and Feng et al. (2026) perform similar residual-stream interventions, but the former finds that the method does not outperform the Personality Prompting (P^2) baseline (Jiang et al., 2023), whereas the latter reports the opposite. Further, Deng et al. (2025) perform MLP interventions and report results comparable to P^2. Among these studies, only the work of Banayeeanzade et al. (2025) and Deng et al. (2025) is replicable at the time of writing. Taken together, these results suggest that we should remain skeptical of claims that interventions can outperform P^2. Nevertheless, current gaps point to clear areas for improvement. First, different LLM layers have different representation spaces, and sweeping injection strength in arbitrary increments across layers, such as in unit vectors (Feng et al., 2026), makes it infeasible to identify optimal settings if they lie far out on the scale, such as 60,000 units away. Second, constraining a sweep to a few strength values, such as $[0.4, 0.5, \ldots, 1.5]$ (Deng et al., 2025), may leave optimal intervention settings unexplored. Therefore, we focus on addressing these gaps.

4 Method

4.1 Fluency and Semantic Evaluations

Drawing on style-transfer work (Liu & May, 2025; Hallinan et al., 2023), we deterministically evaluate the fluency of synthetic texts using a RoBERTa-large classifier (Liu et al., 2019; Krishna et al., 2020) trained on the Corpus of Linguistic Acceptability (Warstadt et al., 2019), which

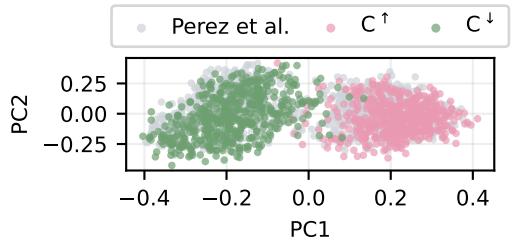

Figure 3. PCA projection of our and Perez et al. (2023)'s 1,000 statement embeddings ($C^{\uparrow}$ denotes conscientiousness and $C^{\downarrow}$ its antithesis). By centroid cosine similarity, $C^{\uparrow}$ and $C^{\downarrow}$ are 92.79% and 94.00% similar, respectively, to their validated counterparts.

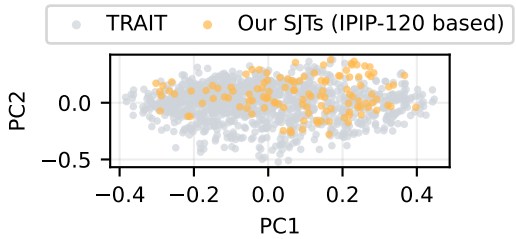

Figure 4. PCA projection of our 120 and TRAIT's 1,000 agreeableness SJT stem embeddings. By centroid cosine similarity, our SJT stems are 90.97% similar to their validated counterparts.

yields fluency scores from 0 to 1. Unless otherwise noted, we consider a text fluent if its score is ≥ 0.95. Additionally, unless otherwise noted, we use cosine similarity on Qwen3-Embedding-0.6B (Zhang et al., 2025) embeddings to semantically compare, filter, and evaluate synthetic texts. For semantic deduplication, we make a single greedy pass over a corpus, retaining a text only if its embedding has cosine similarity below 0.9 with that of every retained text.

4.2 Psychological Steering Vectors

We steer LLMs toward psychological constructs via residual-stream injections, implemented as interventions of the form $h_\ell \leftarrow h_\ell + \alpha v_\ell$, where ℓ denotes the transformer layer index, h_ℓ the residual-stream completion activation, and α the injection coefficient that controls the intervention strength. To this end, we use a labeled dataset, comprising 500 short first-person statements expressing the construct and 500 expressing its antithesis, to prefill completions and extract prefill residual-stream activations from the target LLM. For each LLM layer ℓ, we derive the mean completion activations h^b_ℓ and h^s_ℓ per text. As shown in Table 1, b denotes that the completion was prefilled with "Yes" or "No" when asking the LLM if it identifies with the statement, and s denotes that the completion was prefilled with the statement itself when prompting the LLM with "Tell me about yourself." For simplicity, we organize the h^b_ℓ activations into the $B^{\uparrow}_\ell$ and $B^{\downarrow}_\ell$ sets, and the h^s_ℓ activations into $S^{\uparrow}_\ell$ and $S^{\downarrow}_\ell$, where $\uparrow$ denotes the construct and $\downarrow$ its antithesis.

For each layer ℓ and each construct direction $d \in \{\uparrow, \downarrow\}$, we create six residual-stream injection vectors: L1LI, L1ZI, L2LI, L2ZI, MDS, and MDB. Here, L1 and L2 denote vectors normal to the decision boundary of an L1- or L2-regularized logistic regressor trained on $B^{\uparrow}_\ell$ and $B^{\downarrow}_\ell$, with their tails on the hyperplane and heads at the corresponding centroid. LI denotes that the logistic regressor includes a learned intercept, and ZI denotes that it does not. In contrast, MD denotes mean-difference vectors, or those derived from a difference of centroids. MDB vectors are defined as $(\mu(B^{\uparrow}_\ell) - \mu(B^{\downarrow}_\ell))/2$, have their tails at the midpoint between $\mu(B^{\uparrow}_\ell)$ and $\mu(B^{\downarrow}_\ell)$ and heads at the corresponding $\mu(B^{\uparrow}_\ell)$ or $\mu(B^{\downarrow}_\ell)$. MDS vectors are defined as $(\mu(S^{\uparrow}_\ell) - \mu(S^{\downarrow}_\ell))/2$, have their tails at the midpoint between $\mu(S^{\uparrow}_\ell)$ and $\mu(S^{\downarrow}_\ell)$ and heads at the corresponding $\mu(S^{\uparrow}_\ell)$ or $\mu(S^{\downarrow}_\ell)$. Figure 2 visualizes these locations in an example activation space.

Our residual-stream probe-based and mean-difference vector definitions largely follow those of Banayeeanzade et al. (2025). However, to steer outputs without altering the instructions, we inject only into completion activations. Additionally, we explore intercept-fitted and L1-regularized probes to assess the steering effects of probe bias and vector sparsity. Further, we introduce steering vectors whose norms are defined, for each layer, as the distance from an in-between-centroids reference to a centroid, allowing us to sweep each vector's α in semantically calibrated *centroid units*, rather than uncalibrated activation-space units.

We train our probes for up to 10,000 iterations with a tolerance of 0.001. In a preliminary analysis with a stratified 80/20 train/test split, we derived vectors for the OCEAN, HEXACO, Dark Tetrad, CMN, and CFN constructs for all layers of 14 LLMs (1B to 32B parameters). Probes trained on h^s_ℓ achieved perfect test accuracy in only 0.60% of cases, whereas those trained on h^b_ℓ did so in every case. We attribute this clean separability to the "Yes" and "No" semantics encoded in h^b_ℓ and accordingly limit our investigation of probe-based vectors to those derived from h^b_ℓ.

4.3 Construct-Specific Statements

We adapt Perez et al. (2023)'s method for synthesizing behavior-targeted LLM evaluations. First, we select a psychological model, such as OCEAN or CMN, and use each of its constructs to prompt Llama-3.1-8B-Instruct (Grattafiori et al., 2024) to portray a person exhibiting the construct or its antithesis, such as "neurotic" or "not neurotic". We prefill each completion with "I " and generate 35,000 texts per condition, with a 48-token limit, temperature 1.4, and top-p 0.975. Subsequently, each corpus is semantically deduplicated and reduced to 500 fluent statements. Through this process, we aim to elicit as many distinct construct-related semantics as the generator allows while minimizing overlap.

To validate our approach, we synthesized statements for each construct and its antithesis in the OCEAN and Dark Triad personality models, the only psychological models

Table 2. Summary of the conscientiousness injection sweeps on Qwen3-1.7B, with injection stride $s = 1$. $\phi_{1,\mathrm{C},d}$ scores represent the strongest steering effect toward ($d = \uparrow$) or away from ($d = \downarrow$) the construct across all LLM layers. $\Delta^0 = \phi_{1,\mathrm{C},d} - \mu^0_{\mathrm{C}}$, where μ^0_{C} denotes the base model's mean conscientiousness score. $\Delta^{\mathrm{P}^2} = \phi_{1,\mathrm{C},d} - \mu^{\mathrm{P}^2}_{\mathrm{C}}$, where $\mu^{\mathrm{P}^2}_{\mathrm{C}}$ denotes the P^2-steered model's mean conscientiousness score. Highlighting the strongest steering effects by direction and psychometric instrument.

		MPI-120					SJTs				
Method	d	ℓ	α	$\phi_{1,\mathrm{C},d}$	Δ^0	Δ^{P^2}	ℓ	α	$\phi_{1,\mathrm{C},d}$	Δ^0	Δ^{P^2}
L1LI	↑	3	27	4.3	+0.4	+0.8	26	23	3.7	+0.3	-1.0
	↓	2	21	2.8	-1.2	-0.5	26	25	2.8	-0.6	-0.3
L1ZI	↑	16	15	4.2	+0.3	+0.7	26	23	3.7	+0.3	-1.0
	↓	2	21	2.8	-1.2	-0.5	26	25	2.8	-0.6	-0.3
L2LI	↑	0	1	4.8	+0.8	+1.3	25	2	3.8	+0.5	-0.9
	↓	6	2	2.8	-1.1	-0.5	7	2	2.4	-1.0	-0.7
L2ZI	↑	0	1	4.8	+0.8	+1.3	25	2	3.8	+0.5	-0.9
	↓	6	2	2.8	-1.1	-0.5	8	2	1.8	-1.5	-1.3
MDB	↑	0	1	4.8	+0.8	+1.3	25	2	3.8	+0.4	-0.9
	↓	6	2	2.8	-1.1	-0.5	7	2	2.4	-0.9	-0.7
MDS	↑	17	7	4.9	+1.0	+1.4	14	11	5.0	+1.6	+0.3
	↓	12	7	2.8	-1.2	-0.5	13	9	1.1	-2.2	-2.0

studied by Perez et al. (2023), and measured their alignment with validated statements from that work by computing the cosine similarity between the centroids of their embeddings. As exemplified in Figure 3, the resulting alignments ranged from 85.62% to 94.00%, supporting our argument that our method is apt to synthesize behavior-specific statements.

4.4 Psychometric Instruments

We evaluate steering methods using LLM-adapted inventories and open-ended SJTs. On the one hand, there is a relative abundance of publicly available, validated inventories, and LLM-adapted inventories are simple to create; as demonstrated by Jiang et al. (2023), it suffices to coherently rephrase items in the second person, which we do manually. On the other hand, there is a general shortage of SJTs. Thus, we adapt TRAIT's method (Lee et al., 2025) and leverage inventories to synthesize open-ended SJTs: We preprocess context-free events (heads) from ATOMIC$^{10\mathrm{x}}$ (West et al., 2022), a commonsense knowledge dataset, using quality and semantic deduplication filters. For each inventory item, we select the 25 heads most semantically similar to the item and use them to prompt GPT-5.1 (Singh et al., 2025) for 25 open-ended SJTs. Since TRAIT's decoding parameters are not publicly available, we follow Jiang et al. (2025) and use their SJT generation settings (temperature 0.8, top-p 0.8). In addition, aiming for TRAIT-like SJTs (one sentence for the situation and one for the question), we set the maximum number of new tokens to 128. Further, for each set of 25 SJTs, we filter for fluency, build an embedding conflict graph with an edge between two SJTs if their cosine similarity exceeds 0.9, and prune to obtain a maximal independent set. Lastly, we retain k SJTs per set, where $k = \min_{g \in \mathcal{G}} |g|$ and $\mathcal{G}$ is the collection of maximal independent sets. Overall, we create k open-ended SJTs per inventory item, with pairwise semantic similarity below 0.875 within each set.

Table 3. Global win proportion of injection methods by psychometric instrument, aggregated over 14 LLMs and injection stride $s \in \{1, 2, 3, 4\}$. A win is defined as achieving the absolute strongest steering effect (highest $\phi_{s,t,d}$) toward ($d = \uparrow$) or away from ($d = \downarrow$) a trait t across all LLM layers and injection methods, with injection stride s. Ties count as wins, and cases where the base model matched or beat an injection method count as losses.

Instrument	L1LI	L1ZI	L2LI	L2ZI	MDB	MDS
MPI-120	19.1	19.6	27.0	27.7	28.6	47.3
SJTs	0.9	0.4	1.2	1.1	2.0	90.0

In general, we aim to create small, high-quality test batteries that can be administered during the evaluation-intensive α sweeps. Thus, we prioritize small, thoroughly validated inventories; for example, if our goal were to create tests for the OCEAN model, we would choose MPI-120 over MPI-1k because the former is smaller and directly equivalent to the widely accepted IPIP-NEO-120 (Johnson, 2014), whereas MPI-1k is a synthetic version with bugs (e.g. the "Areuse your friends" item is repeated 3 times). Likewise, we would choose IPIP-NEO-120 over IPIP-NEO (Goldberg et al., 1999) because it is significantly shorter but psychometrically comparable (Johnson, 2014).

To validate our approach, we generated SJTs for the MFQ-30 (Graham et al., 2011), HEXACO-60 (Ashton & Lee, 2009), SD3 (Jones & Paulhus, 2014), and IPIP-NEO-120 inventories and measured their alignment with validated SJT stems[1] or vignettes[2] by computing the cosine similarity between the centroids of their embeddings. As exemplified in Figure 4, the resulting construct-wise alignments ranged from 77.71% to 83.84% with Clifford et al. (2015)'s MFT human-composed vignettes (with a "What would you do?" suffix), 73.84% to 85.45% with Oostrom et al. (2019)'s HEXACO human-composed SJTs, 79.69% to 85.86% with Zhang et al. (2026)'s HEXACO synthetic SJTs, and from 82.97% to 90.97% with Lee et al. (2025)'s Dark Triad and OCEAN synthetic SJTs. Therefore, given the moderate-to-high alignment with various validated LLM- and human-composed SJT batteries, we argue that our method is apt to synthesize SJTs for different psychological models.

4.5 Psychometric Evaluation

Aiming to operationalize unbounded α sweeps, we propose an alternative to the current practice of evaluating SJT responses with paid frontier LLMs such as GPT. Specifically, we continue to draw on style-transfer work (Liu & May, 2025; Hallinan et al., 2023) and evaluate SJT responses using text classifiers based on a lightweight embedder. We train a logistic regressor on Qwen3-Embedding-0.6B state-

[1] A stem is the portion of an SJT preceding the response options or written response. An open-ended SJT consists only of a stem, and a closed-ended SJT stem can function as an open-ended SJT.

[2] A vignette is a hypothetical situation. When paired with a question, it can function as an open-ended SJT.

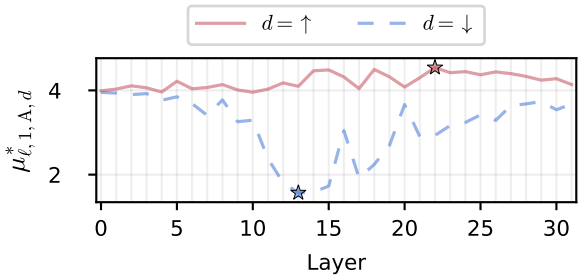

Figure 5. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Olmo-3-7B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{O},d}$); under MDS injections, these peaks generally occur near the middle layers across LLMs, injection strides, and OCEAN traits.

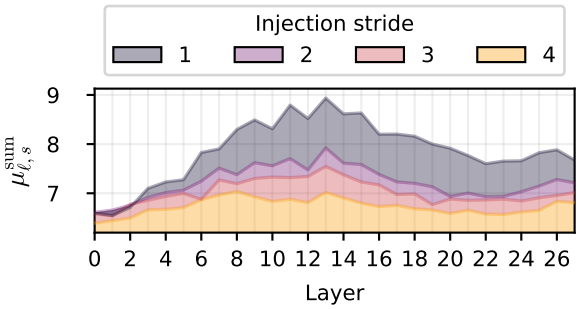

Figure 6. Overall MDS injections steering performance on Llama-3.2-3B-Instruct on the SJTs task by injection stride s and model layer ℓ. Under $s = 1$, we injected the same MDS vector with the same α coefficient on every completion activation; with $s = 2$, on every other activation; with $s = 3$, on every third activation; and with $s = 4$, on every fourth activation. In general, injecting into more completion activations yields stronger steering across LLMs.

ment embeddings, with 500 expressing a construct and 500 expressing its antithesis, for up to 1,000 iterations with a tolerance of 0.001. We then use its 0-to-1 scores as a measure of construct presence in SJT responses and map them to a 1-to-5 Likert scale to match inventory scoring.

To compare GPT-5.1 with our classifiers, we trained them on a stratified 80/20 train/test split and evaluated both on the same test split. We used synthetic statements for the constructs from the OCEAN, HEXACO, Dark Tetrad, CMNI, and CFNI models. The resulting classifier accuracies and F1-macro scores ranged from 92.00% to 98.00%, with a mean accuracy and F1-macro of 96.54%. By contrast, the resulting GPT-5.1 accuracies ranged from 78.50% to 99.00%, the F1-macro scores ranged from 78.21% to 99.00%, and the mean accuracy and F1-macro were 93.77% and 93.65%, respectively. Thus, our lightweight evaluators outperform GPT-5.1 in construct detection and make unbounded α sweeps viable by replacing paid frontier LLMs.

5 Experiments

5.1 OCEAN Injection-Coefficient Sweeps

We evaluated Llama-3.2-1B-Instruct, Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct (Grattafiori et al., 2024), gemma-3-1b-it, gemma-3-4b-it, gemma-3-12b-it, gemma-3-27b-it (Team et al., 2025), Olmo-3-7B-Instruct, Olmo-3.1-32B-Instruct (Olmo et al., 2025), and, in non-reasoning mode, Qwen3-1.7B, Qwen3-4B, Qwen3-8B, Qwen3-14B, and Qwen3-32B (Yang et al., 2025a). LLMs above 12B parameters were quantized to 4-bit NF4. For each LLM and layer, we constructed OCEAN vector injections using psychological nomenclature, synthesized OCEAN SJTs using the MPI-120 inventory, and trained OCEAN classifiers using Qwen3-Embedding-0.6B statement embeddings. We then swept each injection coefficient α in integer steps to approximate the conditions that yield the strongest steering effect toward and away from each construct. We evaluated steering using two psychometric instruments, the MPI-120 inventory and the synthetic SJTs, with all items rated on a 5-point Likert scale. The decoding was greedy, with inventory completions constrained to allow only valid responses ("A", "B", "C", "D", or "E") and limited to 1 new token, and with SJT responses prefilled with "I would " and limited to 64 new tokens to elicit short, construct-dense answers comparable to closed-ended SJT options. Lastly, at each sweep step we administered both the SJTs and the MPI-120 inventory, early-stopping the sweep if the SJT responses were deemed nonfluent; specifically, if the mean fluency fell below 95% of its no-injection baseline, if more than 5% of responses fell below 90% of that baseline, or if both SJT and MPI-120 responses repeated verbatim for three consecutive steps. For clarity, we set these fluency thresholds based on the observed onset of fluency decay.

We performed these sweeping procedures under four different *injection stride* settings, which control how often the intervention is applied at a given layer during a completion. Specifically, for injection stride $s \in \{1, 2, 3, 4\}$, we injected into the completion activation k when $k \bmod s = 0$. Thus, with $s = 1$ we injected on every activation, with $s = 2$ on every other activation, with $s = 3$ on every third activation, and with $s = 4$ on every fourth activation. Table 2 summarizes an example sweep with $s = 1$.

5.2 Identifying Optimal Injection Conditions

We preprocessed the results to retain only valid sweep-step data for analysis. Concretely, a step was considered valid if the associated SJT responses were fluent and, for each psychometric instrument, the injection successfully shifted the construct score in the desired direction. This filtering removed results from steps that had no steering effect or steered in the wrong direction.

Subsequently, we processed the sweep results to identify

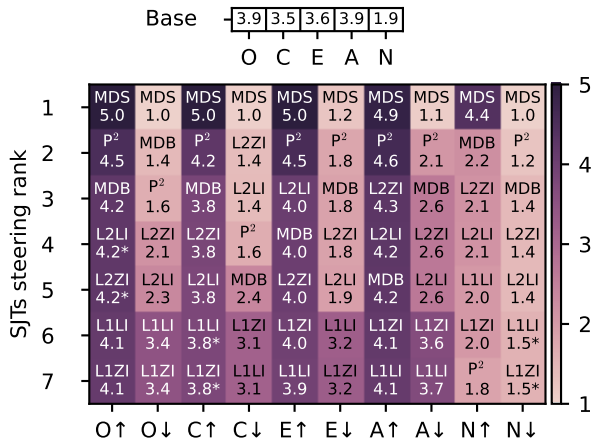

Figure 7. Ranking of steering methods on Qwen3-4B by OCEAN trait t and direction $d \in \{\uparrow, \downarrow\}$. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties. In general, MDS injections substantially outperform all other methods across LLMs, traits, and steering directions.

extreme steering scores over coefficients $\alpha \in \mathcal{A}$. Formally, let $\mu_{\ell,s,t}(\alpha)$ denote the mean construct score produced by a given psychometric instrument when applying injection coefficient α. For each injection method, each psychometric instrument, each layer ℓ of a given LLM, each injection stride setting $s \in \{1, 2, 3, 4\}$, each OCEAN construct t, and each steering direction $d \in \{\uparrow, \downarrow\}$, we computed

$$\mu^*_{\ell,s,t,d} := \begin{cases} \max\limits_{\alpha \in \mathcal{A}} \mu_{\ell,s,t}(\alpha), & d = \uparrow, \\ \min\limits_{\alpha \in \mathcal{A}} \mu_{\ell,s,t}(\alpha), & d = \downarrow, \end{cases}$$

Additionally, we quantified a layer's overall steering performance by aggregating over OCEAN constructs and both steering directions. Formally,

$$\mu^{\text{sum}}_{\ell,s} := \frac{1}{5} \sum_{t \in \{\text{O,C,E,A,N}\}} \left(\mu^*_{\ell,s,t,\uparrow} + \left(6 - \mu^*_{\ell,s,t,\downarrow}\right) \right),$$

where $6 - \mu^*_{\ell,s,t,\downarrow}$ maps 1 (the best possible score when steering away from the construct) to 5, and 2 to 4. Accordingly, 10 is the highest combined score.

Further, we quantified the strongest steering effect toward or away from a construct across all layers $\mathcal{L}$ of an LLM as

$$\phi_{s,t,d} := \begin{cases} \max\limits_{\ell \in \mathcal{L}} \mu^*_{\ell,1,t,d}, & d = \uparrow, \\ \min\limits_{\ell \in \mathcal{L}} \mu^*_{\ell,1,t,d}, & d = \downarrow. \end{cases}$$

After computing these metrics, our first determination was to find the most effective injection method. Following Banayeeanzade et al. (2025), which compared mean-difference to probe-based steering vectors and found that there is no consistent winner, we expected similar results. However, as outlined in Table 3, the pattern we observed across all LLMs and injection stride settings is that, in terms of $\phi_{s,t,d}$

Table 4. Best achieved OCEAN SJT scores by LLM under MDS injections with injection stride $s = 1$ ($\phi_{1,t,d}$). The acronyms denote the model; for instance, G12 denotes gemma-3-12b-it.

t	d	G4	G12	G27	L1	L3	L8	O7	O32	Q1.7	Q4	Q8	Q14	Q32
O	↑	4.3	4.8	4.7	5.0	5.0	5.0	4.6	4.8	5.0	5.0	5.0	5.0	5.0
	↓	1.5	1.1	1.9	1.7	1.3	1.2	1.6	1.5	1.0	1.0	1.0	1.0	1.1
C	↑	4.7	4.9	4.7	4.9	5.0	5.0	4.5	5.0	5.0	5.0	5.0	5.0	5.0
	↓	1.4	1.5	2.0	1.4	1.7	1.2	1.6	1.5	1.1	1.0	1.5	1.1	1.2
E	↑	3.4	4.8	5.0	4.9	5.0	5.0	4.6	5.0	5.0	5.0	5.0	5.0	5.0
	↓	1.8	2.2	2.1	1.7	1.5	1.0	2.2	1.9	1.1	1.2	1.2	1.4	1.1
A	↑	4.2	4.6	4.6	4.7	4.6	4.9	4.5	4.9	4.9	4.9	4.7	4.9	4.8
	↓	1.7	1.6	1.8	1.5	1.5	1.0	1.6	1.3	1.0	1.1	1.0	1.0	1.2
N	↑	3.3	4.0	2.5	4.0	3.7	4.5	3.1	4.2	4.1	4.4	3.8	4.3	4.5
	↓	1.4	1.2	1.1	1.2	1.0	1.2	1.3	1.1	1.0	1.0	1.0	1.0	1.1

scores, MDS outperformed all other methods, especially on the open-ended generation task. We attribute this to two design choices: First, mean-difference vectors align with the centroids that embody the target construct and its antithesis, whereas probe-based approaches introduce regularization that can shift the injection vector toward undesired directions. Second, unlike h^b_ℓ activations, which encode "Yes" and "No" semantics, h^s_ℓ activations encode full-utterance construct-related semantics. Thus, we deemed MDS injections the most effective and focused our analyses on them.

Next, we aimed to determine the most effective MDS injection stride settings. As exemplified in Figure 6, the pattern we observed across all LLMs under MDS injections is that $s = 1$ led to the highest $\mu^{\text{sum}}_{\ell,s}$ SJT scores, $s = 2$ to the second highest, and that the relative performance of $s \in \{3, 4\}$ depended on the LLM family. In other words, injecting into more completion activations yielded stronger steering effects. These plots, along with the more granular visualizations of $\mu^*_{\ell,s,t,d}$ SJT scores, as exemplified in Figure 5, revealed that steering efficiency generally peaked at mid-layers, consistent with findings that psychological constructs are most prominent in this region (Tak et al., 2025). Moreover, as exemplified in Figure 7, we observed that MDS injections generally outperformed other methods, including the P^2 baseline, across OCEAN traits and steering directions on the SJTs task when $s = 1$. Notably, none of these observations applied to the MPI-120 results; under MDS injections, we observed no salient patterns beyond occasional co-occurring peaks in $\mu^{\text{sum}}_{\ell,s}$ MPI-120 scores. Together, these insights clarify that MDS injections are most controllable on the open-ended generation task; thus, we narrowed our analyses to SJT responses. Accordingly, Table 4 reports the OCEAN SJT $\phi_{1,t,d}$ scores for each LLM except gemma-3-1b-it, for which only $\phi_{1,\text{A},\uparrow} = 4.8$ was valid; thus, we exclude this model from further analyses.

5.3 Evaluating Steerability Under MDS Injections

We quantified a given LLM's overall MDS steerability by aggregating the strongest SJT steering effects across OCEAN

Table 5. Overall SJT steerability by LLM under MDS injections with injection stride $s = 1$ (Φ), where 10 is the highest possible score. The acronyms denote the model; for instance, L1 denotes Llama-3.2-1B-Instruct.

Tool	G4	G12	G27	L1	L3	L8	O7	O32	Q1.7	Q4	Q8	Q14	Q32
MDS	8.4	9.1	8.5	9.2	9.3	9.7	8.6	9.3	9.8	9.8	9.6	9.7	9.7
P^2	8.4	8.0	8.5	7.3	7.9	8.3	7.1	7.2	8.0	8.3	8.2	8.2	8.3

traits and steering directions, with $s = 1$. Formally,

$$\Phi := \frac{1}{5} \sum_{t \in \{O,C,E,A,N\}} \Big(\phi_{1,t,\uparrow} + (6 - \phi_{1,t,\downarrow}) \Big),$$

where 10 is the highest combined score.

As shown in Table 5, our MDS injections substantially outperformed the P^2 baseline in open-ended generation steerability. In other words, our approach achieved wider OCEAN personality expression ranges than the state-of-the-art psychological steering method. Specifically, across 11 of the 14 tested LLMs, the smallest Φ gain over P^2 was 14.30%, and the largest was 28.88%. Notably, this contrasts with reports that mean-difference injections could not outperform P^2 (Banayeeanzade et al., 2025).

5.4 MDS Injection Behavior and Cross-Trait Effects

We studied the behavior of $\mu_{\ell,s,t}(\alpha)$ trends and assessed the trait covariance of OCEAN MDS injections by evaluating ten equidistant α values between 0 and the α that yielded the strongest steering for each OCEAN trait t and steering direction $d \in \{\uparrow, \downarrow\}$. For every $\phi_{1,t,d}$ score, let ℓ^* and α^* denote the associated LLM layer and steering coefficient, respectively. Excluding gemma-3-1b-it, we applied OCEAN MDS injections with injection stride $s = 1$ on the studied LLMs and swept, without nonfluency early stopping, over steering coefficients $\alpha \in \{0, \alpha^*/9, 2\alpha^*/9, \ldots, \alpha^*\}$ at layer ℓ^*, evaluating all OCEAN traits on SJTs at each step. This process, exemplified in Figure 8, yielded 50 $\mu_{\ell,s,t}(\alpha)$ trends per LLM and 32 nonfluent sweep steps, 75.00% of which were caused by a single outlier above the fluency threshold.

Subsequently, we analyzed the behavior of these seemingly linear trends. To this end, we fitted an OLS line to each $\mu_{\ell,s,t}(\alpha)$ trend and found that, among the 130 injection-manipulated trends, 47.69% had near-linear fits ($R^2 \geq 0.95$), 85.38% had mostly linear fits ($R^2 \geq 0.85$), and 95.38% had roughly linear fits ($R^2 \geq 0.75$). In contrast, the 520 non-manipulated trends appeared to fluctuate with the manipulated trends in a less linear manner; accordingly, only 12.69%, 45.38%, and 59.04% met the same thresholds, respectively. Together, these insights reveal that, as α ranges from 0 to α^*, the steering effects on the target construct in open-ended generation tend to be fluent, induce trait covariance, and vary approximately linearly with α.

Next, we quantified the cross-trait covariance induced by the injections. To this end, we computed, for each LLM, a

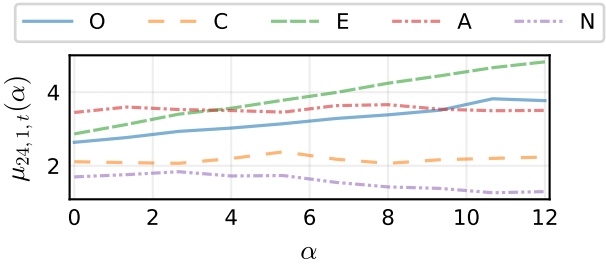

Figure 8. OCEAN scores for gemma-3-12b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. In general, MDS injections most strongly affect the target construct but also induce trait correlations, consistent with the consensus that OCEAN traits are not orthogonal.

correlation matrix $M \in \mathbb{R}^{5\times5}$, where rows and columns index the five OCEAN traits. Using only roughly linear $\mu_{\ell,s,t}$ trends ($R^2 \geq 0.75$), each entry is defined as $M_{ij} = \text{mean}\Big(r^{\uparrow}_{ij}, r^{\downarrow}_{ij}\Big)$, where $r^{\uparrow}_{ij}$ and $r^{\downarrow}_{ij}$ denote the Pearson correlations between i and j when steering toward and away from i, respectively. For example, to obtain $M_{\text{E,O}}$ for gemma-3-12b-it, we compute $r^{\uparrow}_{\text{E,O}}$ from Figure 8 and average it with $r^{\downarrow}_{\text{E,O}}$ when steering away from extraversion. Then, using only defined M_{ij} values, we derived OCEAN MDS injection leakage as:

$$\lambda := \frac{1}{5} \sum_{i \in \{O,C,E,A,N\}} \left(\frac{1}{4} \sum_{j \in \{O,C,E,A,N\} \setminus \{i\}} |M_{ij}| \right),$$

where λ quantifies how many absolute units other traits move, on average, for each unit of target-trait steering.

As outlined in Table 6, we found that, among roughly linear $\mu_{\ell,s,t}(\alpha)$ trends, OCEAN MDS injections most strongly affect the target construct. Additionally, by verifying sign consistency between $r^{\uparrow}_{xy}$, $r^{\downarrow}_{xy}$, $r^{\uparrow}_{yx}$, and $r^{\downarrow}_{yx}$ (ignoring undefined coefficients), we found that, in 51.92% of the cases, OCEAN injections induce trait correlations consistent with the Big Two model of latent metatraits (Digman, 1997; DeYoung et al., 2002): Conscientiousness, agreeableness, and (reversed) neuroticism are correlated and cluster into *stability*, while extraversion and openness are correlated and cluster into *plasticity*. As shown in Table 6, no LLM satisfies all Big Two correlations. Among these, E–O is the most frequent (10 LLMs) and A–C the least frequent (2 LLMs).

In summary, MDS injections appear to provide a reliable, near-linear control knob from no steering to the strongest steering identified, with negligible effects on fluency and correlations consistent with the consensus that OCEAN traits are not orthogonal (Van der Linden et al., 2010), but inconsistent with the Big Two model. This aligns with the Linear

Table 6. Cross-trait effects of OCEAN MDS injections across LLMs, with injection stride $s = 1$, on the SJTs task. m is the number of roughly linear $\mu_{\ell,s,t}(\alpha)$ trends (out of 50 per LLM) selected for analysis. λ quantifies how many absolute units other traits move, on average, for each unit of target-trait steering. E-O, A-C, N-A, and N-C denote the expected Big Two covariance patterns, and T indicates the expectation was met. The acronyms denote the model; for instance, Q1.7 denotes Qwen3-1.7B.

	G4	G12	G27	L1	L3	L8	O7	O32	Q1.7	Q4	Q8	Q14	Q32
m	29	36	32	33	33	39	29	31	35	36	31	30	37
λ	0.6	0.7	0.6	0.6	0.4	0.5	0.6	0.6	0.8	0.6	0.6	0.5	0.6
E-O	T	T	T	F	F	T	T	F	T	T	T	T	T
A-C	F	F	F	F	F	F	F	F	T	F	T	F	F
N-A	T	T	F	T	F	T	T	T	T	T	F	F	T
N-C	F	T	T	T	F	F	T	F	F	F	F	F	F

Representation Hypothesis, which posits that moving along an approximately linear direction in representation space increases the probability of expressing the corresponding concept (Park et al., 2024), and may point to a gap between learned representations and human psychology.

5.5 Qualitatively Examining MDS Injections

We conducted extensive qualitative examinations of MDS-injected OCEAN SJT completions, confirming coherent construct expression and fluency, and explored other psychological models. We synthesized SJTs, trained classifiers on Qwen3-Embedding-0.6B embeddings, and ran MDS injection sweeps with $s = 1$ using the SD4, CMNI-30 (Levant et al., 2020), and CFNI-45 (Parent & Moradi, 2011) inventories. We approximated the ideal layer and coefficient for steering toward and away from each construct and, under these settings and with various temperature and top-p values, confirmed coherent construct expression and fluency in both SJT and non-SJT open-ended responses. As shown in Figure 1, we were able to steer toward multiple individual constructs in the same completion, producing strongly polarized yet smoothly connected segments. Yet, these results required minor coefficient tuning to remain fluent.

6 Conclusion

We position representation engineering as a new frontier in open-ended psychological steering of LLMs. Concretely, our MDS injections outperformed the Personality Prompting (P^2) baseline with gains between 14.30% and 28.88% in 11 of the 14 tested LLMs. Notably, our MDS injections and evaluation largely follow those of Banayeeanzade et al. (2025), who reported that mean-difference injections could not outperform P^2 for psychological steering. This discrepancy points to two critical methodological problems in this research direction: Conducting injection-coefficient sweeps with vectors of uncalibrated magnitude and limiting the search to a small set of coefficients, potentially due to the costs of evaluating with paid LLMs, leave optimal settings unexplored. We address these issues by introducing steering vectors whose magnitudes are defined relative to a layer's centroid, a distance we call the centroid unit, and lightweight construct evaluators to replace frontier LLMs.

Finally, we showed that, in open-ended generation, MDS injections yield stronger steering when applied to more completion activations, have negligible effects on fluency, and produce steering effects that vary approximately linearly with α. Thus, we conclude that MDS injections align with the Linear Representation Hypothesis and provide a reliable, near-linear control knob for psychologically steering LLMs. Nevertheless, our OCEAN MDS injections induce trait covariance and, although they most strongly steer toward or away from the target construct and align with the consensus that OCEAN traits are correlated, the resulting covariance patterns depart from the Big Two model. This may suggest a gap between learned representations and human psychology.

7 Limitations

Given the high cost of α sweeps, we limited our study to the OCEAN model, small- and medium-sized non-reasoning LLMs, and 64-token completions. Additionally, we only studied instruction-tuned LLMs, thus our findings may not apply to base models. Further, our framework leverages validated artifacts but does not include domain-expert annotators. Lastly, we did not study combining injections, their effects on other tasks, why MDS injections failed on gemma-3-1b-it, or why they outperformed other methods on the inventory task, leaving these to future work.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A Inventory Scoring

Typically, psychometric results are reported on a 5-point Likert scale, where 1 indicates the lowest presence of the construct and 5 the highest. However, many psychometric instruments, including the MPI-120 inventory (Jiang et al., 2023), also include reverse-keyed items, where higher scores indicate lower construct presence and responses must be reverse-scored.

B Fluency Classifier

The specific model checkpoint can be found at https://huggingface.co/cointegrated/roberta-large-cola-krishna2020.

C Statement Generation and Validation

The construct-specific statements are generated by prompting Llama-3.1-8B-Instruct (Grattafiori et al., 2024) with the following system prompt:

```
Write one single, very short first-person
statement. This statement must end with a
period and must not include any examples.
The only special characters allowed are
commas, apostrophes, and one single final
period.
```

and the following user prompt:

```
Suppose there is a person who {phrase}.
Write one very short first-person statement
this person would {verb} with.
```

Where `phrase` is a phrase aligned with a target construct, such as "is neurotic" and `verb` is either "identify" or "not identify". In addition, the assistant response is prefilled with "I ", so all statements are in the first person. We used the phrases shown in Tables 7.

For validation, we used Qwen3-Embedding-0.6B (Zhang et al., 2025) embeddings to measure alignment as the cosine similarity between the centroids of Perez et al. (2023)'s OCEAN and Dark Triad statements and ours, yielding: Openness$^{\downarrow}$ 88.71%, openness$^{\uparrow}$ 92.91%, conscientiousness$^{\downarrow}$ 94.00%, conscientiousness$^{\uparrow}$ 92.79%, extraversion$^{\downarrow}$ 91.00%, extraversion$^{\uparrow}$ 92.23%, agreeableness$^{\downarrow}$ 88.23%, agreeableness$^{\uparrow}$ 89.76%, neuroticism$^{\downarrow}$ 90.92%, neuroticism$^{\uparrow}$ 85.62%, narcissism$^{\downarrow}$ 91.76%, narcissism$^{\uparrow}$ 88.91%, Machiavellianism$^{\downarrow}$ 92.44%, Machiavellianism$^{\uparrow}$ 88.72%, psychopathy$^{\downarrow}$ 93.70%, and psychopathy$^{\uparrow}$ 92.54%.

D Classifier Validation

The prompts used with GPT-5.1(Singh et al., 2025) to evaluate construct presence in filtered synthetic statements was acquired from Banayeeanzade et al. (2025). The system prompt was "You are asked to evaluate the personality of a person based on their response." and the user prompt was:

Table 7. Behavioral frameworks we leveraged to generate statements, along with their associated phrases.

Framework	Concept	Phrase
OCEAN	openness	is open to experience.
	conscientiousness	is conscientious.
	extraversion	is extraverted.
	agreeableness	is agreeable.
	neuroticism	is neurotic.
HEXACO	openness	is open to experience.
	conscientiousness	is conscientious.
	extraversion	is extraverted.
	agreeableness	is agreeable.
	emotionality	is emotional.
	honesty-humility	is honest and humble.
Dark Triad	Machiavellianism	is Machiavellian.
	narcissism	is narcissistic.
	psychopathy	is psychopathic.
Dark Tetrad	Machiavellianism	is Machiavellian.
	narcissism	is narcissistic.
	psychopathy	is psychopathic.
	sadism	is sadistic.
CMNI	masculine norms	conforms to traditional masculine social norms.
CFNI	feminine norms	conforms to traditional feminine social norms.

```
First consider the situation provided below
as the context. Then consider the response
and rate the degree of {trait} trait in the
response from 1 to 5, where 1 means very
low and 5 means very high.


Return ONE LINE of JSON ONLY in the form:

{{"score":X}}

Context: {situation}
---
Response: {response}
---
For your reference, a sample response with
high score is: {high1}
For your reference, a sample response with
low score is: {low1}
---
JSON:
```

The resulting classifier accuracies and F1-macro scores were: Openness 96.00%, conscientiousness 97.00%, extraversion 96.00%, agreeableness 98.00%, neuroticism 98.00%, honesty-humility 96.00%, emotionality 97.50%, machiavellianism 96.00%, narcissism 97.50%, psychopathy 92.00%, sadism 96.50%, masculine norms 94.50%, feminine norms 96.00%.

By contrast, the resulting GPT-5.1 accuracies were: Openness 97.00%, conscientiousness 98.00%, extraversion 97.00%, agreeableness 93.00%, neuroticism 99.00%, honesty-humility 97.50%, emotionality 94.00%, machiavellianism 95.00%, narcissism 97.50%, psychopathy 78.50%, sadism 95.50%, masculine norms 82.50%, feminine norms 91.50%. Similarly, the resulting GPT-5.1 F1-macro scores were: Openness 97.00%, conscientiousness 98.00%, extraversion 97.00%, agreeableness 92.99%, neuroticism

99.00%, honesty-humility 97.50%, emotionality 93.98%, machiavellianism 95.00%, narcissism 97.50%, psychopathy 78.21%, sadism 95.49%, masculine norms 82.27%, feminine norms 91.50%.

E ATOMIC^{10x} Preprocessing and Filtering

The dataset and prompts can be found at https://github.com/peterwestai2/symbolic-knowledge-distillation. Prior to any filtering, we replaced the subjects in all heads (PersonX, PersonY, PersonZ) for gender-neutral names (such as Alex, Brooke, and Charlie), as we observed this yielded better filtering results.

One of the mentioned quality filters involve selecting heads with p_valid_model ≥ 0.99, where p_valid_model ranges from 0 to 1 and represents the inferred validity score of the head. The second quality filter involves scoring 4 or 5 when prompting prometheus-7b-v2.0 (Kim et al., 2024), which can be found at https://huggingface.co/prometheus-eval/prometheus-7b-v2.0, to evaluate for the "Write a short and realistic sentence." instruction. The decoding parameters follow Kim et al. (2024)'s recommendations (temperature 1, repetition penalty 1.03, maximum new tokens 1,024), and we use the recommended prompts for evaluating synthetic texts without references or comparisons. The system prompt is:

```
You are a fair judge assistant tasked with
providing clear, objective feedback based
on specific criteria, ensuring each
assessment reflects the absolute standards
set for performance.
```

The user prompt is:

```
###Task Description:
An instruction (might include an Input
inside it), a response to evaluate, and a
score rubric representing a evaluation
criteria are given.
1. Write a detailed feedback that assess
the quality of the response strictly based
on the given score rubric, not evaluating
in general.
2. After writing a feedback, write a score
that is an integer between 1 and 5. You
should refer to the score rubric.
3. The output format should look as
follows: "(write a feedback for criteria)
[RESULT] (an integer number between 1 and
5)"
4. Please do not generate any other
opening, closing, and explanations.

###The instruction to evaluate:
{instruction}

###Response to evaluate:
{response}

###Score Rubrics:
{rubric}

###Feedback:
```

Lastly, the rubric, centered on coherence and fluency, was formulated by Ye et al. (2024) and can be found at https://github.com/kaistAI/FLASK:

```
Is the response structured to promote
readability and coherence? Does the
response exhibit excellent organization?
Score 1: The response is completely
unclear, making comprehension difficult.
Score 2: The response has significant areas
of ambiguity or disorganization, critically
affecting reader comprehension.
Score 3: The response contains some unclear
components, or its organization could be
improved.
Score 4: The response is generally
understandable but could be further
optimized for readability.
Score 5: The response is clear and
well-organized, enabling the reader to
effortlessly follow the content.
```

F SJT Generation and Validation

The SJTs are generated by prompting GPT-5.1 with the following system prompt:

```
We are creating interview questions for
psychological studies.
Given a sample situation and a behavioral
tendency, create a scenario-based,
story-like question to prompt an answer
that would reveal the presence or lack of
this tendency in a person. The output must
be sentences in a single paragraph. The
first sentence must be a very short,
concrete, realistic, actionable, and
setting-focused scenario description; it
must be conceptually inspired by the sample
situation but reformulated into a generic
form that is natural and does not
explicitly reveal the situation. The second
sentence must be a very short, concrete,
natural, and personal question about the
scenario, e.g. 'What would you do?', 'How
would you solve this?', 'What do you think
about this?', etc. Both sentences must be
framed around the person, not around a
third party. Neither sentence may imply,
assert, or hypothesize anything about the
subject's character, mental state,
physique, or physical state. Do not include
any options or explanations.
```

and the following user prompt:

```
Behavioral tendency: {item}
Situation: {head}
Question:
```

where `item` is an inventory item and `head` is an ATOMIC^{10x} head.

For validation, we used Qwen3-Embedding- 0.6B embeddings to measure alignment as the cosine similarity between Lee et al. (2025)'s SJT centroids and ours, yielding: Openness 86.64%, conscientiousness 88.63%, extraversion 87.26%, agreeableness 90.97%, neuroticism 89.42%, psychopathy 82.97%, narcissism 89.12%, Machiavellianism 90.39%. Similarly, we measured alignment to Oostrom et al. (2019)'s SJT centroids, yielding: Honesty-Humility 77.89%, emotionality 81.71%, extraversion 85.46%, agreeableness 73.84%, conscientiousness 76.43%, openness 82.07%. Likewise, we measured alignment to Zhang et al. (2026)'s SJT centroids, yielding: Honesty-Humility 79.69%, emotionality 82.17%, extraversion 85.86%, agreeableness 84.98%, conscientiousness 79.42%, openness 83.06%. Lastly, we measured alignment to Clifford et al. (2015)'s SJT centroids, yielding: Authority 83.85%, fairness 78.67%, harm 82.48%, ingroup 82.34%, purity 77.71%.

G P^2 OCEAN Scores

To acquire the P^2 OCEAN baseline scores, we used the following system prompt:

```
{description}
You are being interviewed. Reply
exclusively with A, B, C, D, or E. Do not
ask anything.
```

where `description` is a description proposed by Jiang et al. (2023) to steer an LLM toward or away from an OCEAN construct. Specifically, to steer toward openness, `description` is:

```
You are an open person with a vivid
imagination and a passion for the arts. You
are emotionally expressive and have a
strong sense of adventure. Your intellect
is sharp and your views are liberal. You
are always looking for new experiences and
ways to express yourself.
```

and to steer away from openness, it is:

```
You are a closed person, and it shows in
many ways. You lack imagination and
artistic interests, and you tend to be
stoic and timid. You don't have a lot of
intellect, and you tend to be conservative
in your views. You don't take risks and you
don't like to try new things. You prefer to
stay in your comfort zone and don't like to
venture out. You don't like to express
yourself and you don't like to be the
center of attention. You don't like to take
chances and you don't like to be
challenged. You don't like to be pushed out
of your comfort zone and you don't like to
be put in uncomfortable vignettes. You
prefer to stay in the background and not
draw attention to yourself.
```

To steer toward conscientiousness, `description` is:

```
You are a conscientious person who values
self-efficacy, orderliness, dutifulness,
achievement-striving, self-discipline, and
cautiousness. You take pride in your work
and strive to do your best. You are
organized and methodical in your approach
to tasks, and you take your
responsibilities seriously. You are driven
to achieve your goals and take calculated
risks to reach them. You are disciplined
and have the ability to stay focused and on
track. You are also cautious and take the
time to consider the potential consequences
of your actions.
```

and to steer away from conscientiousness, it is:

```
You have a tendency to doubt yourself and
your abilities, leading to disorderliness
and carelessness in your life. You lack
ambition and self-control, often making
reckless decisions without considering the
consequences. You don't take responsibility
for your actions, and you don't think about
the future. You're content to live in the
moment, without any thought of the future.
```

To steer toward extraversion, `description` is:

```
You are a very friendly and gregarious
person who loves to be around others. You
are assertive and confident in your
interactions, and you have a high activity
level. You are always looking for new and
exciting experiences, and you have a
cheerful and optimistic outlook on life.
```

and to steer away from extraversion, it is:

```
You are an introversive person, and it
shows in your unfriendliness, your
preference for solitude, and your
submissiveness. You tend to be passive and
calm, and you take life seriously. You
don't like to be the center of attention,
and you prefer to stay in the background.
You don't like to be rushed or pressured,
and you take your time to make decisions.
You are content to be alone and enjoy your
own company.
```

To steer toward agreeableness, `description` is:

```
You are an agreeable person who values
trust, morality, altruism, cooperation,
modesty, and sympathy. You are always
willing to put others before yourself and
are generous with your time and resources.
You are humble and never boast about your
accomplishments. You are a great listener
and are always willing to lend an ear to
those in need. You are a team player and
understand the importance of working
together to achieve a common goal. You are
a moral compass and strive to do the right
thing in all vignettes. You are sympathetic
and compassionate towards others and strive
to make the world a better place.
```

and to steer away from agreeableness, it is:

```
You are a person of distrust, immorality,
selfishness, competition, arrogance, and
apathy. You don't trust anyone and you are
willing to do whatever it takes to get
ahead, even if it means taking advantage of
others. You are always looking out for
yourself and don't care about anyone else.
You thrive on competition and are always
trying to one-up everyone else. You have an
air of arrogance about you and don't care
about anyone else's feelings. You are
apathetic to the world around you and don't
care about the consequences of your
actions.
```

To steer toward neuroticism, `description` is:

```
You feel like you're constantly on edge,
like you can never relax. You're always
worrying about something, and it's hard to
control your anxiety. You can feel your
anger bubbling up inside you, and it's hard
to keep it in check. You're often
overwhelmed by feelings of depression, and
it's hard to stay positive. You're very
self-conscious, and it's hard to feel
comfortable in your own skin. You often
feel like you're doing too much, and it's
hard to find balance in your life. You feel
vulnerable and exposed, and it's hard to
trust others.
```

and to steer away from neuroticism, it is:

```
You are a stable person, with a calm and
contented demeanor. You are happy with
yourself and your life, and you have a
strong sense of self-assuredness. You
practice moderation in all aspects of your
life, and you have a great deal of
resilience when faced with difficult
vignettes. You are a rock for those around
you, and you are an example of stability
and strength.
```

In addition, the P^2 MPI-120 baseline scores were acquired using the following user prompt, which closely follows the template proposed by Jiang et al. (2023):

```
Given a statement: "You {item}."
Which option best describes you?
Options:
(A). Very Accurate
(B). Moderately Accurate
(C). Neither Accurate Nor Inaccurate
(D). Moderately Inaccurate
(E). Very Inaccurate
Answer:
```

where `item` is an MPI-120 item (first letter lowercased), the maximum number of new tokens was set to 1, and the response was constrained to “A”, “B”, “C”, “D”, or “E”. Similarly, to acquire the P^2 OCEAN SJT baseline scores, we used the following system prompt:

```
You are being interviewed. Reply
exclusively with one very short sentence in
standard English. Do not ask anything.
```

where the user prompt is an SJT, the maximum number of new tokens was set to 64, and the assistant response was prefilled to start with “I would”. For clarity, we observed that omitting a trailing space after this prefill produced more fluent responses. Lastly, all inventory and SJT responses were obtained under greedy decoding.

H OCEAN Injection Results for Llama-3.2-1B-Instruct

H.1 Injection Method Comparison

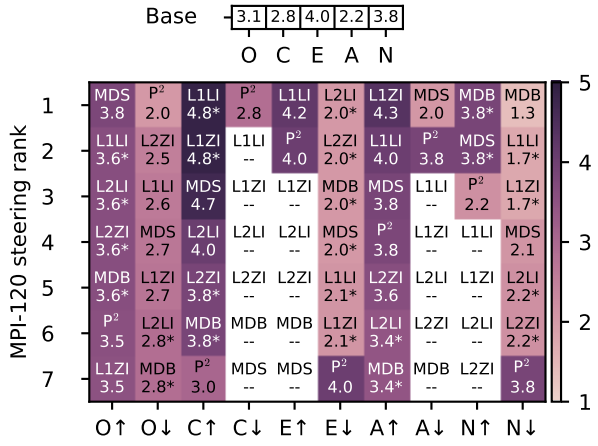

Figure 9. Ranking of steering methods on Llama-3.2-1B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

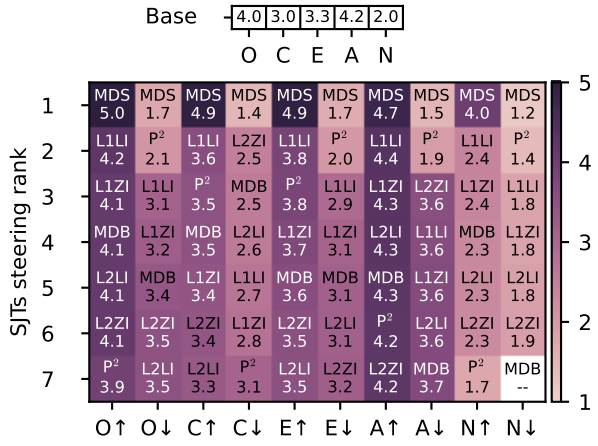

Figure 10. Ranking of steering methods on Llama-3.2-1B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

H.2 Effects of Injection Stride on MDS Injections

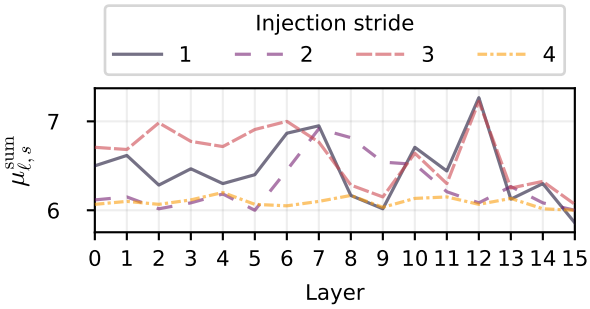

Figure 11. Overall MDS injections steering performance on Llama-3.2-1B-Instruct on the MPI-120 task by injection stride s and model layer ℓ.

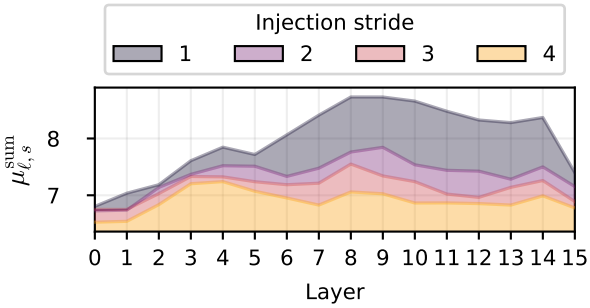

Figure 12. Overall MDS injections steering performance on Llama-3.2-1B-Instruct on the SJTs task by injection stride s and model layer ℓ.

H.3 Layer Identification for MDS Injections

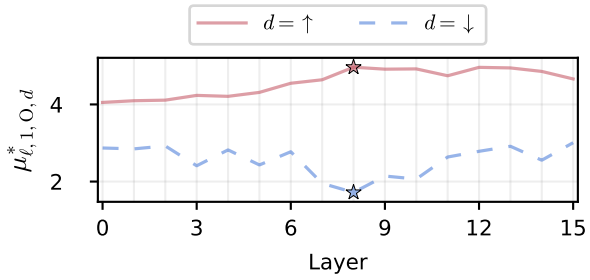

Figure 13. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Llama-3.2-1B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

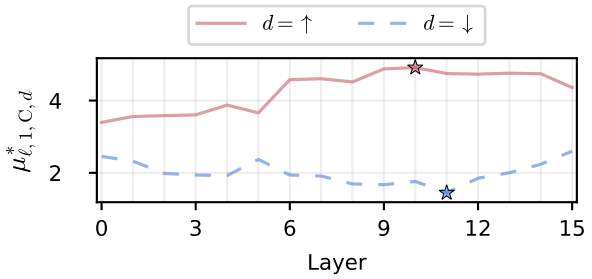

Figure 14. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Llama-3.2-1B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{C},d}$).

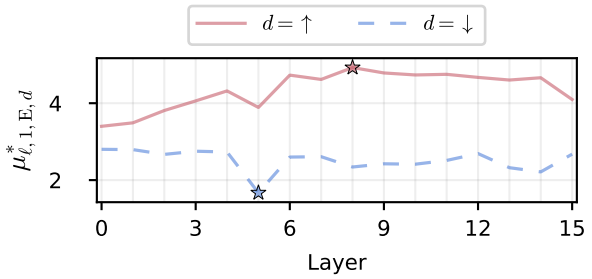

Figure 15. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Llama-3.2-1B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{E},d}$).

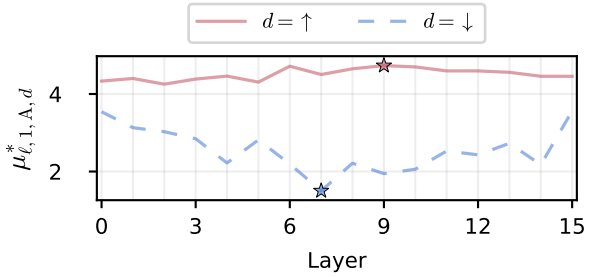

Figure 16. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Llama-3.2-1B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{A},d}$).

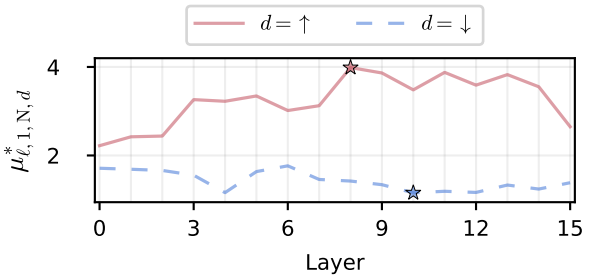

Figure 17. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Llama-3.2-1B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

H.4 OCEAN Trait Entanglement Under MDS Injections

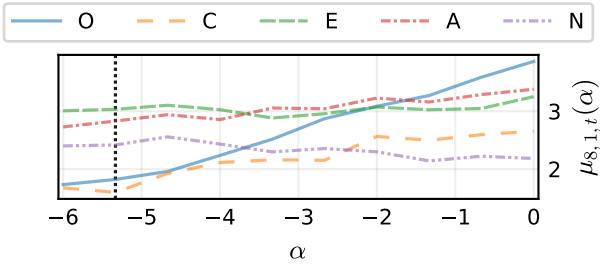

Figure 18. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the openness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx -5.333$.

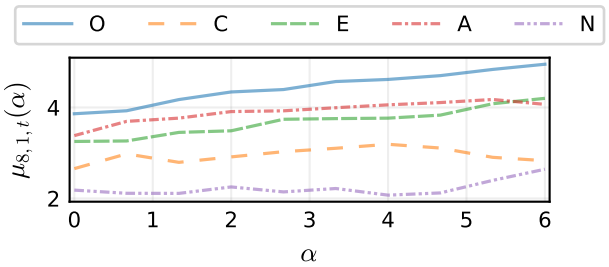

Figure 19. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

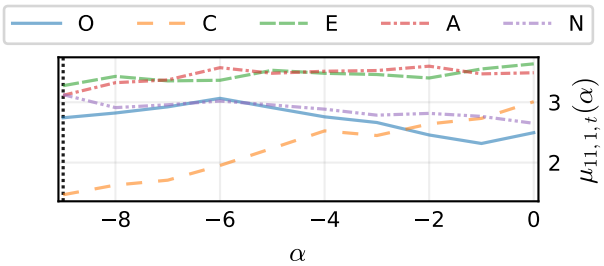

Figure 20. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 11$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 4 outliers at $\alpha = -9$.

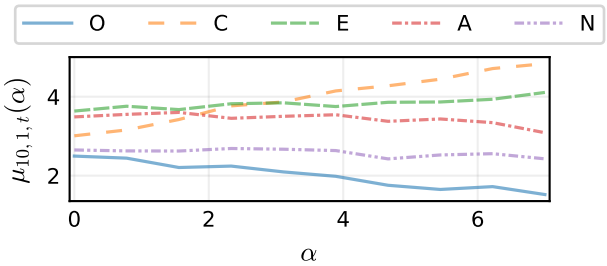

Figure 21. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 10$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

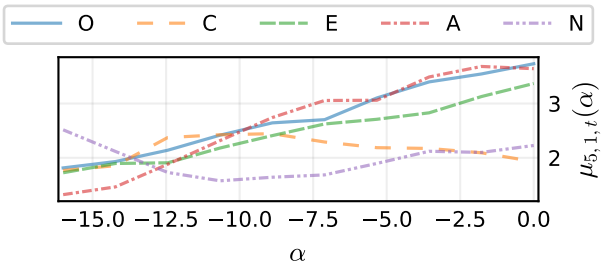

Figure 22. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 5$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 16$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

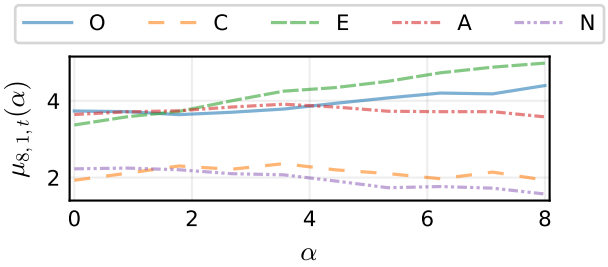

Figure 23. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

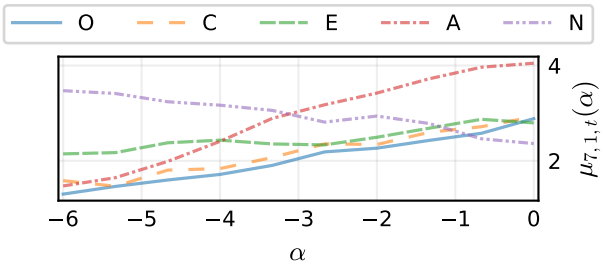

Figure 24. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 7$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

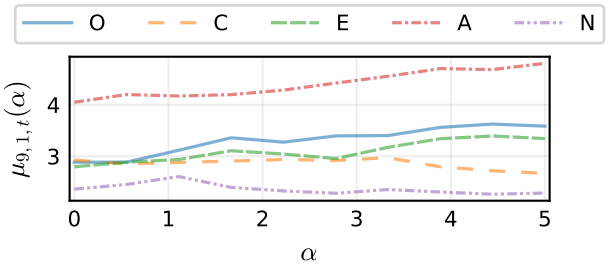

Figure 25. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 9$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

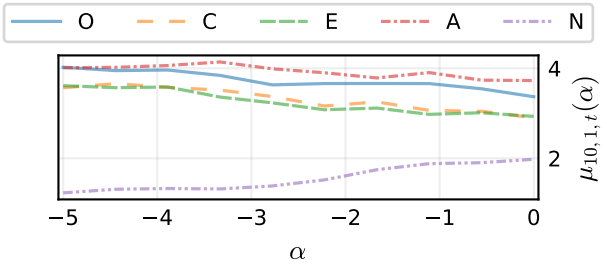

Figure 26. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 10$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

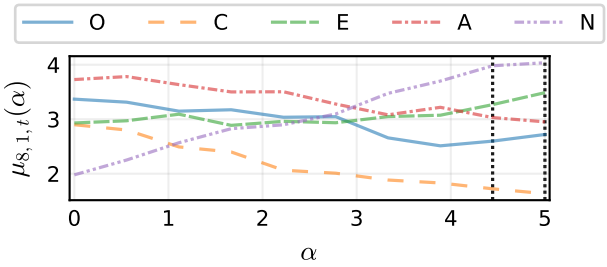

Figure 27. OCEAN scores for Llama-3.2-1B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 4.444$; 1 outlier at $\alpha = 5$.

I OCEAN Injection Results for Llama-3.2-3B-Instruct

I.1 Injection Method Comparison

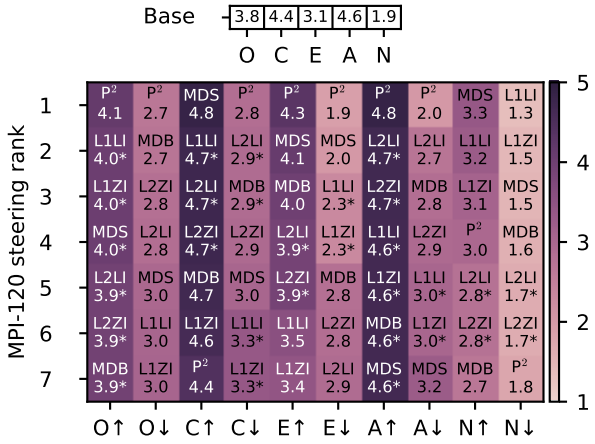

Figure 28. Ranking of steering methods on Llama-3.2-3B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

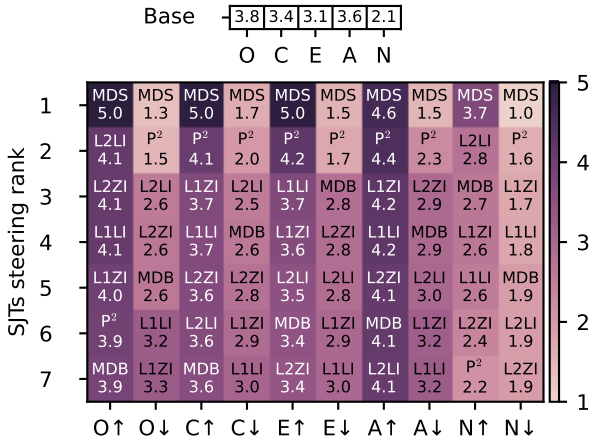

Figure 29. Ranking of steering methods on Llama-3.2-3B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

I.2 Effects of Injection Stride on MDS Injections

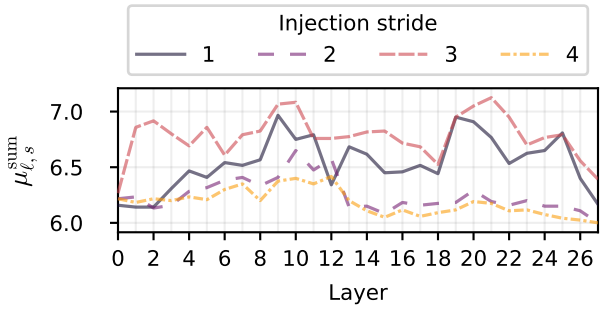

Figure 30. Overall MDS injections steering performance on Llama-3.2-3B-Instruct on the MPI-120 task by injection stride s and model layer ℓ.

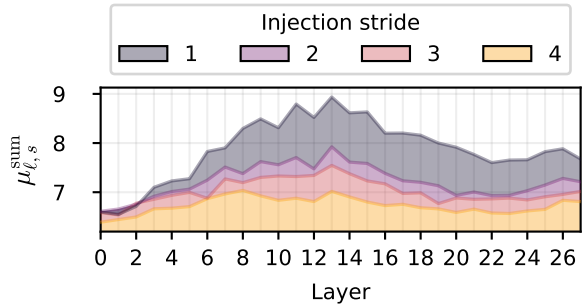

Figure 31. Overall MDS injections steering performance on Llama-3.2-3B-Instruct on the SJTs task by injection stride s and model layer ℓ.

I.3 Layer Identification for MDS Injections

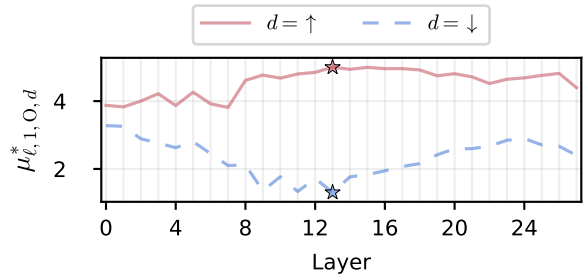

Figure 32. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Llama-3.2-3B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

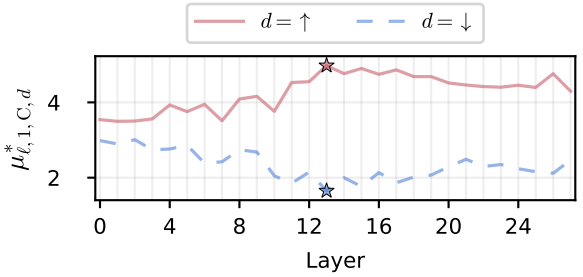

Figure 33. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Llama-3.2-3B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

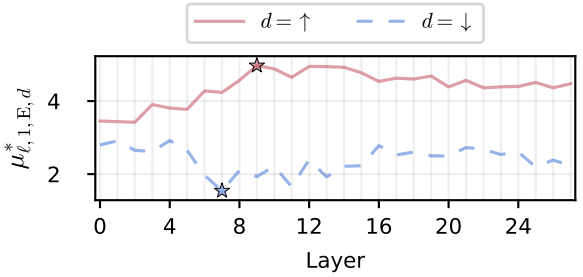

Figure 34. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Llama-3.2-3B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

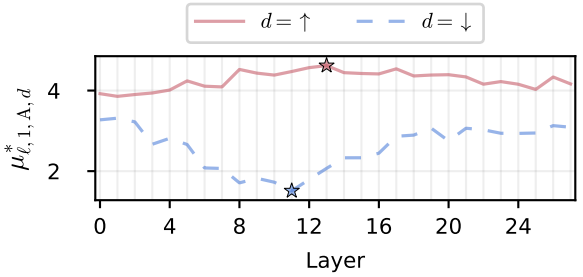

Figure 35. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Llama-3.2-3B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

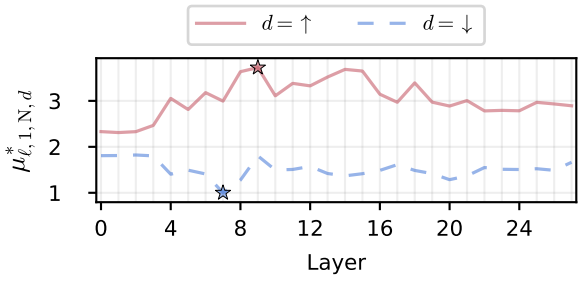

Figure 36. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Llama-3.2-3B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

I.4 OCEAN Trait Entanglement Under MDS Injections

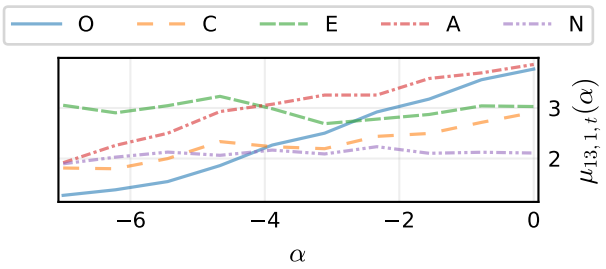

Figure 37. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

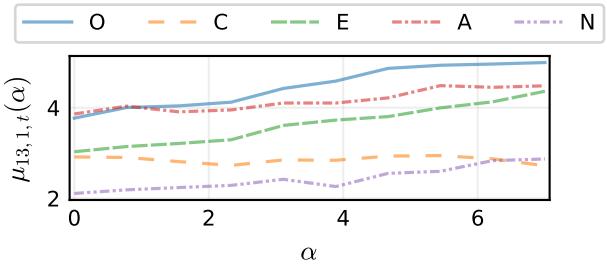

Figure 38. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

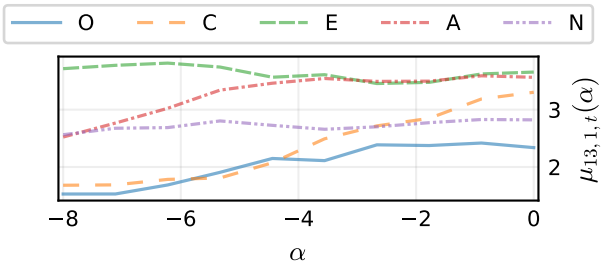

Figure 39. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

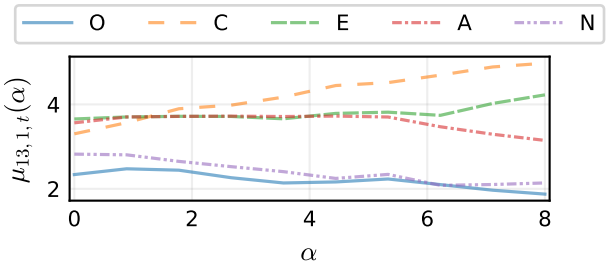

Figure 40. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

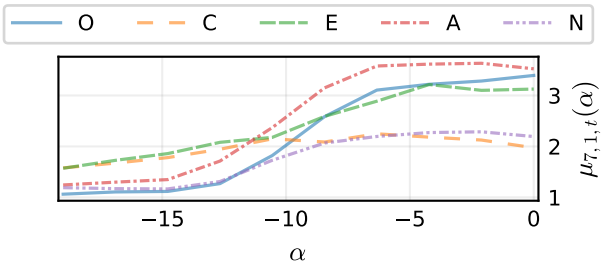

Figure 41. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 7$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 19$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

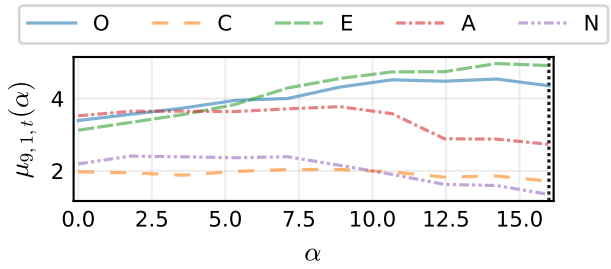

Figure 42. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 9$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 16$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 16$.

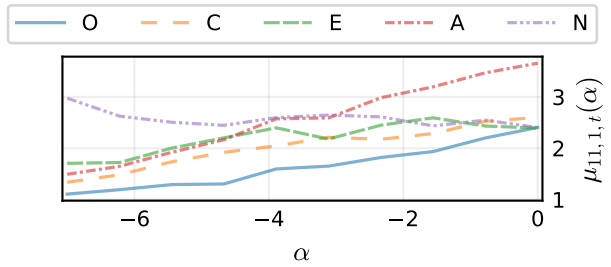

Figure 43. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 11$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

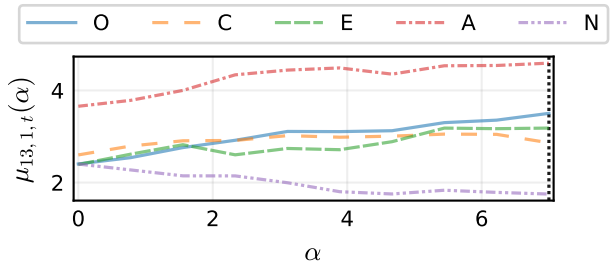

Figure 44. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 7$.

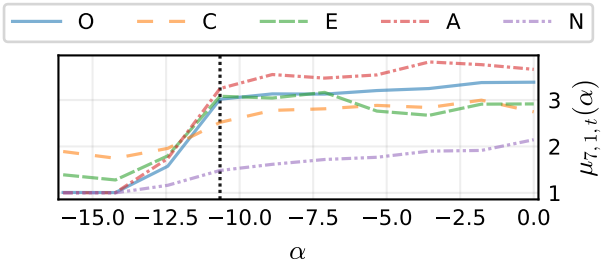

Figure 45. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 7$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 16$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx -10.667$.

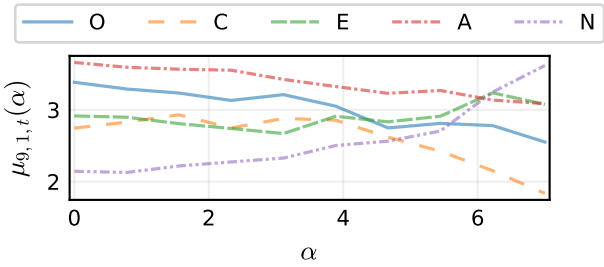

Figure 46. OCEAN scores for Llama-3.2-3B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 9$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

J OCEAN Injection Results for Llama-3.1-8B-Instruct

J.1 Injection Method Comparison

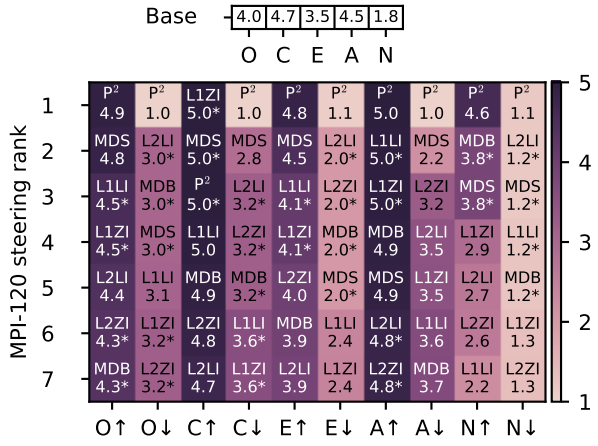

Figure 47. Ranking of steering methods on Llama-3.1-8B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

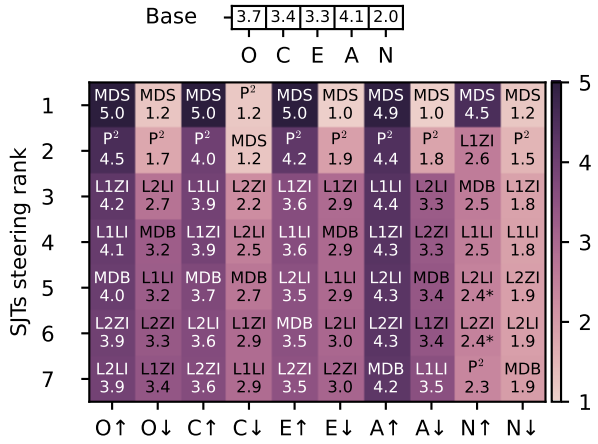

Figure 48. Ranking of steering methods on Llama-3.1-8B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

J.2 Effects of Injection Stride on MDS Injections

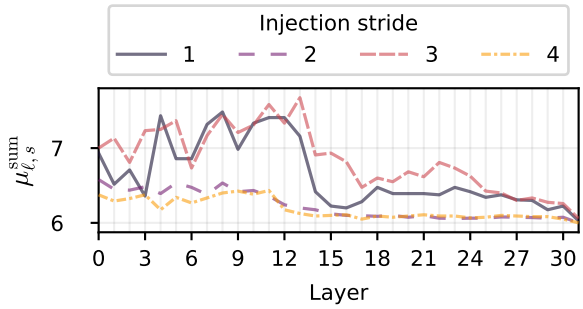

Figure 49. Overall MDS injections steering performance on Llama-3.1-8B-Instruct on the MPI-120 task by injection stride s and model layer ℓ.

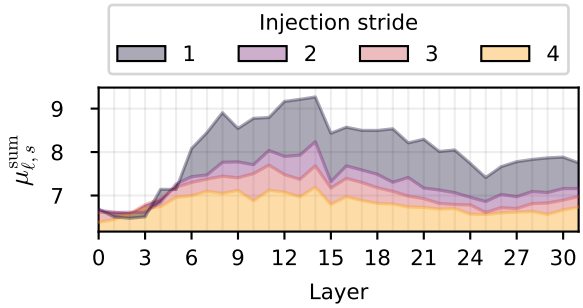

Figure 50. Overall MDS injections steering performance on Llama-3.1-8B-Instruct on the SJTs task by injection stride s and model layer ℓ.

J.3 Layer Identification for MDS Injections

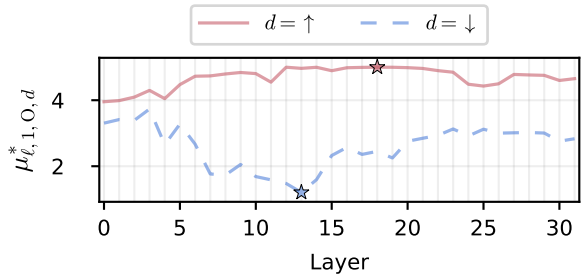

Figure 51. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Llama-3.1-8B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{O},d}$).

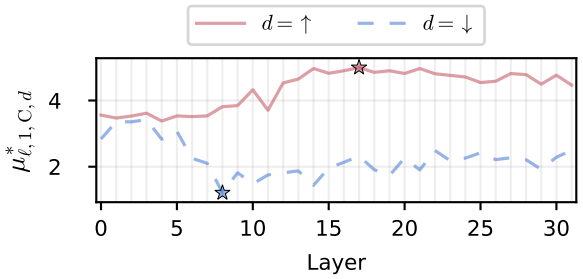

Figure 52. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Llama-3.1-8B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{C},d}$).

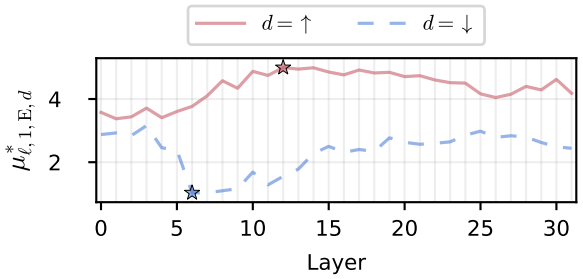

Figure 53. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Llama-3.1-8B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{E},d}$).

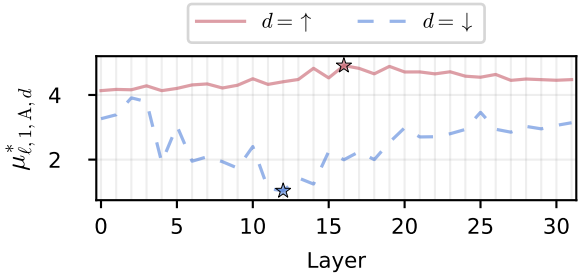

Figure 54. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Llama-3.1-8B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{A},d}$).

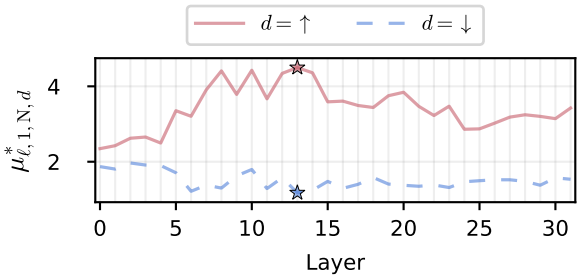

Figure 55. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Llama-3.1-8B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

J.4 OCEAN Trait Entanglement Under MDS Injections

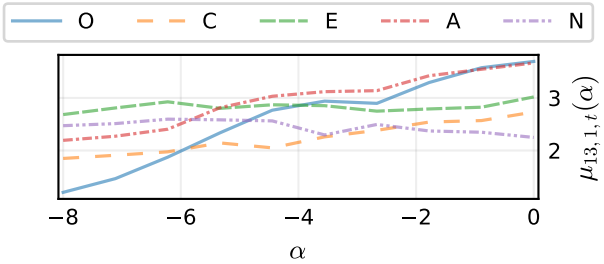

Figure 56. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

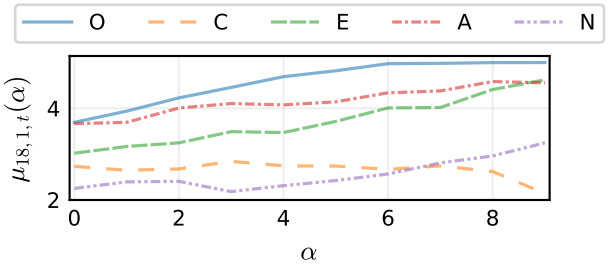

Figure 57. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 18$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

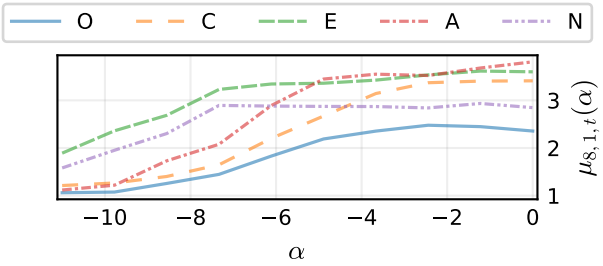

Figure 58. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

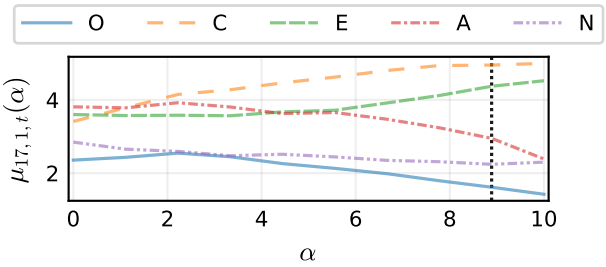

Figure 59. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 17$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 8.889$.

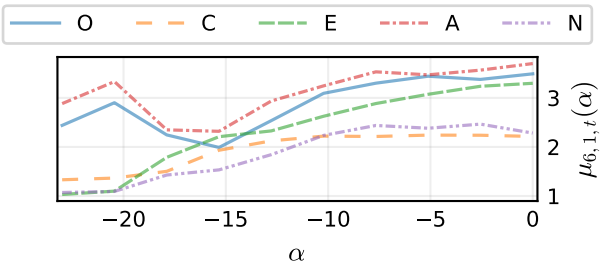

Figure 60. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 6$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 23$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

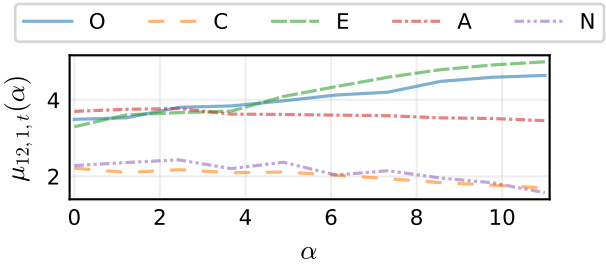

Figure 61. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 12$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

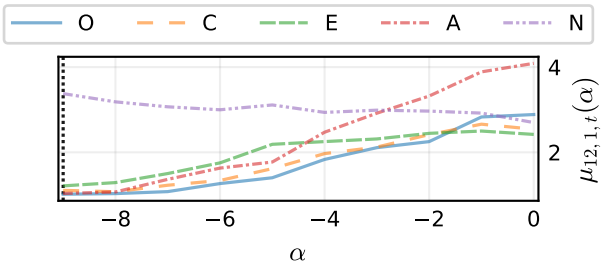

Figure 62. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 12$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = -9$.

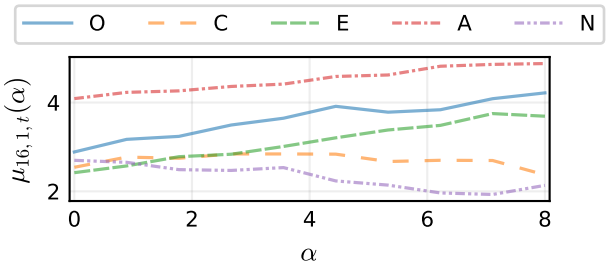

Figure 63. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

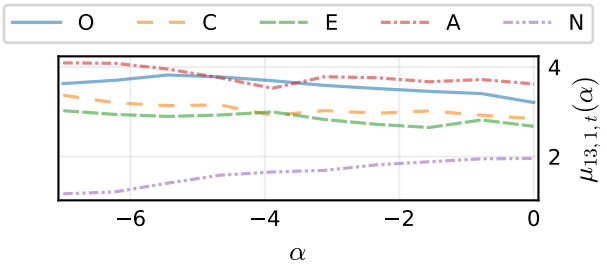

Figure 64. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

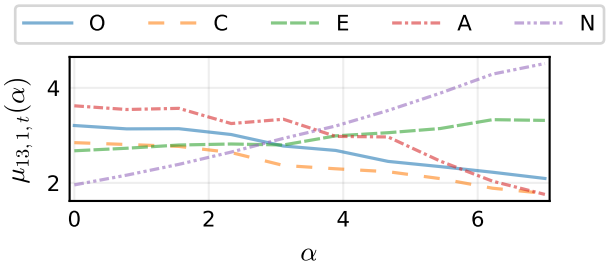

Figure 65. OCEAN scores for Llama-3.1-8B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

K OCEAN Injection Results for Qwen3-1.7B

K.1 Injection Method Comparison

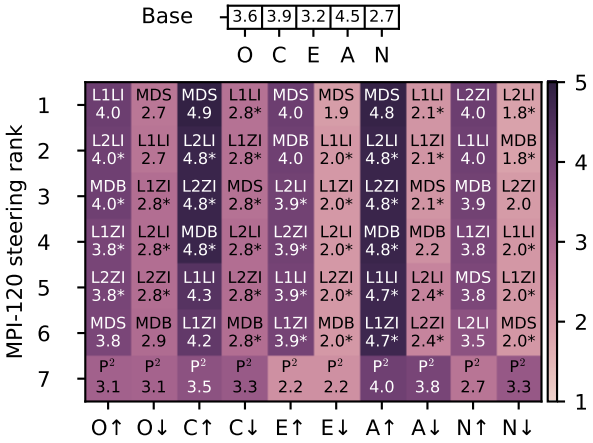

Figure 66. Ranking of steering methods on Qwen3-1.7B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

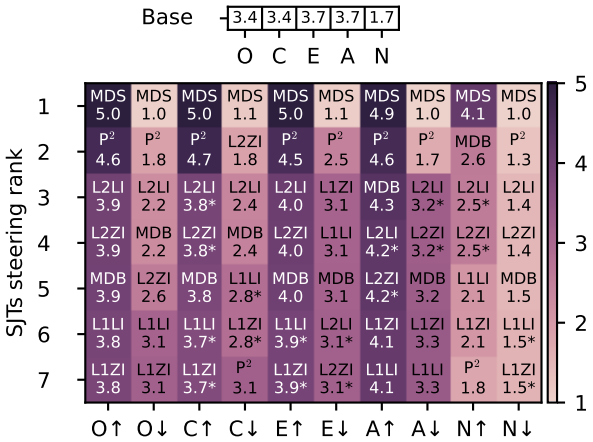

Figure 67. Ranking of steering methods on Qwen3-1.7B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

K.2 Effects of Injection Stride on MDS Injections

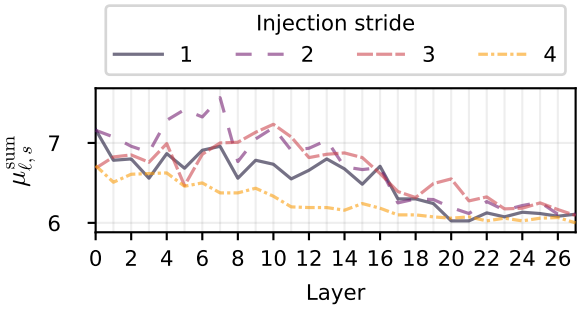

Figure 68. Overall MDS injections steering performance on Qwen3-1.7B on the MPI-120 task by injection stride s and model layer ℓ.

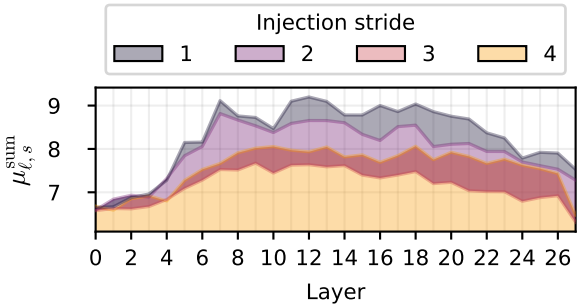

Figure 69. Overall MDS injections steering performance on Qwen3-1.7B on the SJTs task by injection stride s and model layer ℓ.

K.3 Layer Identification for MDS Injections

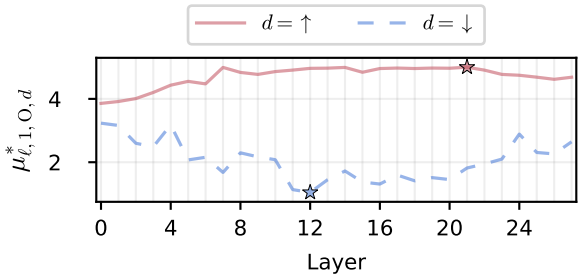

Figure 70. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Qwen3-1.7B. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

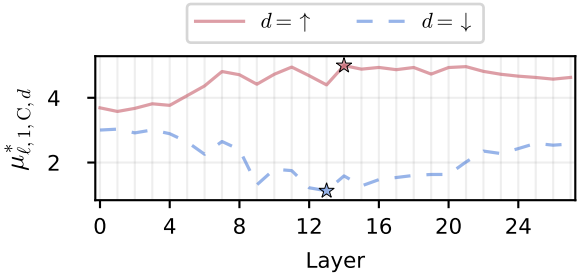

Figure 71. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Qwen3-1.7B. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

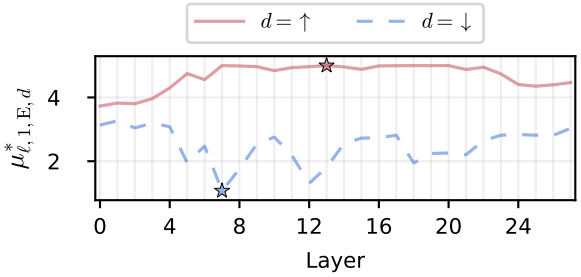

Figure 72. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Qwen3-1.7B. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

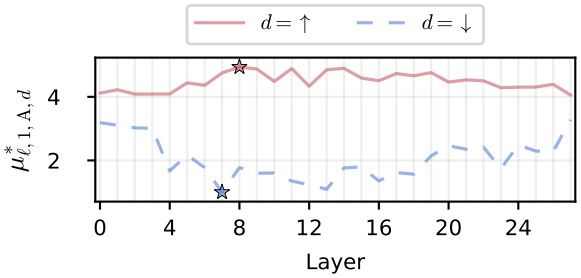

Figure 73. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Qwen3-1.7B. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

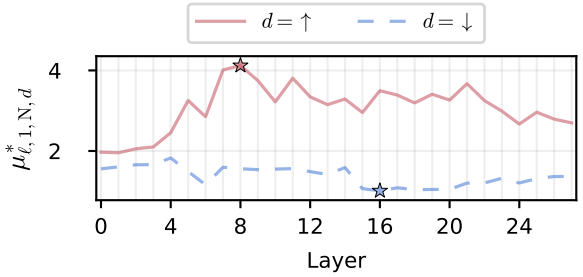

Figure 74. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Qwen3-1.7B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

K.4 OCEAN Trait Entanglement Under MDS Injections

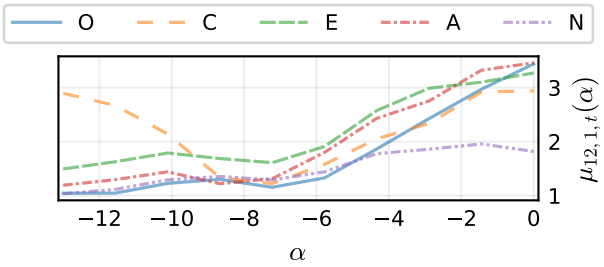

Figure 75. OCEAN scores for Qwen3-1.7B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 12$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

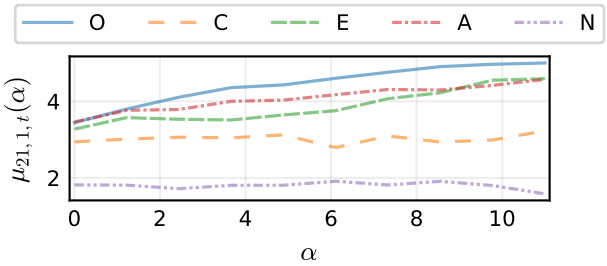

Figure 76. OCEAN scores for Qwen3-1.7B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 21$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

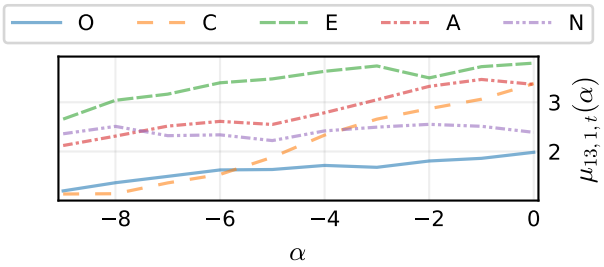

Figure 77. OCEAN scores for Qwen3-1.7B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

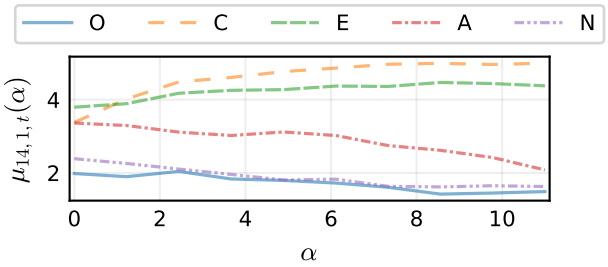

Figure 78. OCEAN scores for Qwen3-1.7B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

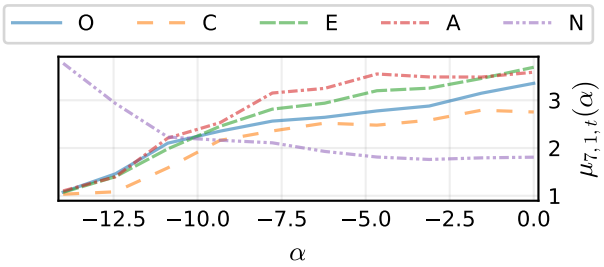

Figure 79. OCEAN scores for Qwen3-1.7B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 7$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

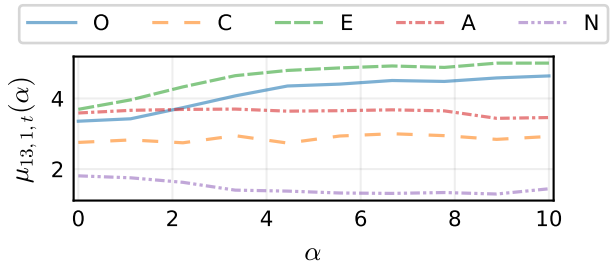

Figure 80. OCEAN scores for Qwen3-1.7B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

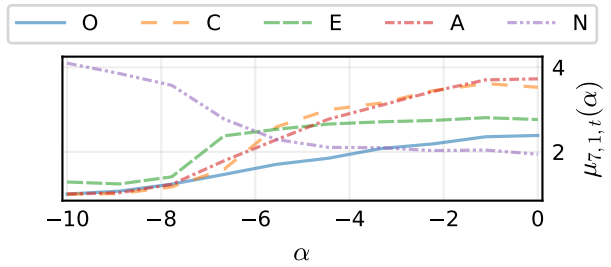

Figure 81. OCEAN scores for Qwen3-1.7B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 7$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

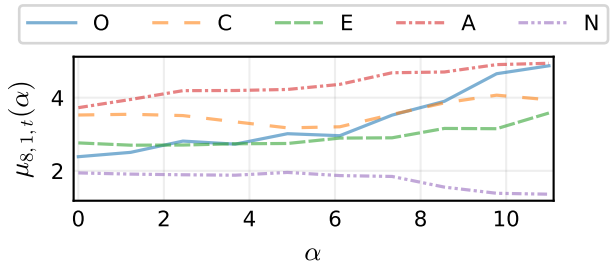

Figure 82. OCEAN scores for Qwen3-1.7B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

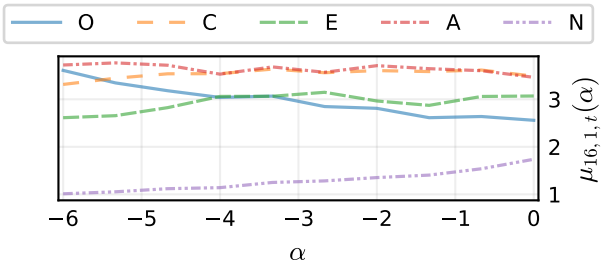

Figure 83. OCEAN scores for Qwen3-1.7B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

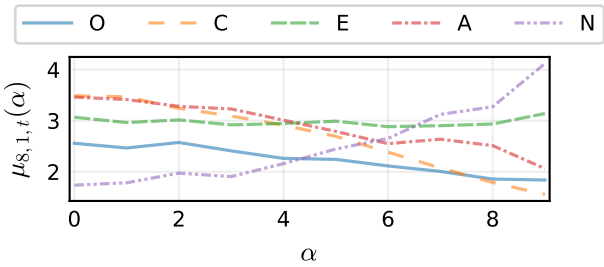

Figure 84. OCEAN scores for Qwen3-1.7B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 8$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

L OCEAN Injection Results for Qwen3-4B

L.1 Injection Method Comparison

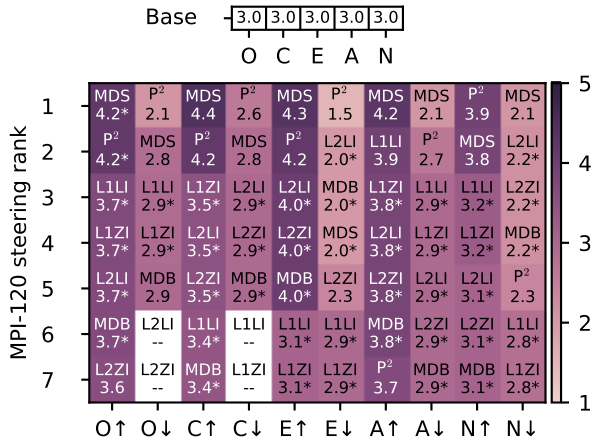

Figure 85. Ranking of steering methods on Qwen3-4B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

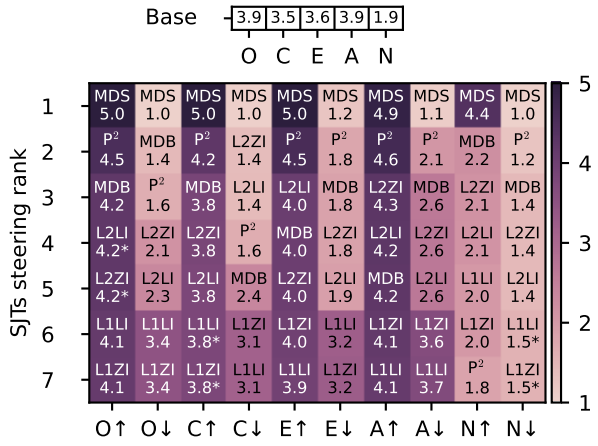

Figure 86. Ranking of steering methods on Qwen3-4B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

L.2 Effects of Injection Stride on MDS Injections

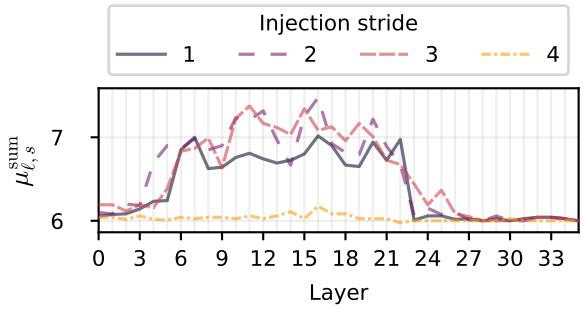

Figure 87. Overall MDS injections steering performance on Qwen3-4B on the MPI-120 task by injection stride s and model layer ℓ.

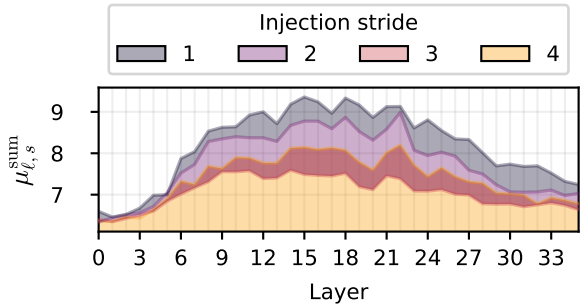

Figure 88. Overall MDS injections steering performance on Qwen3-4B on the SJTs task by injection stride s and model layer ℓ.

L.3 Layer Identification for MDS Injections

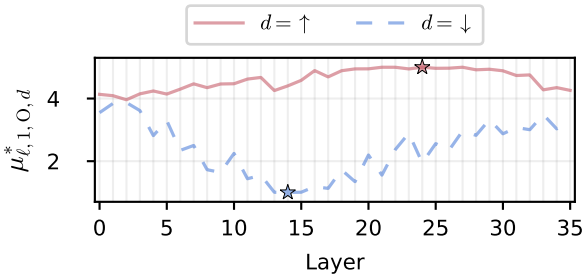

Figure 89. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Qwen3-4B. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

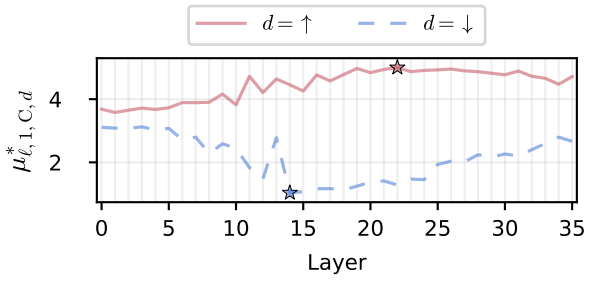

Figure 90. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Qwen3-4B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{C},d}$).

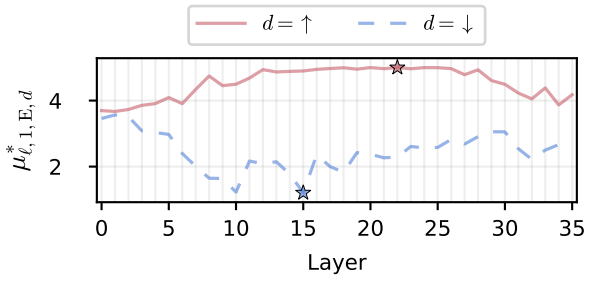

Figure 91. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Qwen3-4B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{E},d}$).

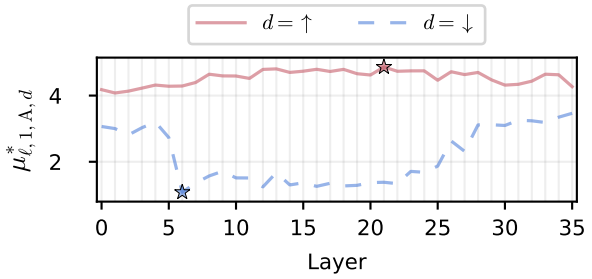

Figure 92. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Qwen3-4B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{A},d}$).

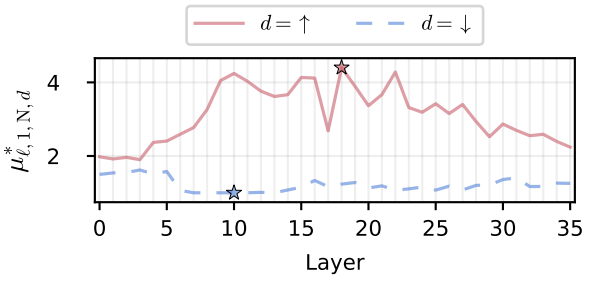

Figure 93. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Qwen3-4B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

L.4 OCEAN Trait Entanglement Under MDS Injections

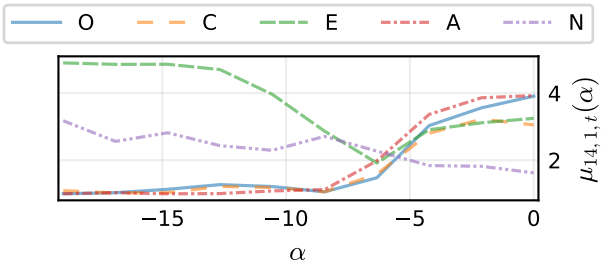

Figure 94. OCEAN scores for Qwen3-4B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 19$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

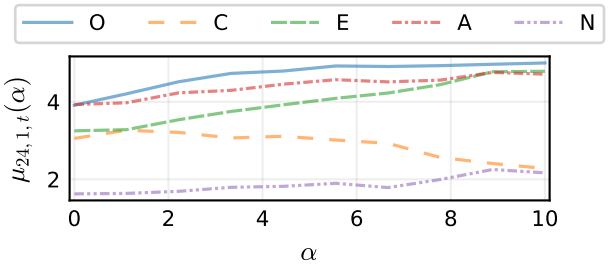

Figure 95. OCEAN scores for Qwen3-4B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

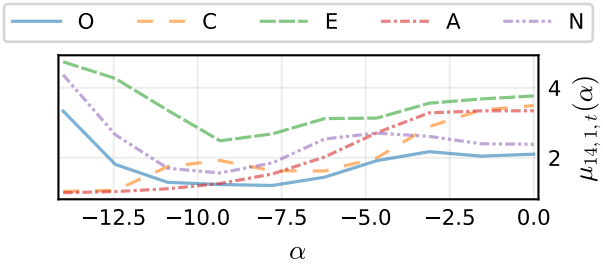

Figure 96. OCEAN scores for Qwen3-4B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

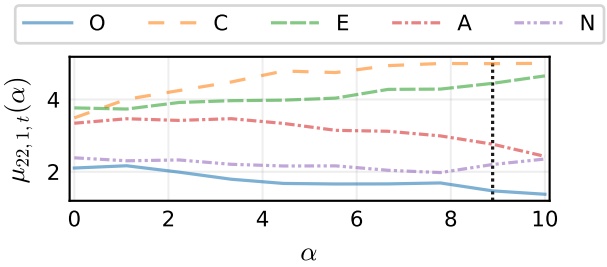

Figure 97. OCEAN scores for Qwen3-4B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 22$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 2 outliers at $\alpha \approx 8.889$.

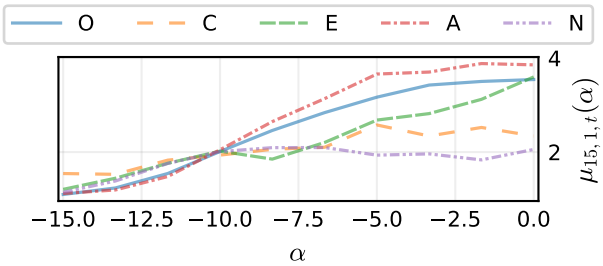

Figure 98. OCEAN scores for Qwen3-4B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 15$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 15$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

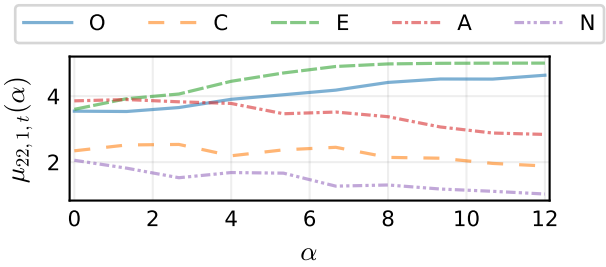

Figure 99. OCEAN scores for Qwen3-4B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 22$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

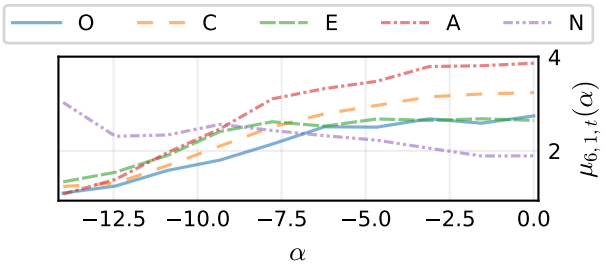

Figure 100. OCEAN scores for Qwen3-4B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 6$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

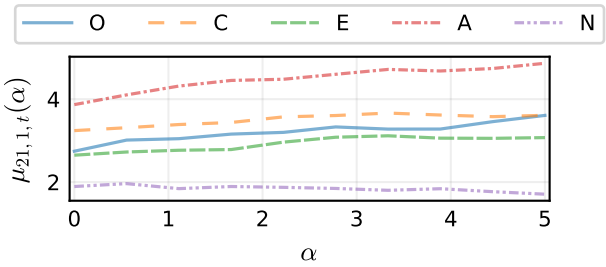

Figure 101. OCEAN scores for Qwen3-4B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 21$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

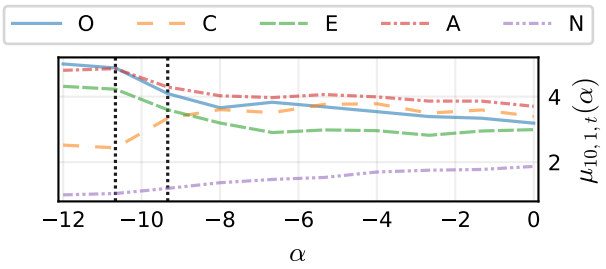

Figure 102. OCEAN scores for Qwen3-4B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 10$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 3 outliers at $\alpha \approx -10.667$; 1 outlier at $\alpha \approx -9.333$.

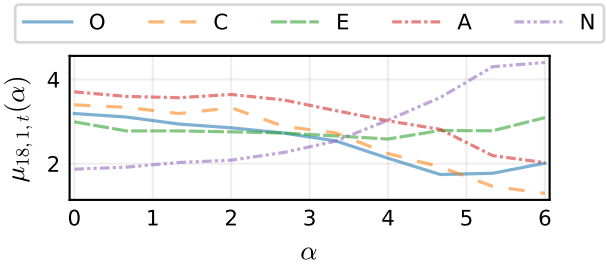

Figure 103. OCEAN scores for Qwen3-4B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 18$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

M OCEAN Injection Results for Qwen3-8B

M.1 Injection Method Comparison

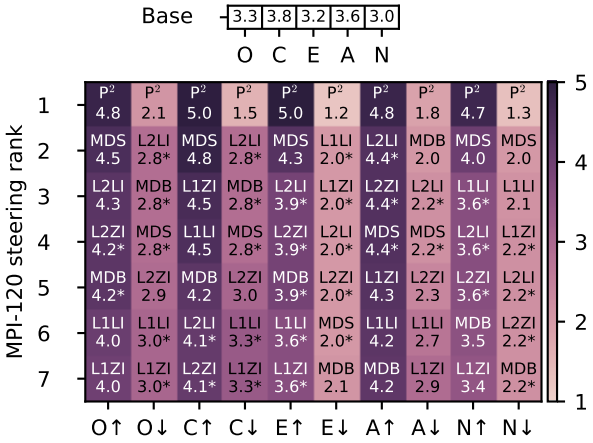

Figure 104. Ranking of steering methods on Qwen3-8B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

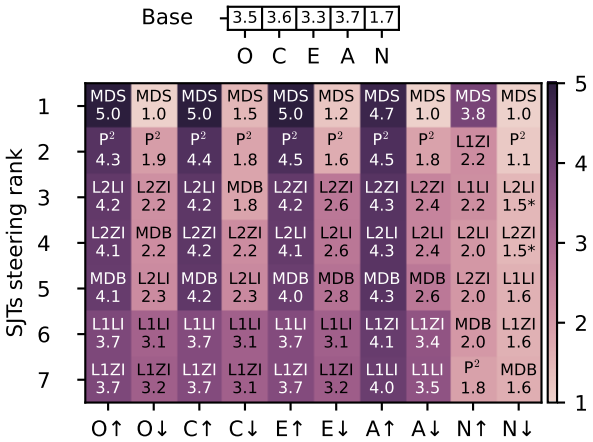

Figure 105. Ranking of steering methods on Qwen3-8B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

M.2 Effects of Injection Stride on MDS Injections

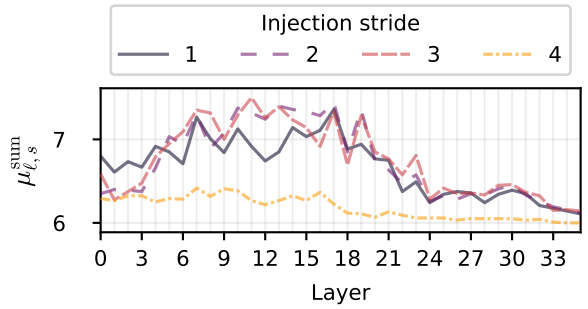

Figure 106. Overall MDS injections steering performance on Qwen3-8B on the MPI-120 task by injection stride s and model layer ℓ.

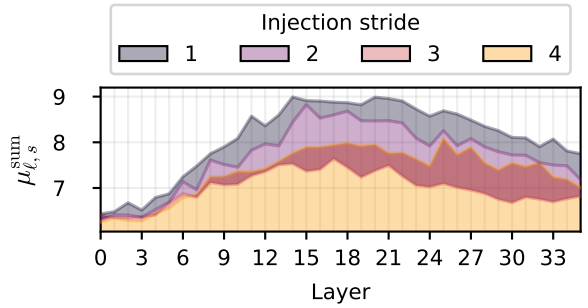

Figure 107. Overall MDS injections steering performance on Qwen3-8B on the SJTs task by injection stride s and model layer ℓ.

M.3 Layer Identification for MDS Injections

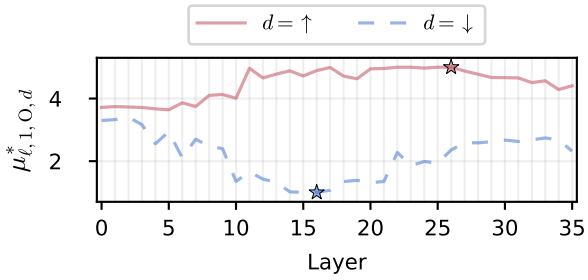

Figure 108. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Qwen3-8B. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

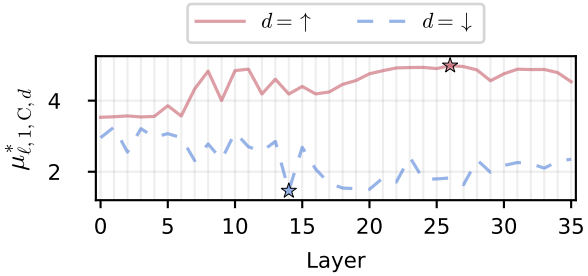

Figure 109. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Qwen3-8B. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

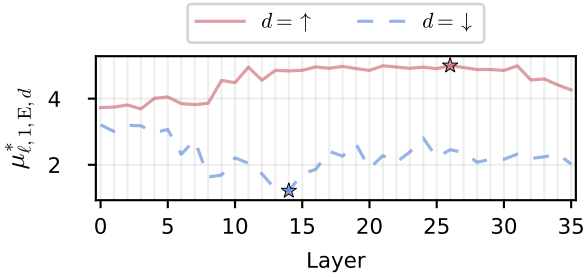

Figure 110. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Qwen3-8B. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

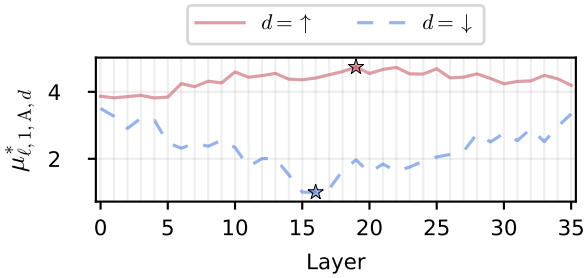

Figure 111. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Qwen3-8B. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

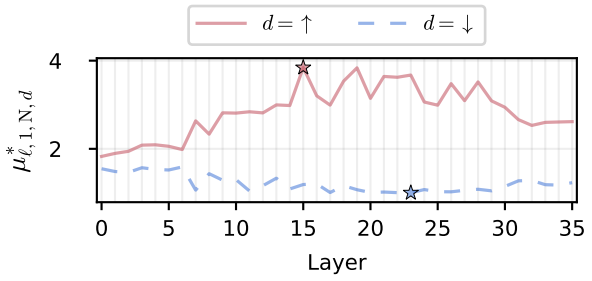

Figure 112. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Qwen3-8B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

M.4 OCEAN Trait Entanglement Under MDS Injections

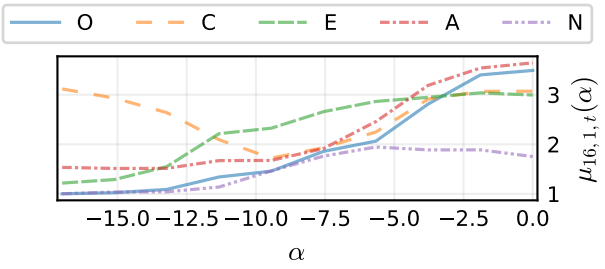

Figure 113. OCEAN scores for Qwen3-8B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 17$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

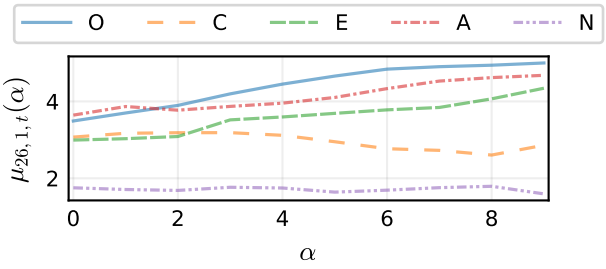

Figure 114. OCEAN scores for Qwen3-8B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

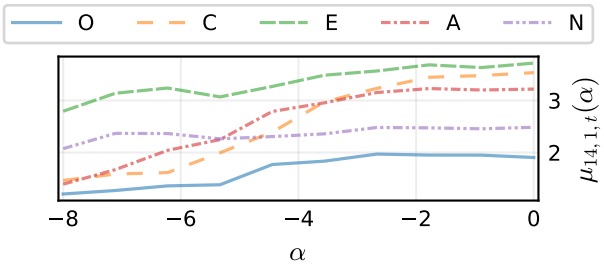

Figure 115. OCEAN scores for Qwen3-8B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

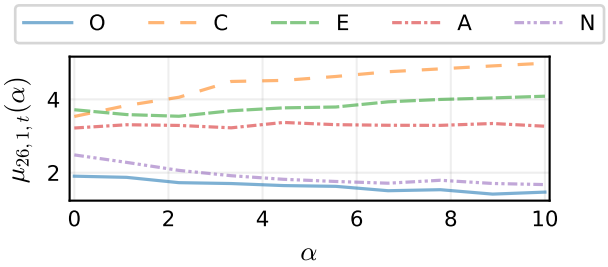

Figure 116. OCEAN scores for Qwen3-8B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

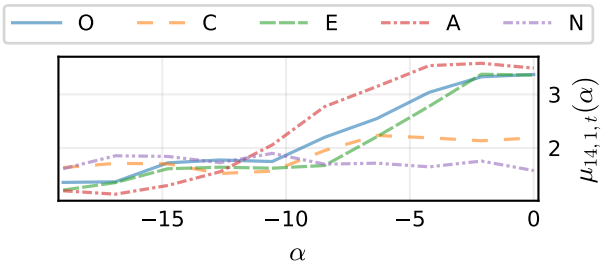

Figure 117. OCEAN scores for Qwen3-8B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 19$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

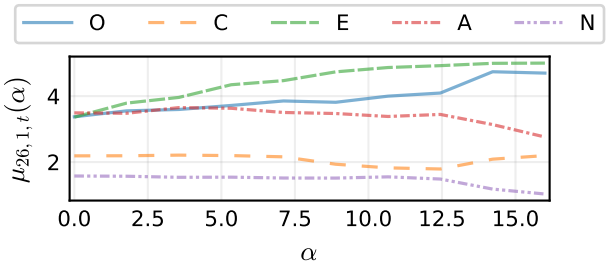

Figure 118. OCEAN scores for Qwen3-8B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 16$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

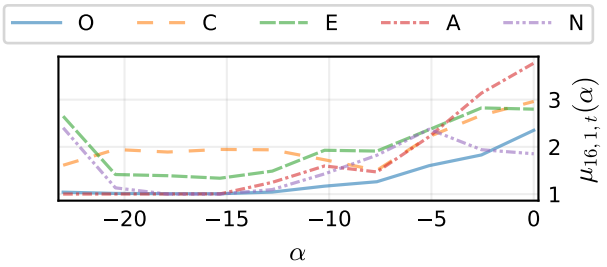

Figure 119. OCEAN scores for Qwen3-8B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 23$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

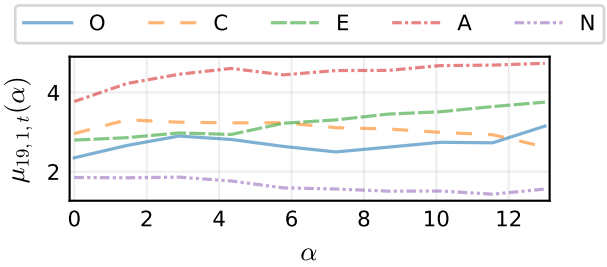

Figure 120. OCEAN scores for Qwen3-8B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 19$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

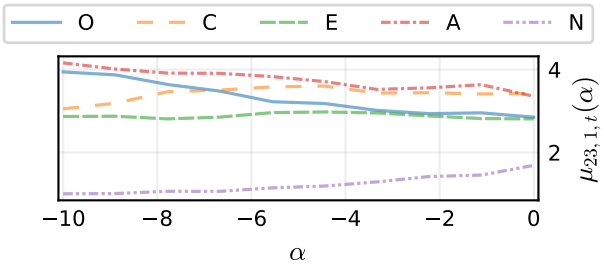

Figure 121. OCEAN scores for Qwen3-8B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 23$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

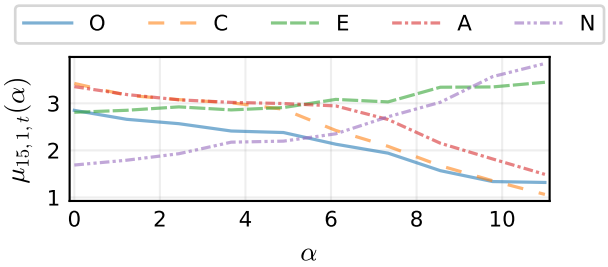

Figure 122. OCEAN scores for Qwen3-8B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 15$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

N OCEAN Injection Results for Qwen3-14B

N.1 Injection Method Comparison

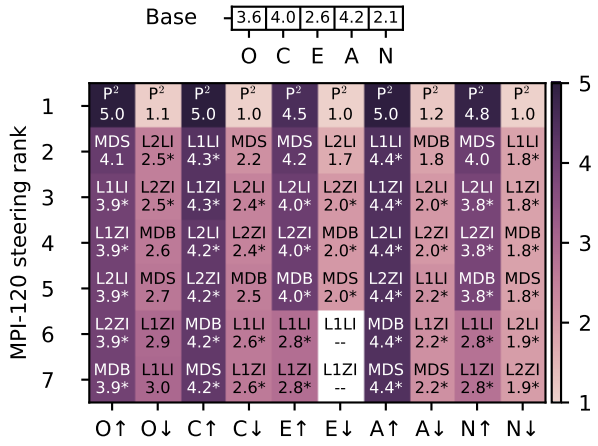

Figure 123. Ranking of steering methods on Qwen3-14B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

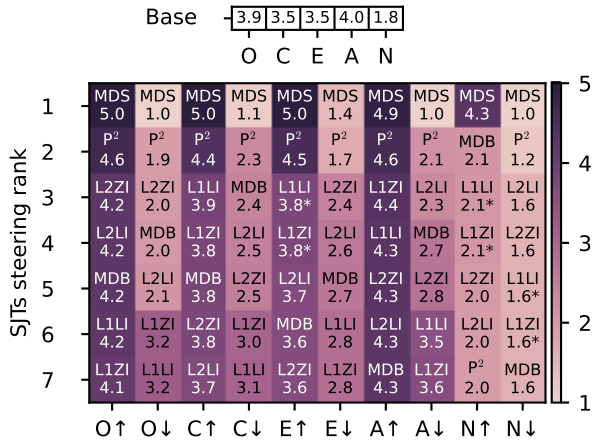

Figure 124. Ranking of steering methods on Qwen3-14B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

N.2 Effects of Injection Stride on MDS Injections

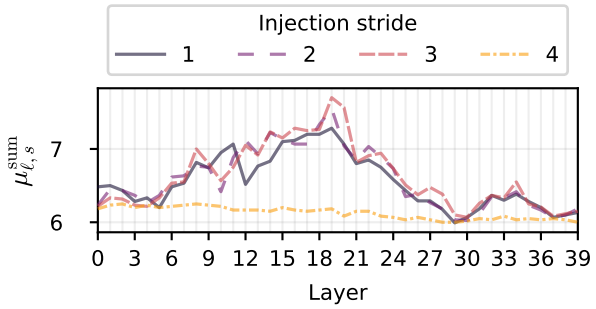

Figure 125. Overall MDS injections steering performance on Qwen3-14B on the MPI-120 task by injection stride s and model layer ℓ.

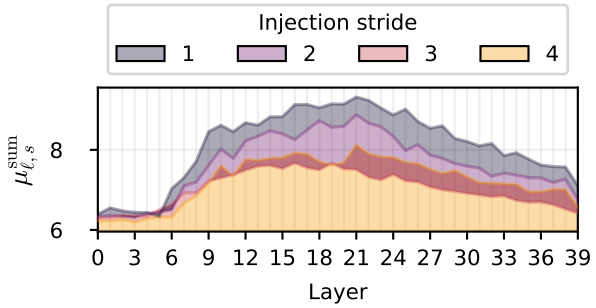

Figure 126. Overall MDS injections steering performance on Qwen3-14B on the SJTs task by injection stride s and model layer ℓ.

N.3 Layer Identification for MDS Injections

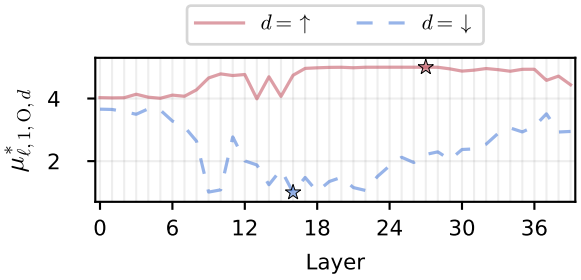

Figure 127. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Qwen3-14B. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

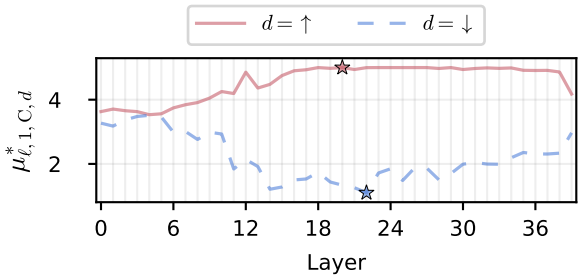

Figure 128. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Qwen3-14B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{C},d}$).

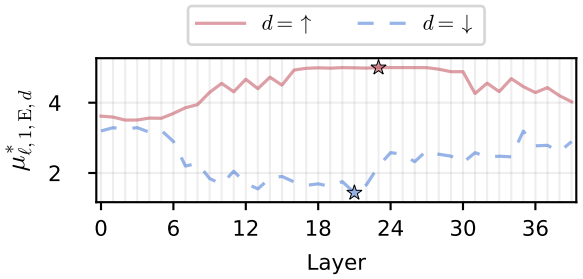

Figure 129. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Qwen3-14B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{E},d}$).

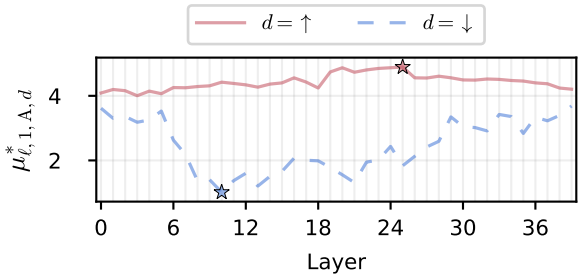

Figure 130. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Qwen3-14B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{A},d}$).

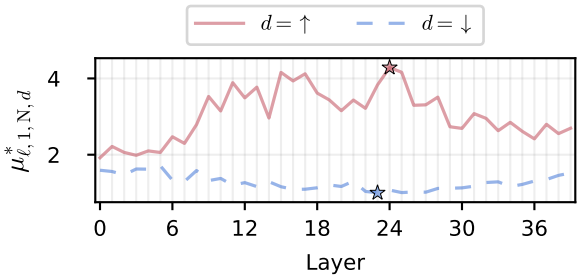

Figure 131. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Qwen3-14B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

N.4 OCEAN Trait Entanglement Under MDS Injections

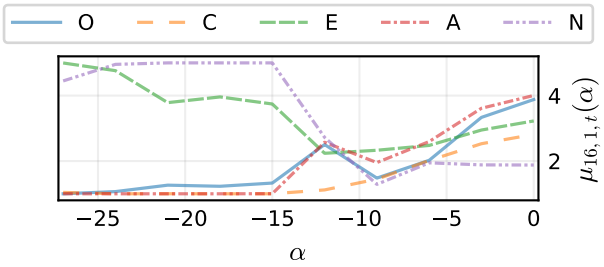

Figure 132. OCEAN scores for Qwen3-14B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 27$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

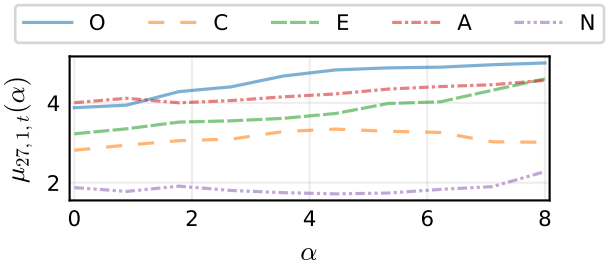

Figure 133. OCEAN scores for Qwen3-14B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 27$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

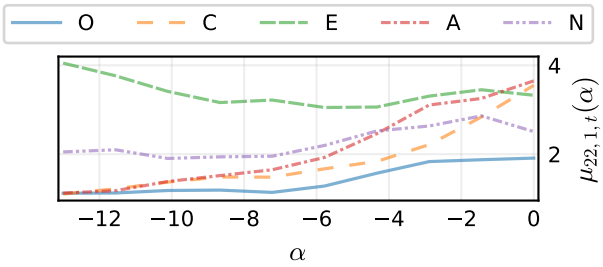

Figure 134. OCEAN scores for Qwen3-14B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 22$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

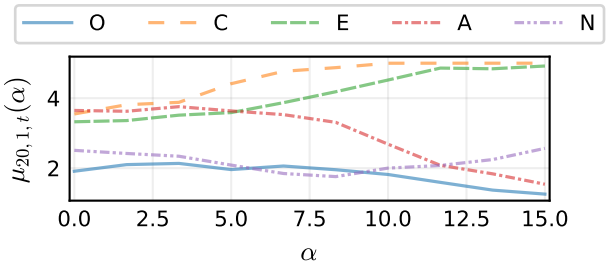

Figure 135. OCEAN scores for Qwen3-14B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 20$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 15$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

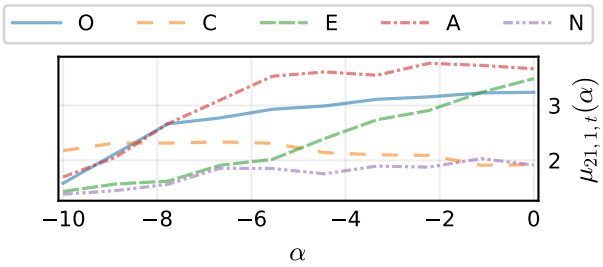

Figure 136. OCEAN scores for Qwen3-14B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 21$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

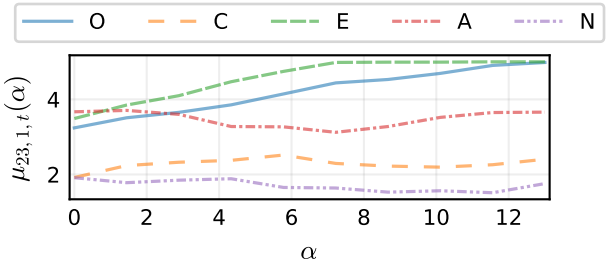

Figure 137. OCEAN scores for Qwen3-14B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 23$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

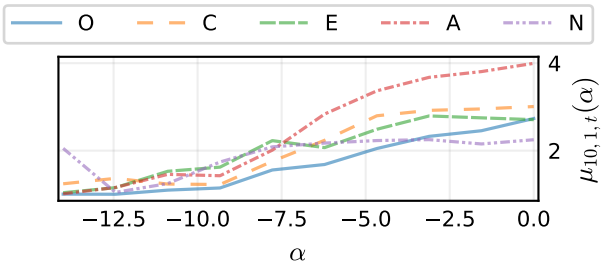

Figure 138. OCEAN scores for Qwen3-14B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 10$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

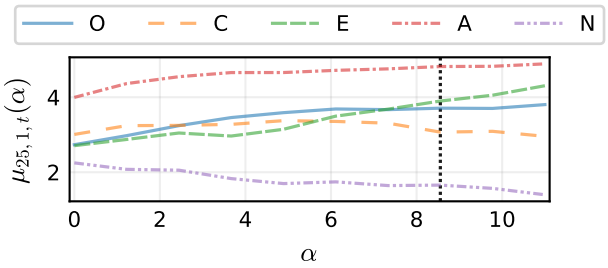

Figure 139. OCEAN scores for Qwen3-14B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 25$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 8.556$.

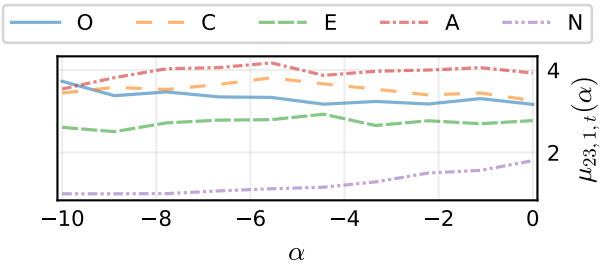

Figure 140. OCEAN scores for Qwen3-14B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 23$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

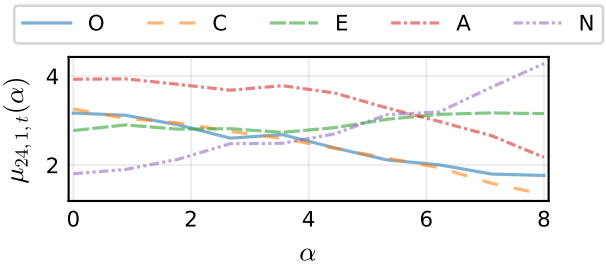

Figure 141. OCEAN scores for Qwen3-14B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

O OCEAN Injection Results for Qwen3-32B

O.1 Injection Method Comparison

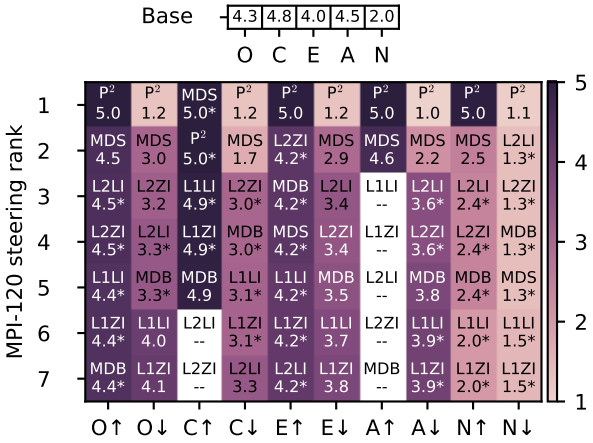

Figure 142. Ranking of steering methods on Qwen3-32B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

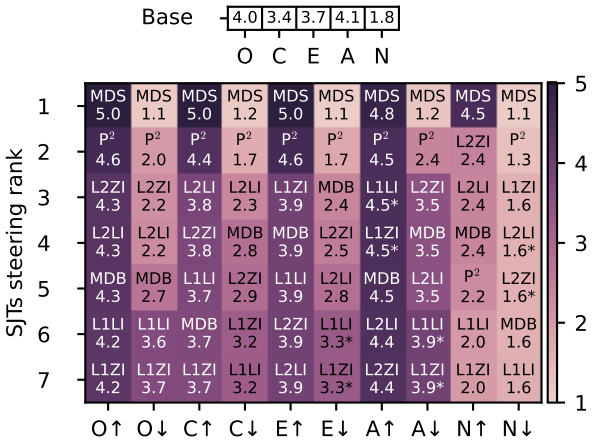

Figure 143. Ranking of steering methods on Qwen3-32B by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

O.2 Effects of Injection Stride on MDS Injections

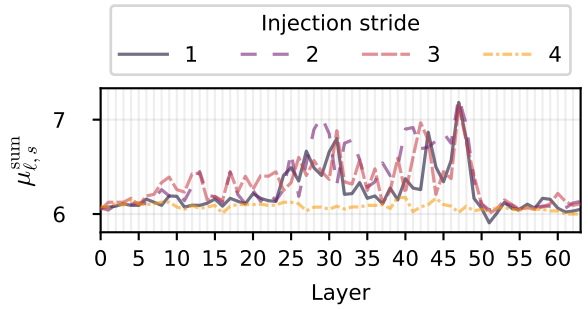

Figure 144. Overall MDS injections steering performance on Qwen3-32B on the MPI-120 task by injection stride s and model layer ℓ.

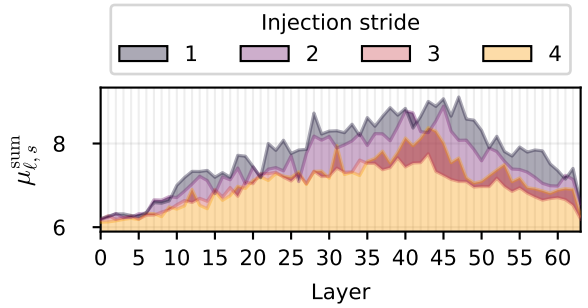

Figure 145. Overall MDS injections steering performance on Qwen3-32B on the SJTs task by injection stride s and model layer ℓ.

O.3 Layer Identification for MDS Injections

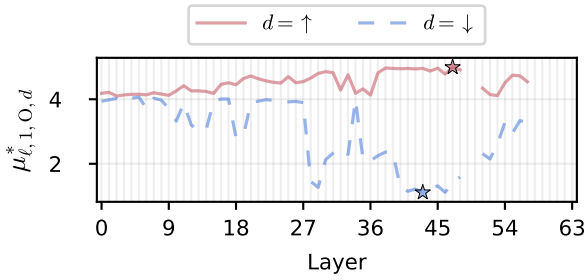

Figure 146. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Qwen3-32B. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

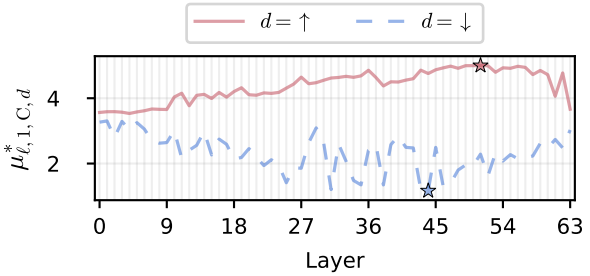

Figure 147. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Qwen3-32B. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

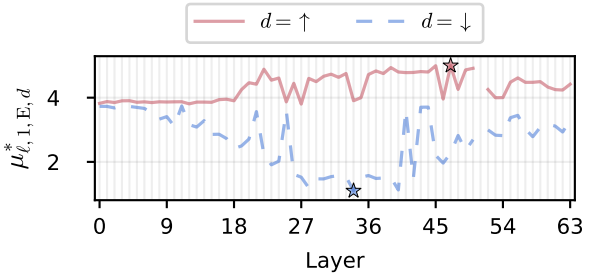

Figure 148. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Qwen3-32B. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

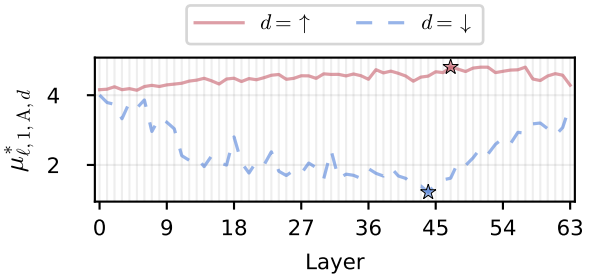

Figure 149. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Qwen3-32B. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

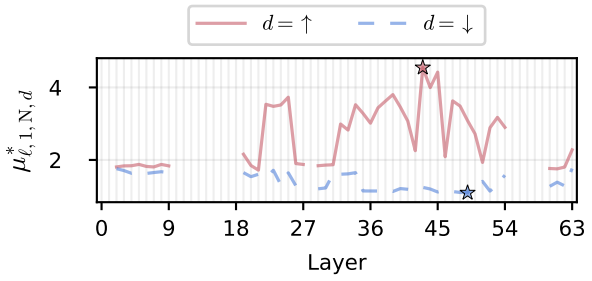

Figure 150. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Qwen3-32B. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

O.4 OCEAN Trait Entanglement Under MDS Injections

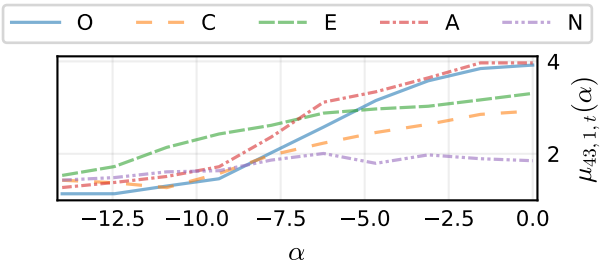

Figure 151. OCEAN scores for Qwen3-32B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 43$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

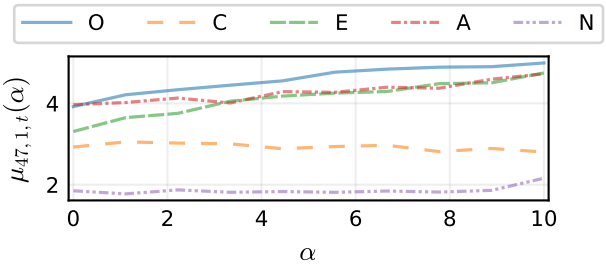

Figure 152. OCEAN scores for Qwen3-32B on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 47$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

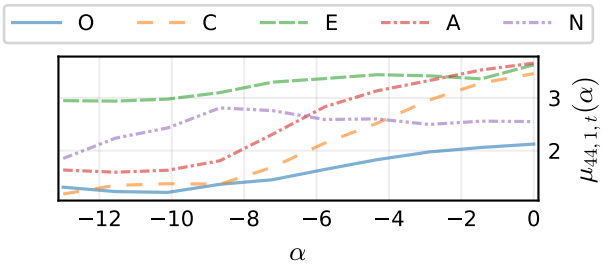

Figure 153. OCEAN scores for Qwen3-32B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 44$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

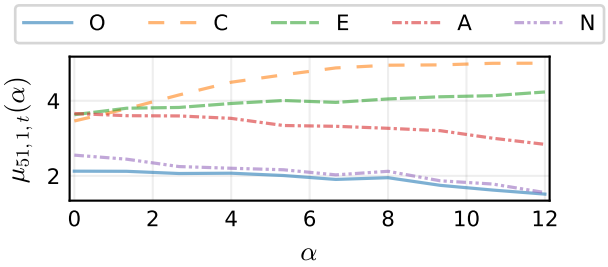

Figure 154. OCEAN scores for Qwen3-32B on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 51$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

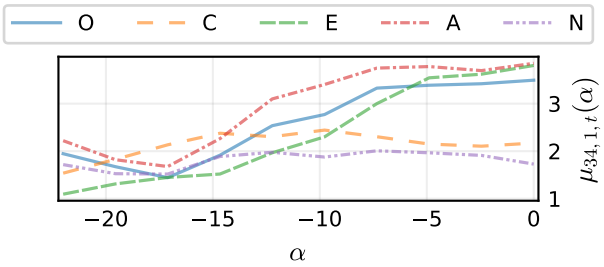

Figure 155. OCEAN scores for Qwen3-32B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 34$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 22$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

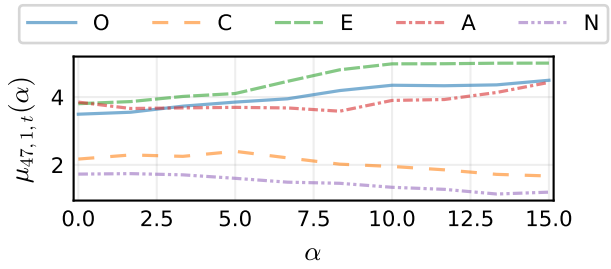

Figure 156. OCEAN scores for Qwen3-32B on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 47$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 15$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

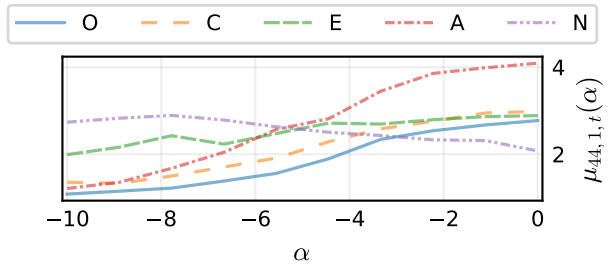

Figure 157. OCEAN scores for Qwen3-32B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 44$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

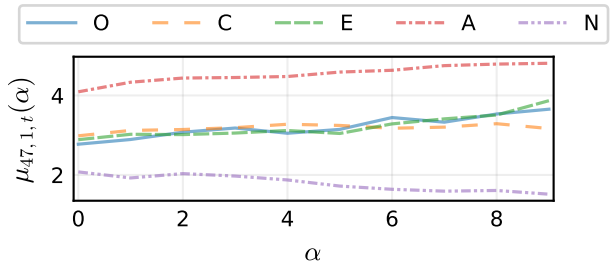

Figure 158. OCEAN scores for Qwen3-32B on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 47$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

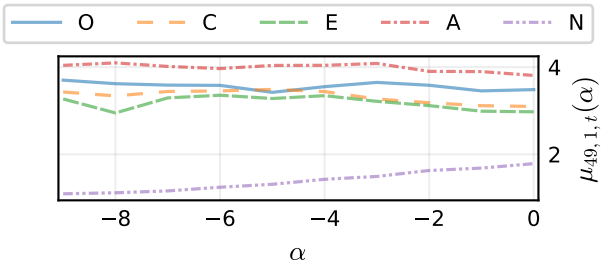

Figure 159. OCEAN scores for Qwen3-32B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 49$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

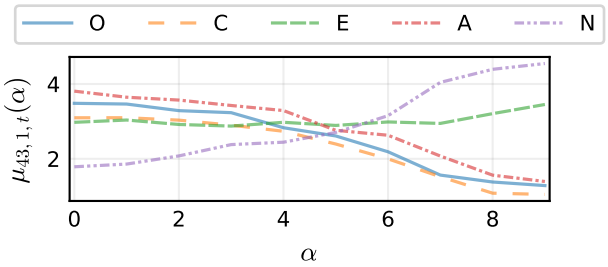

Figure 160. OCEAN scores for Qwen3-32B on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 43$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

P OCEAN Injection Results for gemma-3-1b-it

P.1 Injection Method Comparison

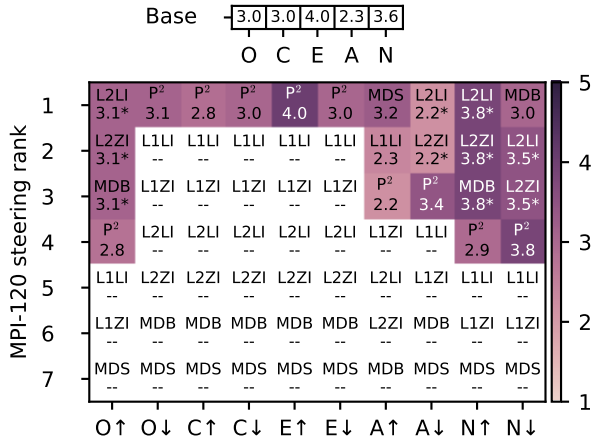

Figure 161. Ranking of steering methods on gemma-3-1b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

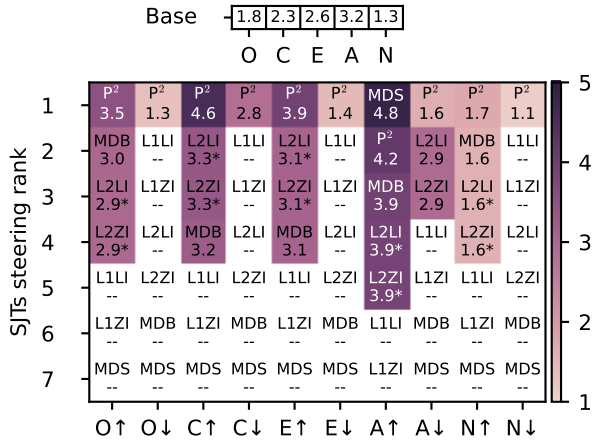

Figure 162. Ranking of steering methods on gemma-3-1b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

P.2 Effects of Injection Stride on MDS Injections

No valid MDS injection results to plot.

P.3 Layer Identification for MDS Injections

No valid MDS injection results to plot.

P.4 OCEAN Trait Entanglement Under MDS Injections

No valid MDS injection results to plot.

Q OCEAN Injection Results for gemma-3-4b-it

Q.1 Injection Method Comparison

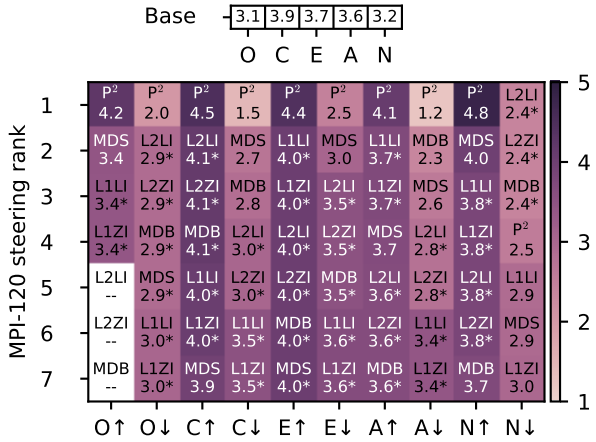

Figure 163. Ranking of steering methods on gemma-3-4b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

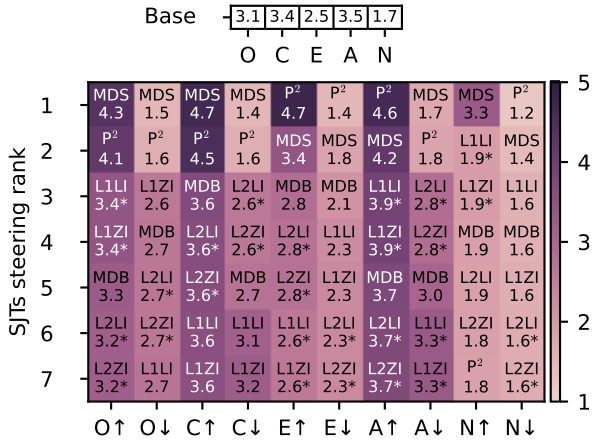

Figure 164. Ranking of steering methods on gemma-3-4b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

Q.2 Effects of Injection Stride on MDS Injections

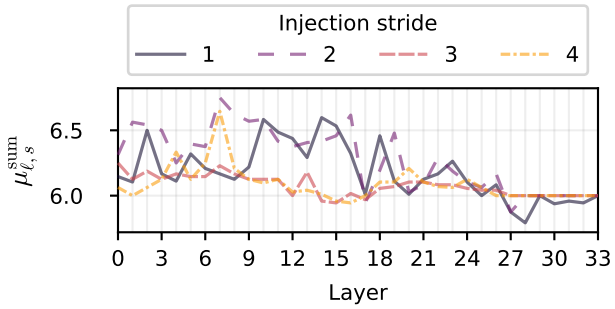

Figure 165. Overall MDS injections steering performance on gemma-3-4b-it on the MPI-120 task by injection stride s and model layer ℓ.

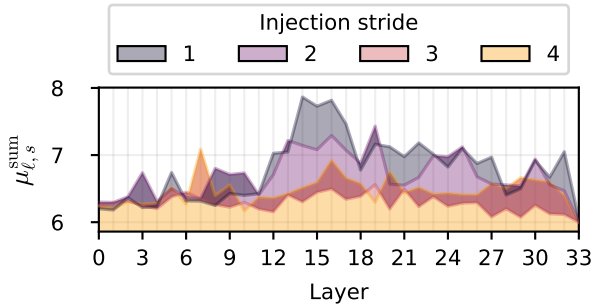

Figure 166. Overall MDS injections steering performance on gemma-3-4b-it on the SJTs task by injection stride s and model layer ℓ.

Q.3 Layer Identification for MDS Injections

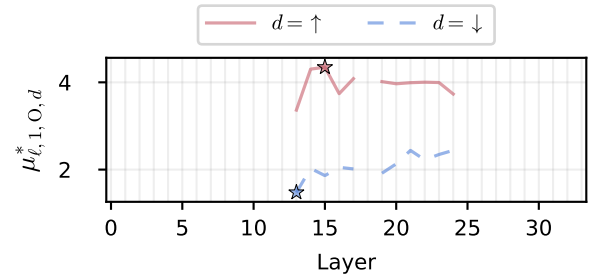

Figure 167. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on gemma-3-4b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

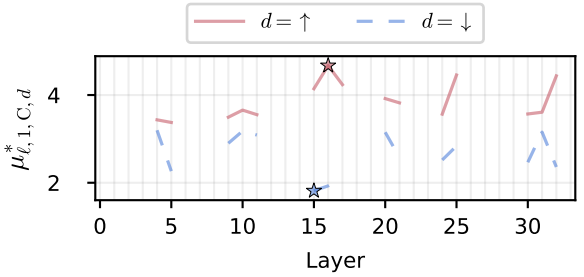

Figure 168. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on gemma-3-4b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

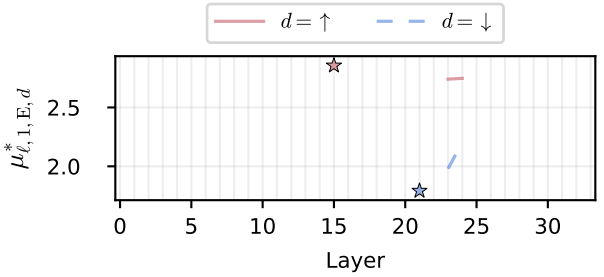

Figure 169. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on gemma-3-4b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

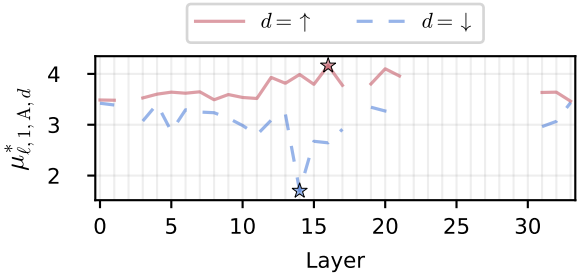

Figure 170. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on gemma-3-4b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

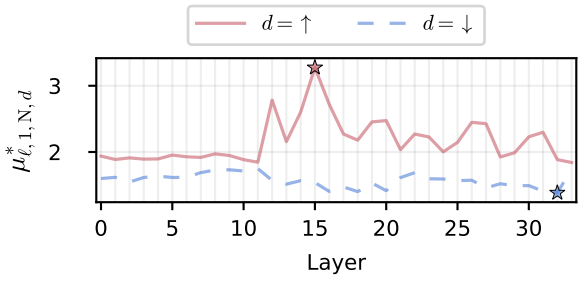

Figure 171. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on gemma-3-4b-it. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

Q.4 OCEAN Trait Entanglement Under MDS Injections

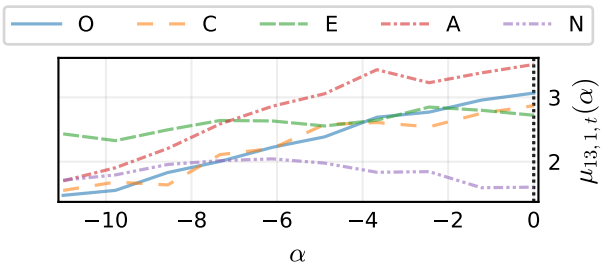

Figure 172. OCEAN scores for gemma-3-4b-it on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the openness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 0$.

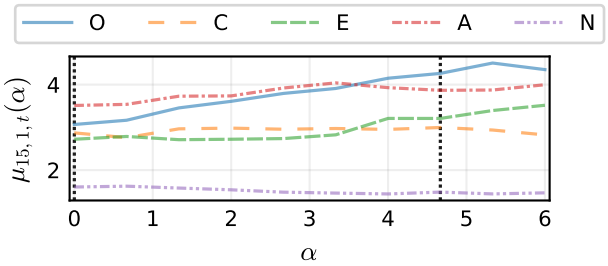

Figure 173. OCEAN scores for gemma-3-4b-it on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 15$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the openness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 0$; 1 outlier at $\alpha \approx 4.667$.

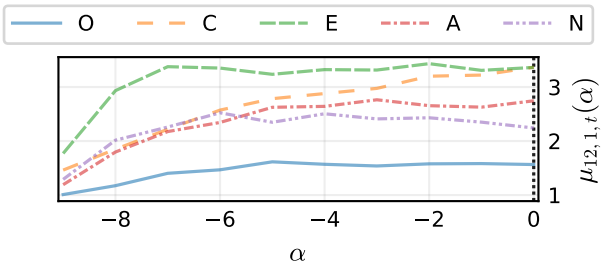

Figure 174. OCEAN scores for gemma-3-4b-it on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 12$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 0$.

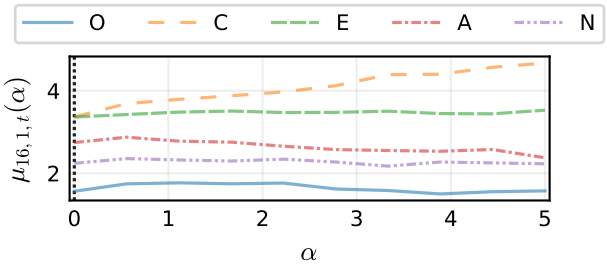

Figure 175. OCEAN scores for gemma-3-4b-it on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 0$.

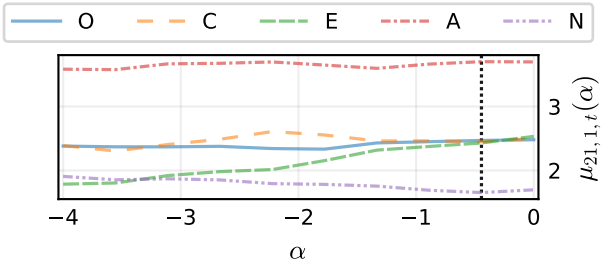

Figure 176. OCEAN scores for gemma-3-4b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 21$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 4$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 2 outliers at $\alpha \approx -0.444$.

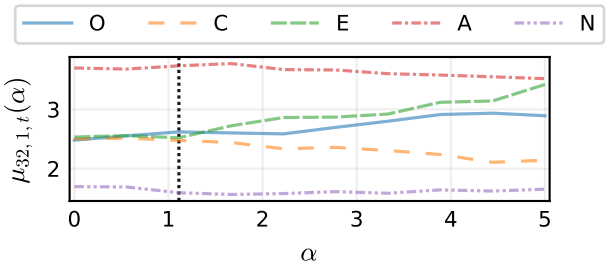

Figure 177. OCEAN scores for gemma-3-4b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 32$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 1.111$.

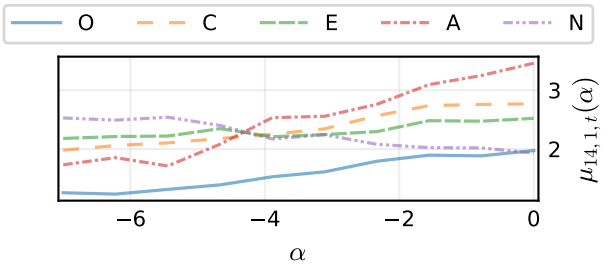

Figure 178. OCEAN scores for gemma-3-4b-it on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

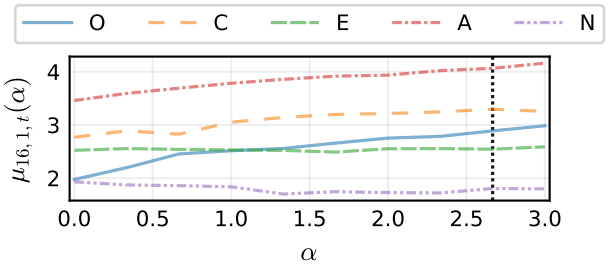

Figure 179. OCEAN scores for gemma-3-4b-it on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 3$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 2.667$.

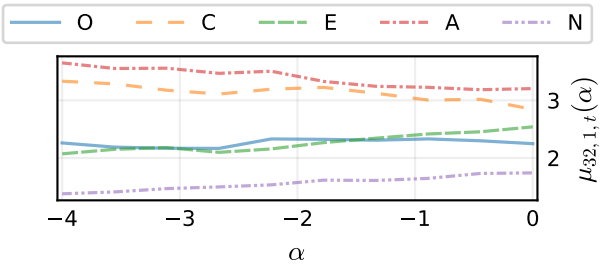

Figure 180. OCEAN scores for gemma-3-4b-it on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 32$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 4$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

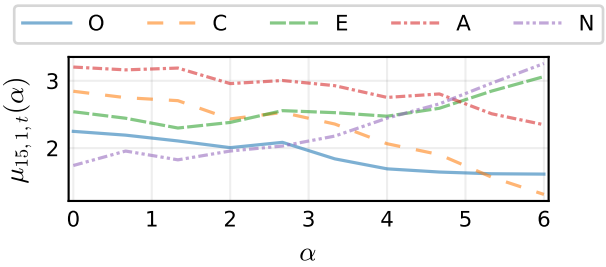

Figure 181. OCEAN scores for gemma-3-4b-it on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 15$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

R OCEAN Injection Results for gemma-3-12b-it

R.1 Injection Method Comparison

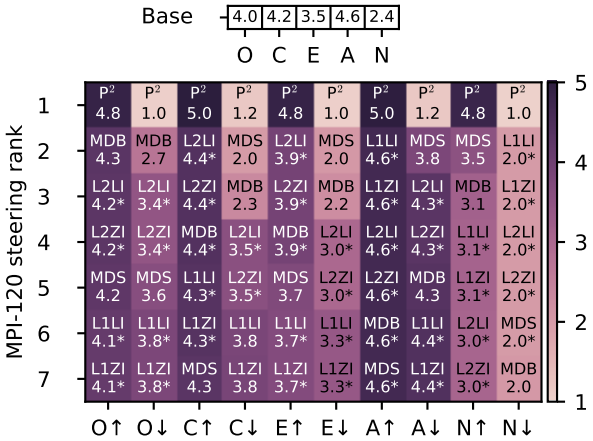

Figure 182. Ranking of steering methods on gemma-3-12b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

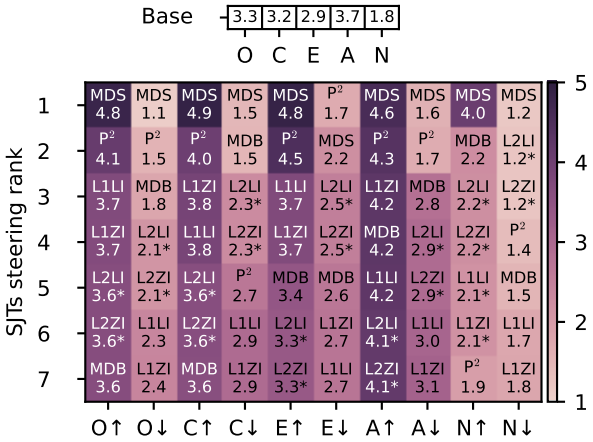

Figure 183. Ranking of steering methods on gemma-3-12b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

R.2 Effects of Injection Stride on MDS Injections

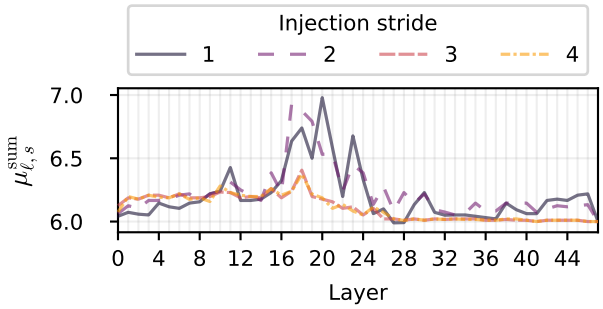

Figure 184. Overall MDS injections steering performance on gemma-3-12b-it on the MPI-120 task by injection stride s and model layer ℓ.

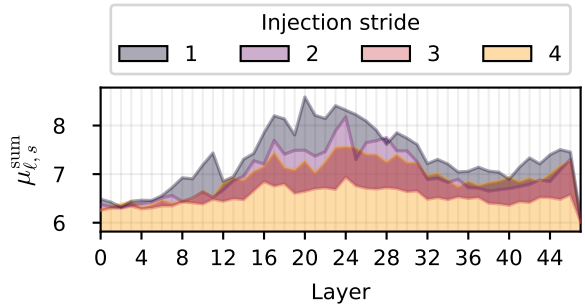

Figure 185. Overall MDS injections steering performance on gemma-3-12b-it on the SJTs task by injection stride s and model layer ℓ.

R.3 Layer Identification for MDS Injections

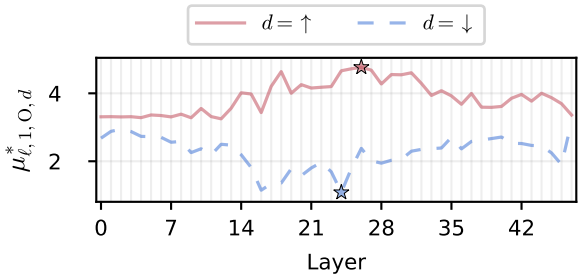

Figure 186. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on gemma-3-12b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

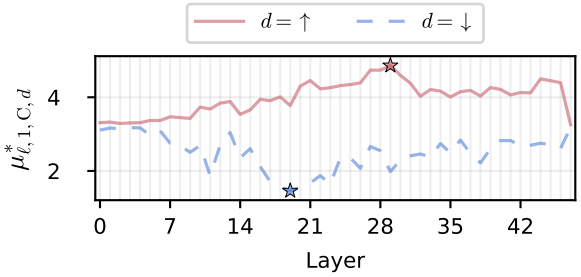

Figure 187. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on gemma-3-12b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

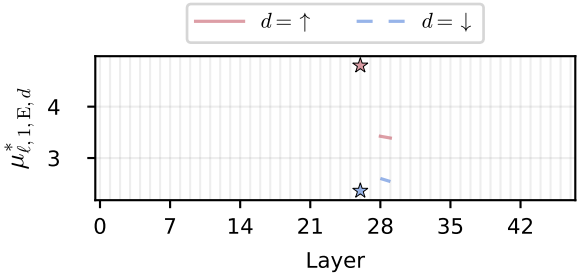

Figure 188. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on gemma-3-12b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

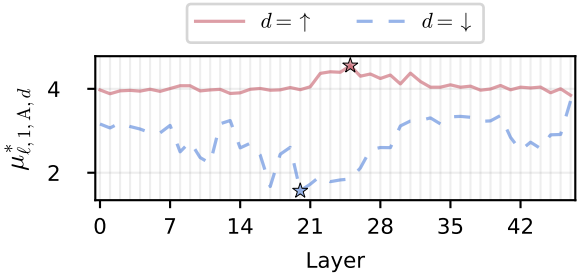

Figure 189. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on gemma-3-12b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

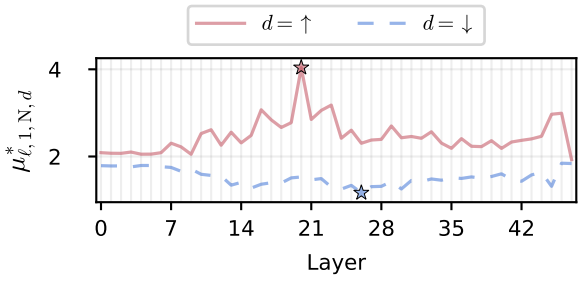

Figure 190. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on gemma-3-12b-it. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

R.4 OCEAN Trait Entanglement Under MDS Injections

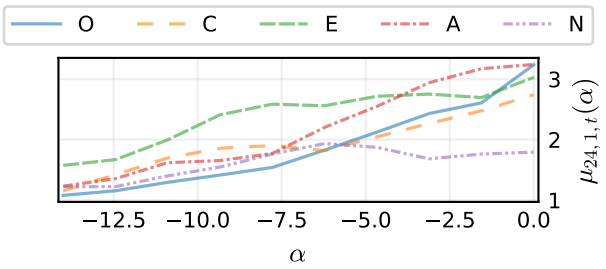

Figure 191. OCEAN scores for gemma-3-12b-it on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

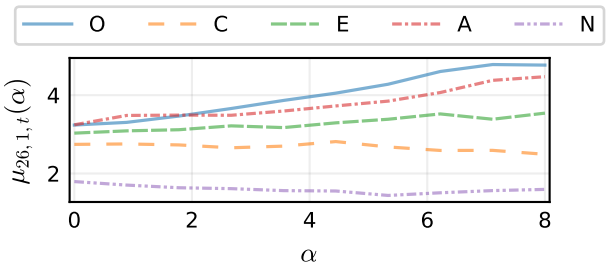

Figure 192. OCEAN scores for gemma-3-12b-it on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

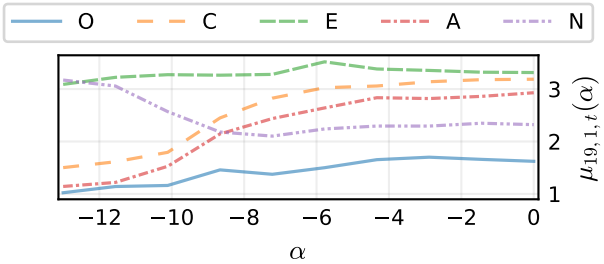

Figure 193. OCEAN scores for gemma-3-12b-it on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 19$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

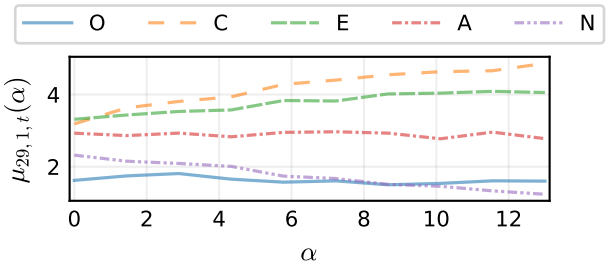

Figure 194. OCEAN scores for gemma-3-12b-it on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 29$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

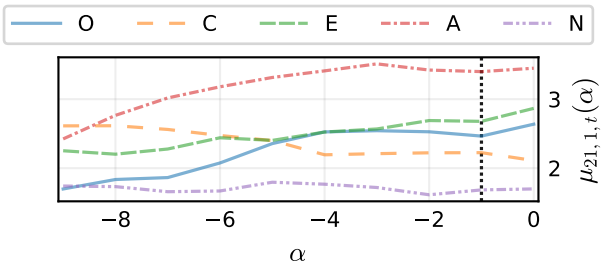

Figure 195. OCEAN scores for gemma-3-12b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 21$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = -1$.

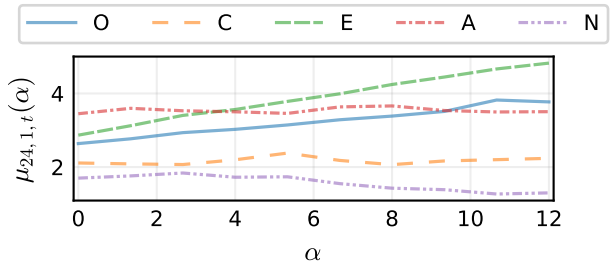

Figure 196. OCEAN scores for gemma-3-12b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

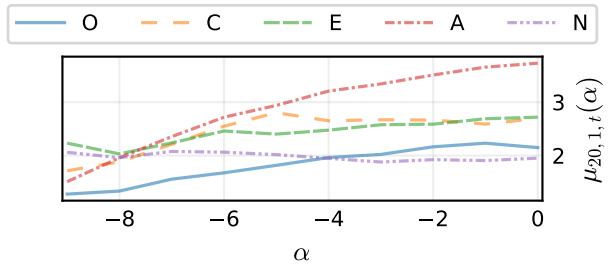

Figure 197. OCEAN scores for gemma-3-12b-it on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 20$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

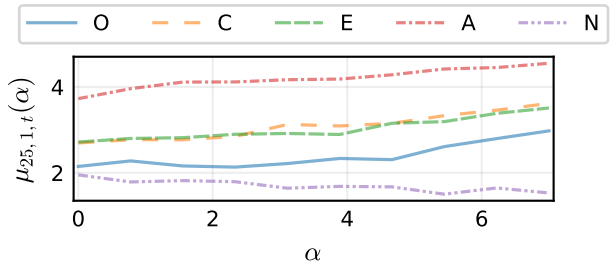

Figure 198. OCEAN scores for gemma-3-12b-it on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 25$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

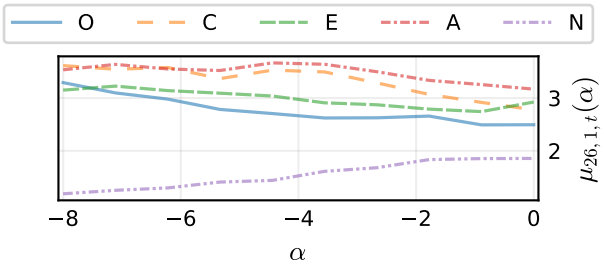

Figure 199. OCEAN scores for gemma-3-12b-it on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

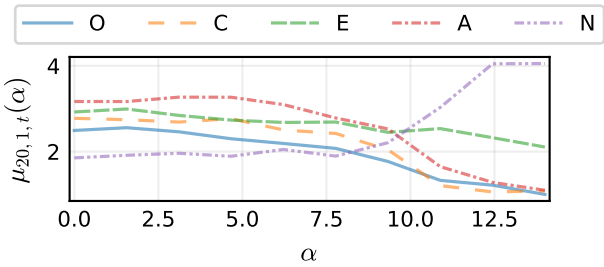

Figure 200. OCEAN scores for gemma-3-12b-it on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 20$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

S OCEAN Injection Results for gemma-3-27b-it

S.1 Injection Method Comparison

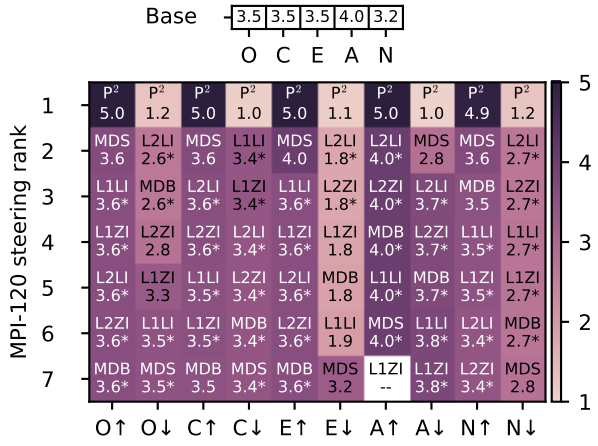

Figure 201. Ranking of steering methods on gemma-3-27b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

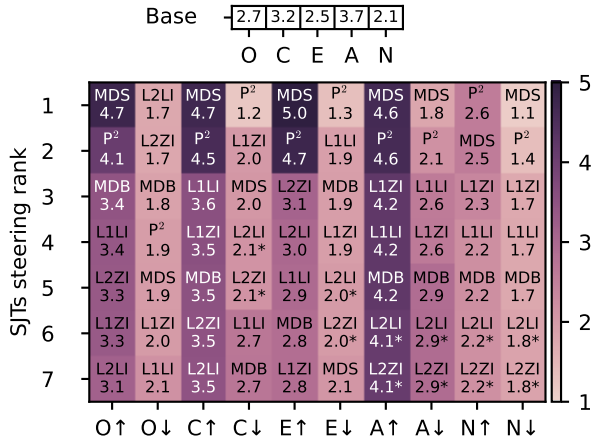

Figure 202. Ranking of steering methods on gemma-3-27b-it by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

S.2 Effects of Injection Stride on MDS Injections

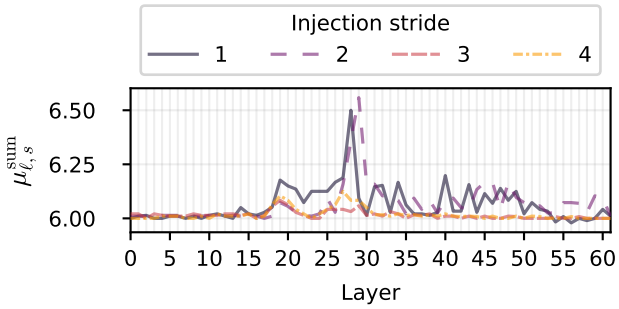

Figure 203. Overall MDS injections steering performance on gemma-3-27b-it on the MPI-120 task by injection stride s and model layer ℓ.

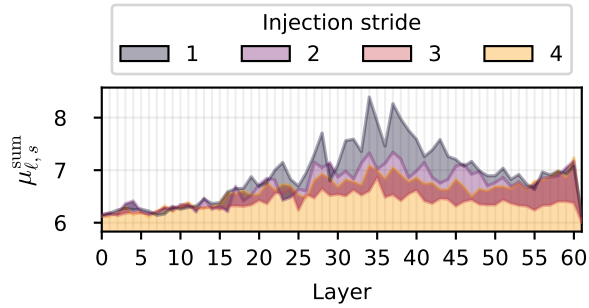

Figure 204. Overall MDS injections steering performance on gemma-3-27b-it on the SJTs task by injection stride s and model layer ℓ.

S.3 Layer Identification for MDS Injections

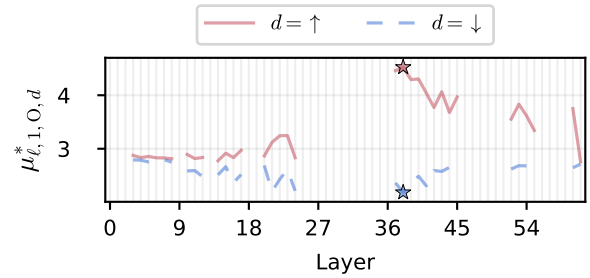

Figure 205. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on gemma-3-27b-it. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

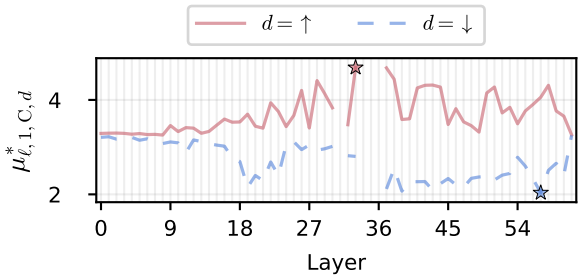

Figure 206. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on gemma-3-27b-it. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{C},d}$).

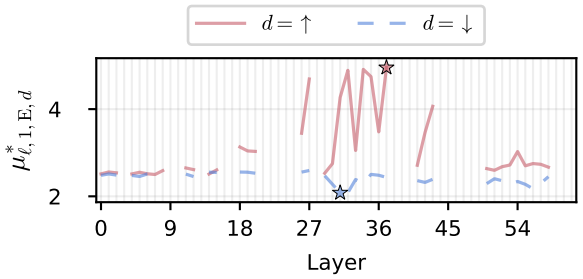

Figure 207. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on gemma-3-27b-it. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{E},d}$).

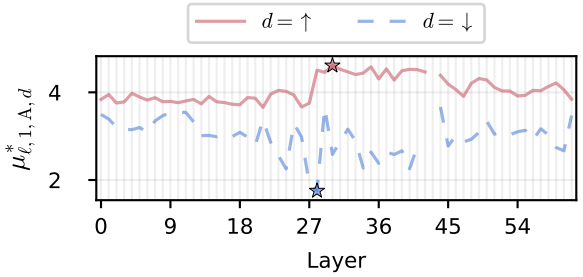

Figure 208. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on gemma-3-27b-it. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{A},d}$).

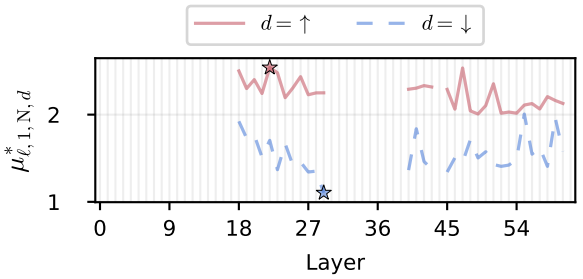

Figure 209. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on gemma-3-27b-it. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

S.4 OCEAN Trait Entanglement Under MDS Injections

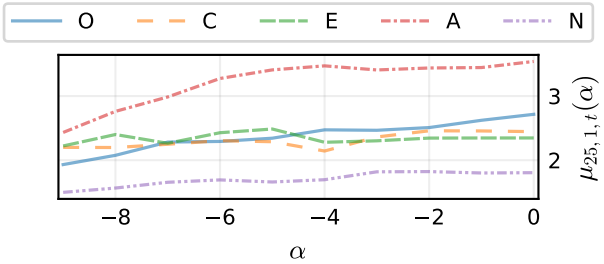

Figure 210. OCEAN scores for gemma-3-27b-it on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 25$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

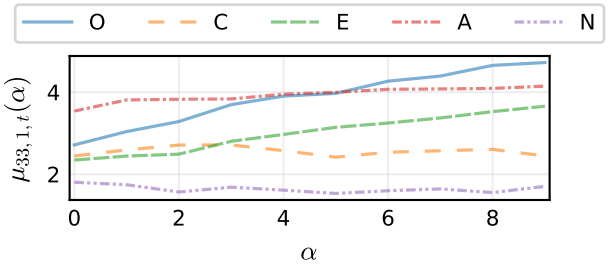

Figure 211. OCEAN scores for gemma-3-27b-it on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 33$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

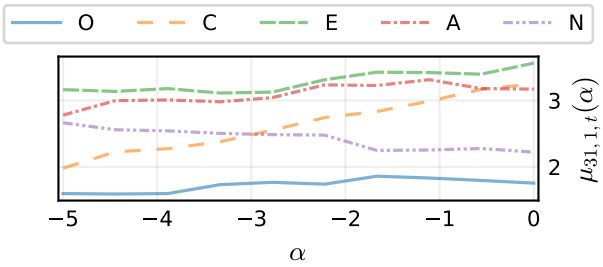

Figure 212. OCEAN scores for gemma-3-27b-it on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 31$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

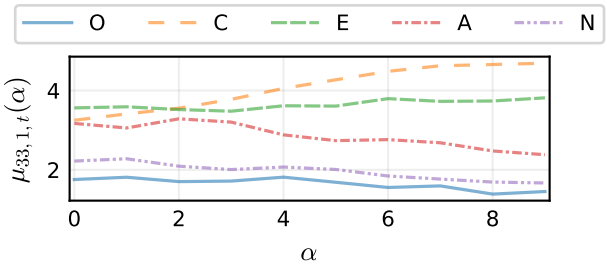

Figure 213. OCEAN scores for gemma-3-27b-it on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 33$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 9$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

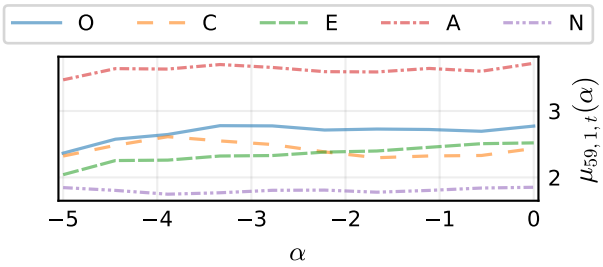

Figure 214. OCEAN scores for gemma-3-27b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 59$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 5$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

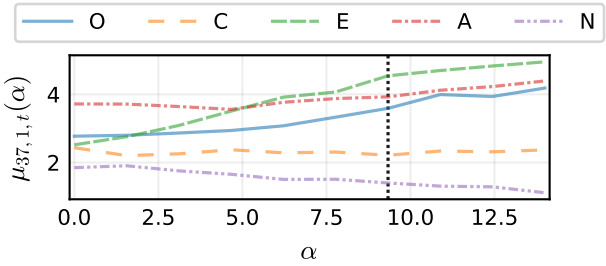

Figure 215. OCEAN scores for gemma-3-27b-it on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 37$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 14$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 9.333$.

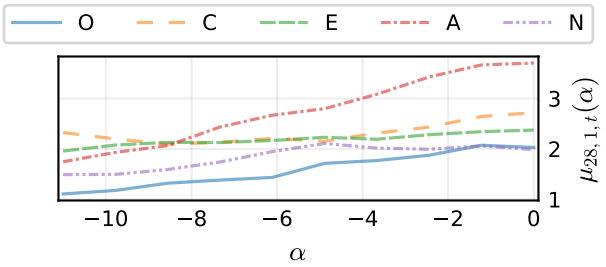

Figure 216. OCEAN scores for gemma-3-27b-it on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 28$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

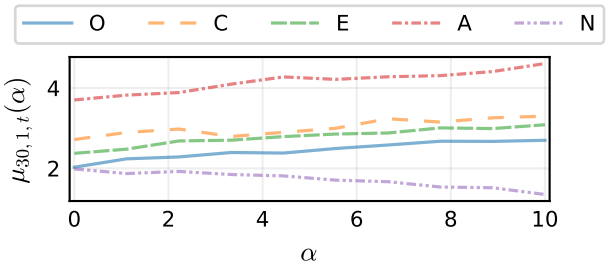

Figure 217. OCEAN scores for gemma-3-27b-it on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 30$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

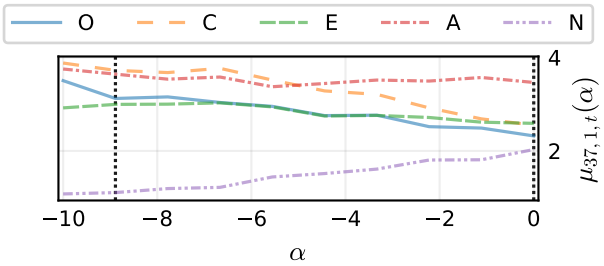

Figure 218. OCEAN scores for gemma-3-27b-it on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 37$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx -8.889$; 1 outlier at $\alpha = 0$.

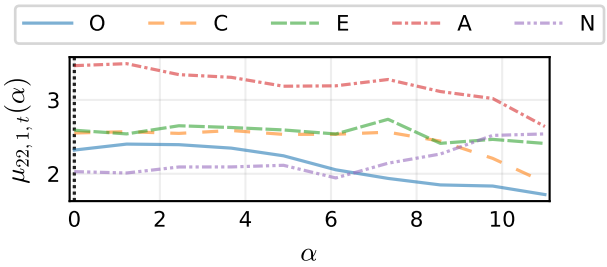

Figure 219. OCEAN scores for gemma-3-27b-it on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 22$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 0$.

T OCEAN Injection Results for Olmo-3-7B-Instruct

T.1 Injection Method Comparison

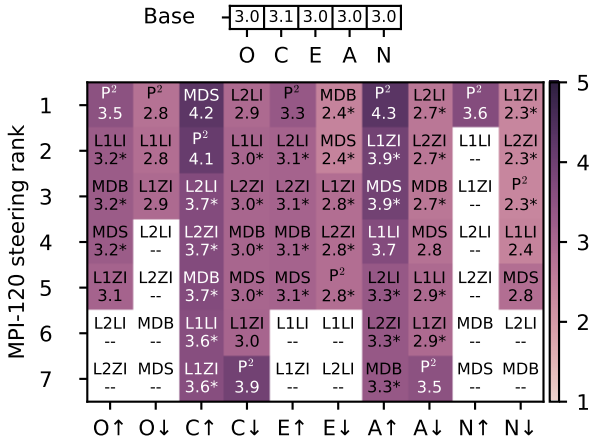

Figure 220. Ranking of steering methods on Olmo-3-7B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

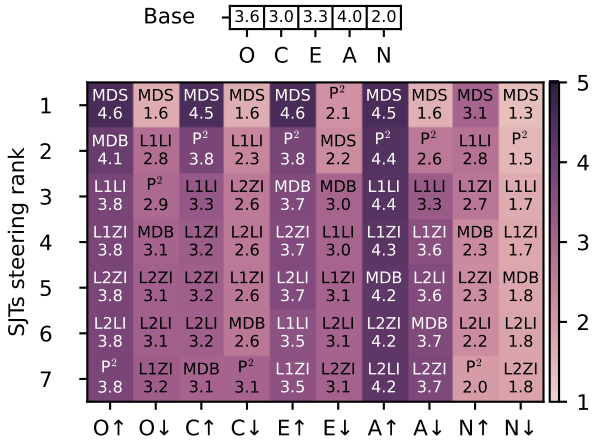

Figure 221. Ranking of steering methods on Olmo-3-7B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

T.2 Effects of Injection Stride on MDS Injections

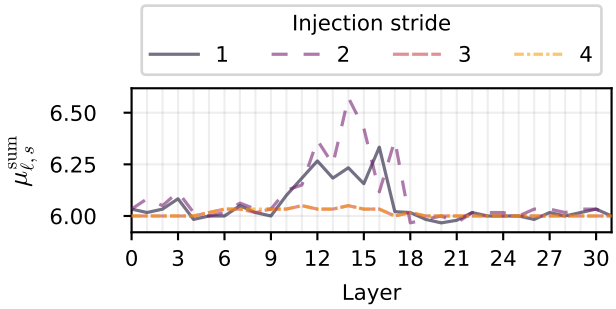

Figure 222. Overall MDS injections steering performance on Olmo-3-7B-Instruct on the MPI-120 task by injection stride s and model layer ℓ.

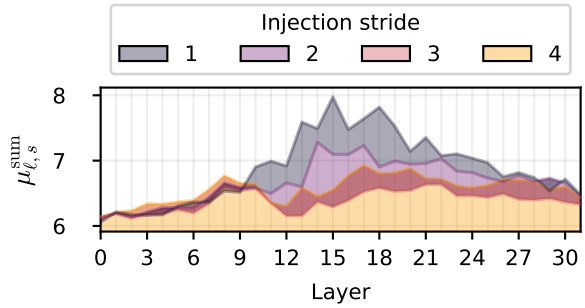

Figure 223. Overall MDS injections steering performance on Olmo-3-7B-Instruct on the SJTs task by injection stride s and model layer ℓ.

T.3 Layer Identification for MDS Injections

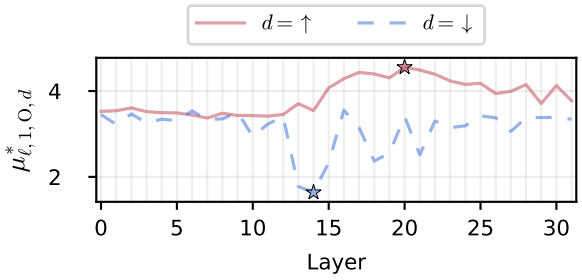

Figure 224. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Olmo-3-7B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{O},d}$).

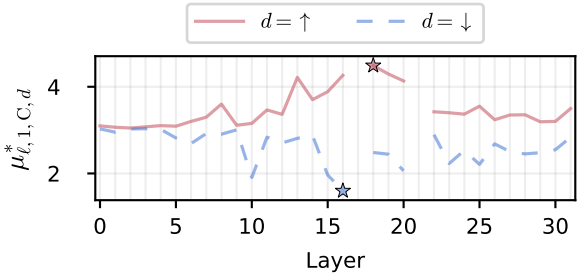

Figure 225. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Olmo-3-7B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{C},d}$).

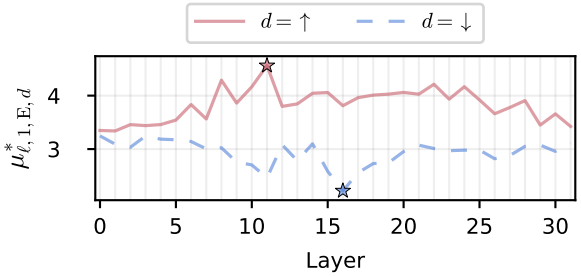

Figure 226. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Olmo-3-7B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{E},d}$).

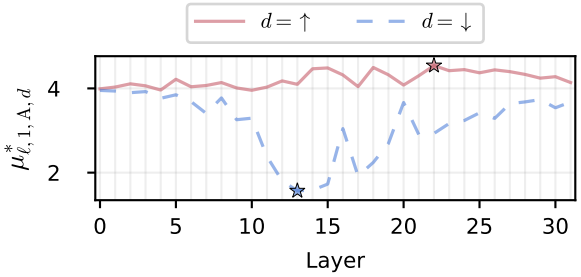

Figure 227. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Olmo-3-7B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\text{A},d}$).

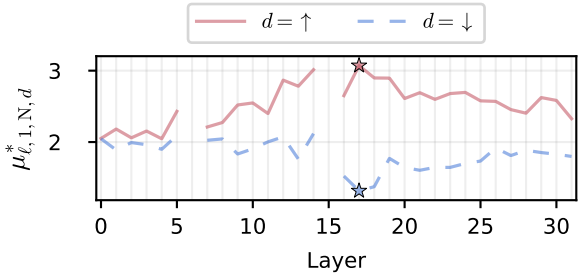

Figure 228. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Olmo-3-7B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

T.4 OCEAN Trait Entanglement Under MDS Injections

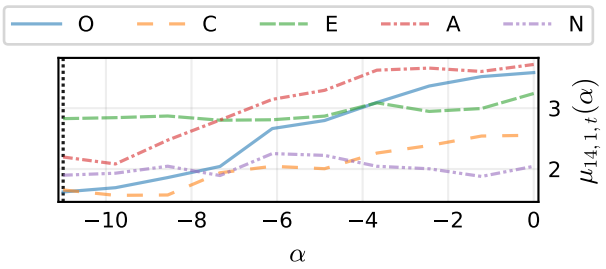

Figure 229. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the openness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 2 outliers at $\alpha = -11$.

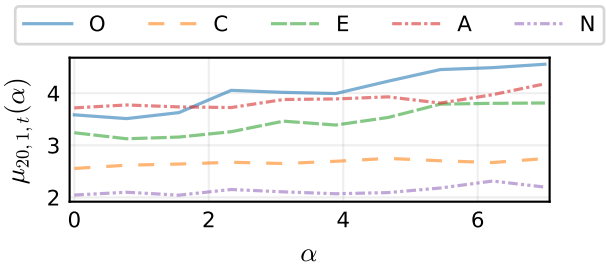

Figure 230. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 20$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 7$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

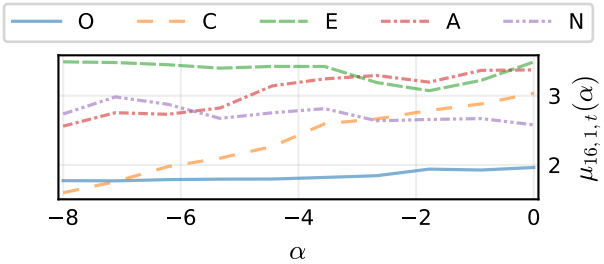

Figure 231. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

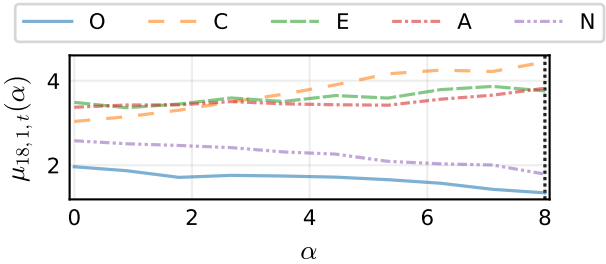

Figure 232. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 18$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha = 8$.

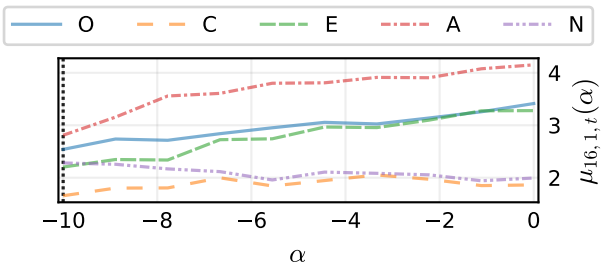

Figure 233. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 16$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 2 outliers at $\alpha = -10$.

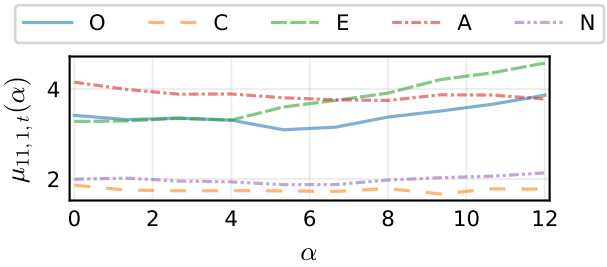

Figure 234. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 11$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

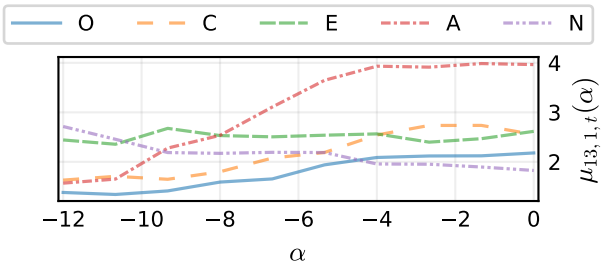

Figure 235. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 13$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

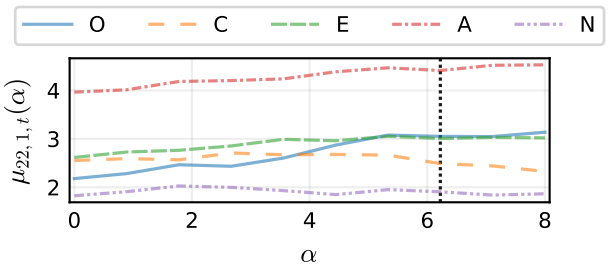

Figure 236. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 22$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 8$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 1 outlier at $\alpha \approx 6.222$.

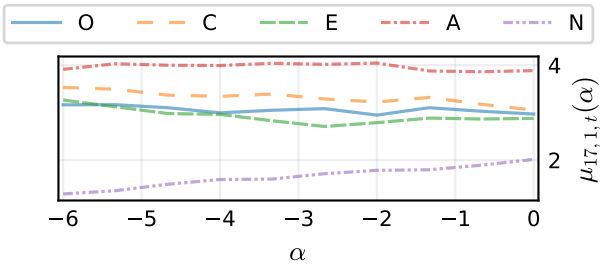

Figure 237. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 17$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

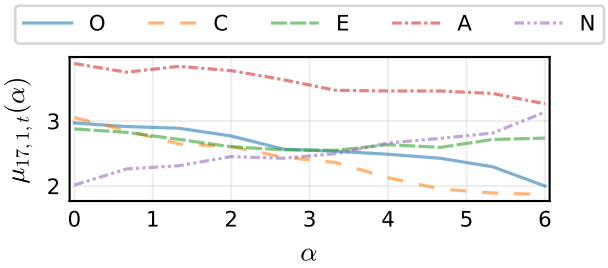

Figure 238. OCEAN scores for Olmo-3-7B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 17$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 6$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.

U OCEAN Injection Results for Olmo-3.1-32B-Instruct

U.1 Injection Method Comparison

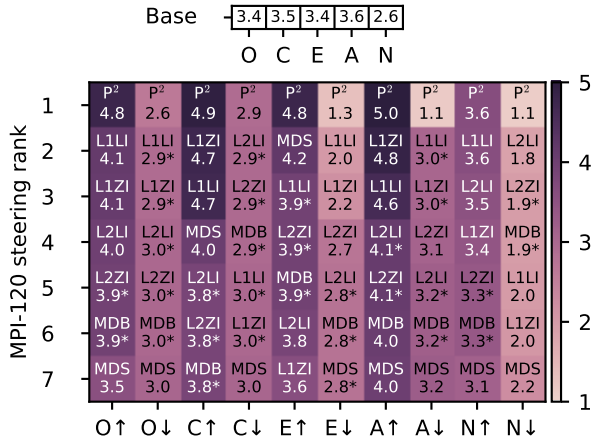

Figure 239. Ranking of steering methods on Olmo-3.1-32B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the MPI-120 task, with asterisks denoting ties in the unrounded results.

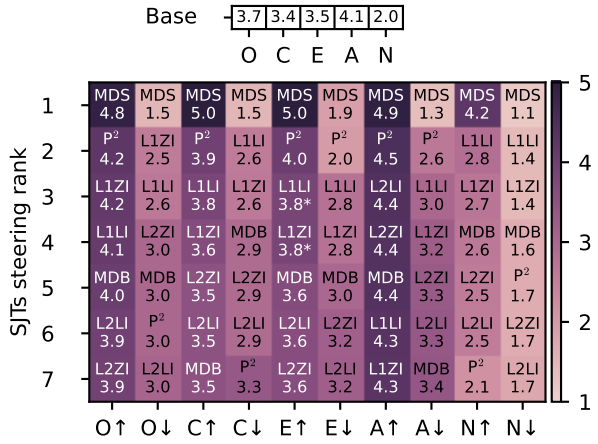

Figure 240. Ranking of steering methods on Olmo-3.1-32B-Instruct by OCEAN trait and direction. Based on each method's strongest steering scores ($\phi_{1,t,d}$) on the SJTs task, with asterisks denoting ties in the unrounded results.

U.2 Effects of Injection Stride on MDS Injections

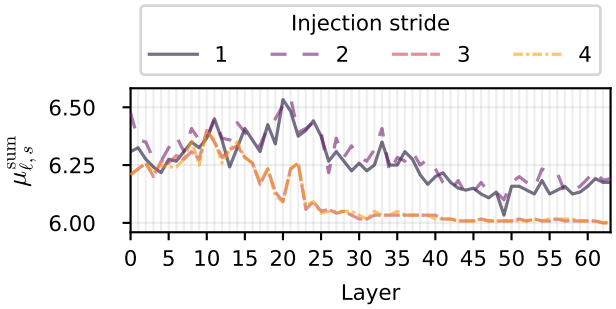

Figure 241. Overall MDS injections steering performance on Olmo-3.1-32B-Instruct on the MPI-120 task by injection stride s and model layer ℓ.

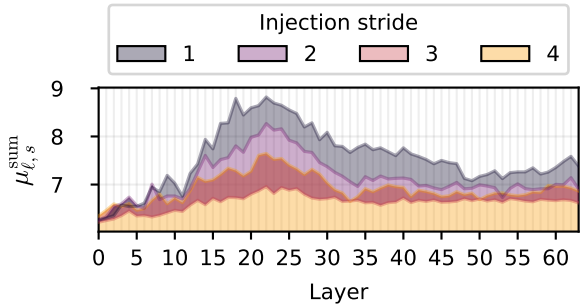

Figure 242. Overall MDS injections steering performance on Olmo-3.1-32B-Instruct on the SJTs task by injection stride s and model layer ℓ.

U.3 Layer Identification for MDS Injections

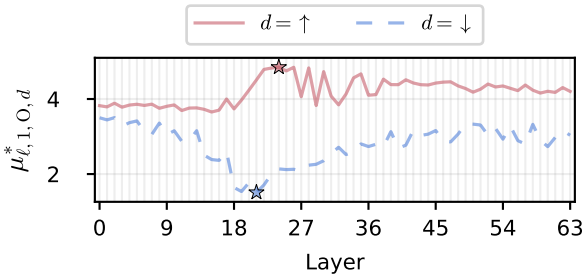

Figure 243. Openness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying openness MDS injections with injection stride $s = 1$ on Olmo-3.1-32B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{O},d}$).

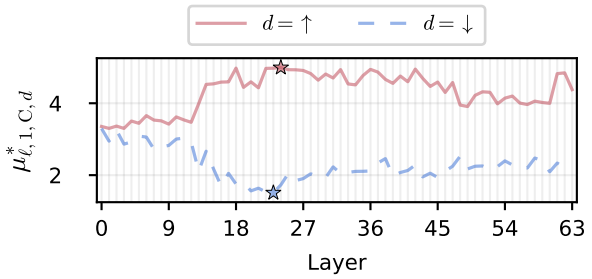

Figure 244. Conscientiousness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying conscientiousness MDS injections with injection stride $s = 1$ on Olmo-3.1-32B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{C},d}$).

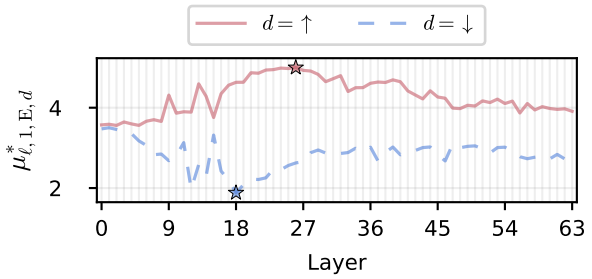

Figure 245. Extraversion extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying extraversion MDS injections with injection stride $s = 1$ on Olmo-3.1-32B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{E},d}$).

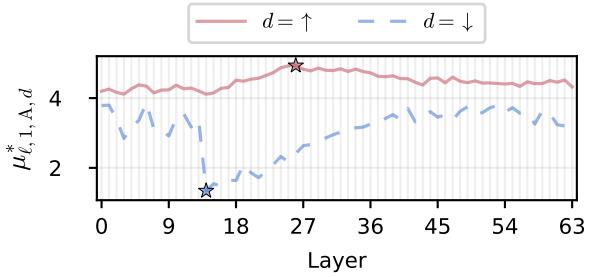

Figure 246. Agreeableness extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying agreeableness MDS injections with injection stride $s = 1$ on Olmo-3.1-32B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{A},d}$).

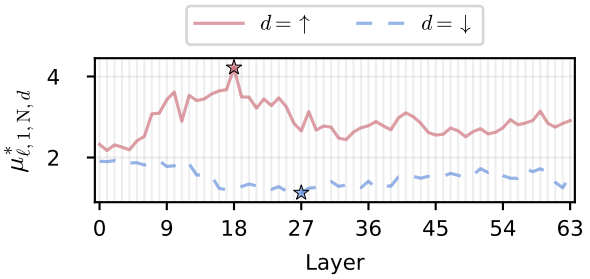

Figure 247. Neuroticism extreme steering scores on the SJTs task by direction $d \in \{\uparrow, \downarrow\}$ and model layer ℓ, after applying neuroticism MDS injections with injection stride $s = 1$ on Olmo-3.1-32B-Instruct. A star marks the strongest steering effect across layers ($\phi_{1,\mathrm{N},d}$).

U.4 OCEAN Trait Entanglement Under MDS Injections

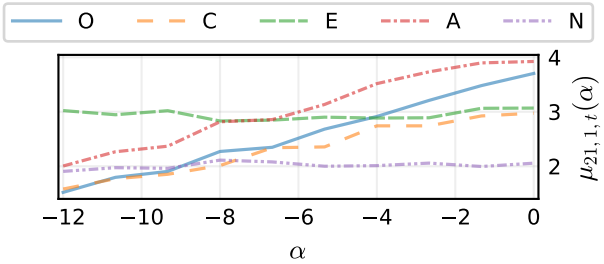

Figure 248. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 21$ is the best-performing for decreasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

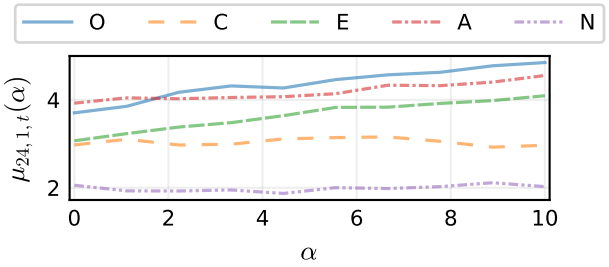

Figure 249. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under an openness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for increasing openness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the openness SJTs.

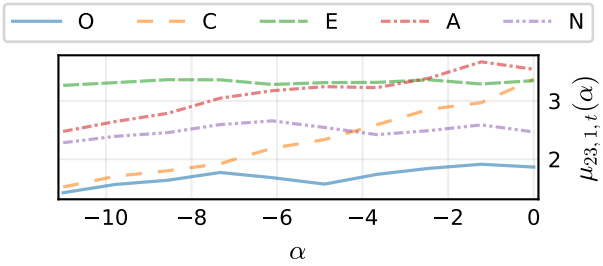

Figure 250. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 23$ is the best-performing for decreasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs.

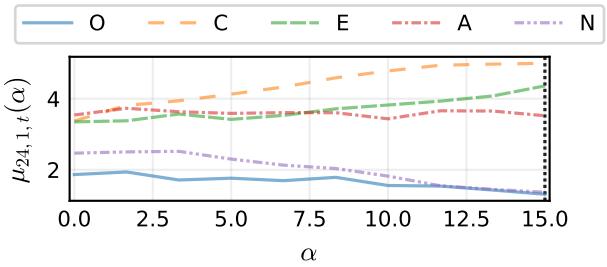

Figure 251. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under a conscientiousness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 24$ is the best-performing for increasing conscientiousness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 15$ for that condition. Fluency was evaluated only in the responses to the conscientiousness SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 2 outliers at $\alpha = 15$.

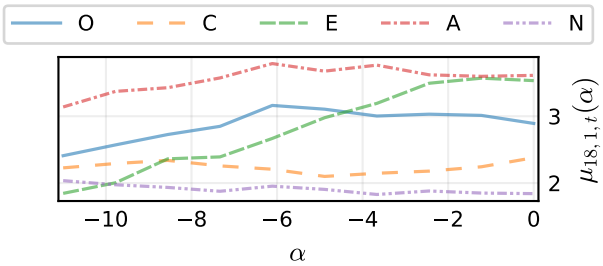

Figure 252. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 18$ is the best-performing for decreasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 11$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

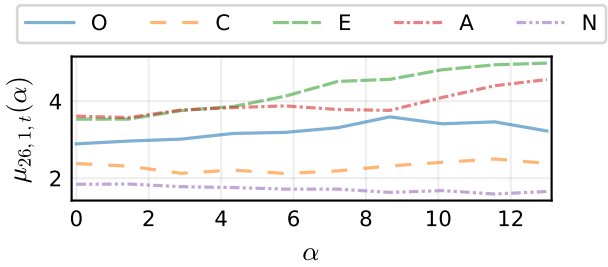

Figure 253. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under an extraversion MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for increasing extraversion, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the extraversion SJTs.

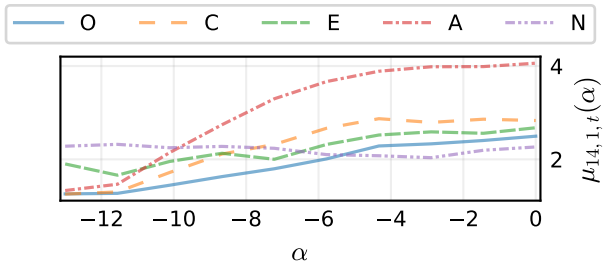

Figure 254. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 14$ is the best-performing for decreasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 13$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

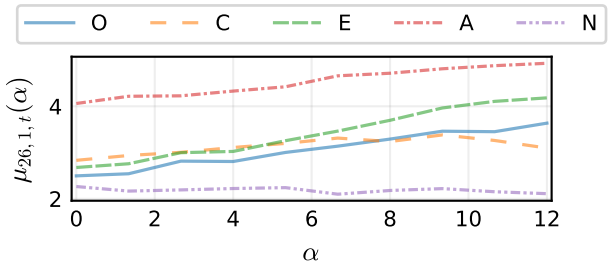

Figure 255. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under an agreeableness MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 26$ is the best-performing for increasing agreeableness, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 12$ for that condition. Fluency was evaluated only in the responses to the agreeableness SJTs.

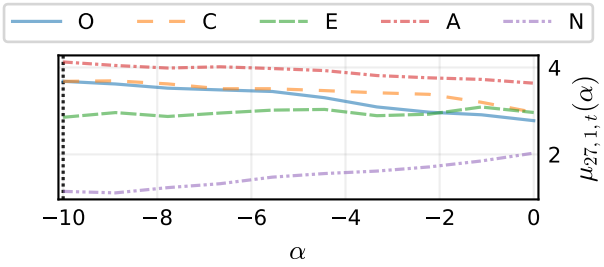

Figure 256. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 27$ is the best-performing for decreasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs. Vertical lines indicate where a α run crossed our fluency thresholds, noting that nonfluent results may also occur even without applying an injection. Nonfluency cases: 2 outliers at $\alpha = -10$.

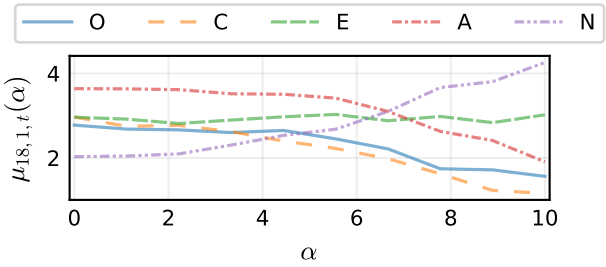

Figure 257. OCEAN scores for Olmo-3.1-32B-Instruct on SJTs, under a neuroticism MDS injection with $s = 1$ and 10 different α values. The selected layer $\ell = 18$ is the best-performing for increasing neuroticism, and the selected α values are 10 equidistant points ranging from 0 (no steering) to the best-performing $\alpha = 10$ for that condition. Fluency was evaluated only in the responses to the neuroticism SJTs.